BIOMASS FUTURES

Biomass role in achieving the Climate Change & Renewables EU policy targets. Demand and Supply dynamics under the perspective of stakeholders . **IEE 08 653 SI2. 529 241**

Atlas of EU biomass potentials

Deliverable 3.3:

Spatially detailed and quantified overview of EU biomass potential taking into account the main criteria determining biomass availability from different sources

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Preface

This publication is part of the BIOMASS FUTURES project (Biomass role in achieving the Climate Change & Renewables EU policy targets. Supply dynamics under the perspective of stakeholders - IEE 08 653 SI2. 529 241, www.biomassfutures.eu) funded by the European Union's Intelligent Energy Programme.

In this publication a mapped and quantified overview is given of different biomass feedstocks. This information has been further combined with cost information to derive cost-supply curves at national and EU wide scale. This report should serve as a first basis for further discussion and guidance from project partners and stakeholders as to the further elaboration of the environmentally and technically constrained biomass potentials in 2010, 2020 and 2030.

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List of Abbreviations

BEE Biomass Energy Europe project

BL Black Liquor

BMW Biodegradable Municipal Waste

CAP Common Agricultural Policy

CAPRI Common Agricultural Policy Regionalised Impact model

CEEC Central and Eastern European Countries

CLC CORINE Land Cover

CO_2 Carbon dioxide emission (a greenhouse gas)

CORINE Coordination of Information on the Environment

DEM Digital Elevation Model

DM Dry matter

EC European Commission

EEA European Environment Agency

EO Earth Observation

EU European Union

Eurostat Statistical institute of the EU

FAO Food and Agriculture Organisation of the United Nations

GHG Greenhouse gas

GJ GigaJoule

GLOBIOM Global Biomass Optimization Model

GMES Global Monitoring for Environment and Security

GSD Ground Sampling Distance

Ha Hectare

HNV High Nature Value

IPCC Intergovernmental Panel on Climate Change

Ktoe Kilo Tonnes Oil Equivalents

LHV Lower heating value

MSW Municipal Solid Waste

Mtoe Million tonnes of oil equivalent

NUTS Nomenclature of Territorial Units for Statistics

PRIMES Energy System Model

RESolve RESolve model kit consists of 3 models: RESolve-E (Electricity), RESolve-T (Transport model) and RESolve-H (heat)

RES Renewable Energy Sources

RS Remote Sensing

SRC Short rotation coppice

SRF Short Rotation Forestry

Toe Tonnes of oil equivalent

UNFCCC United Nations Framework Convention on Climate Change

1 Introduction

1.1 Objective of this report

Within the Biomass Futures project (www.biomassfutures.eu) we aim within work packages 3 and 4 to provide a comprehensive strategic analysis of biomass supply options and their availability in response to different demands in a timeframe from 2010- 2030. This is done in different steps. The steps presented in this report relate to:

1) Identifying different biomass feedstocks and make an inventory of data to quantify and map the technically constrained biomass potentials. This also includes estimates of alternative uses of by- and waste in order to estimate the share that is available for bioenergy purposes and the share that competes with other uses.

2) Map present technically constrained potentials of the different feedstock as spatially explicit as possible (regional level)

3) Determine scenario specifications according to which future 2020 potentials can be estimated

4) Quantify actual, 2020 and 2030 potentials according to scenarios from step 3)

5) Identifying information on which basis cost levels for the different feedstocks can be established taking into account competing uses and costs for production, yielding and transport. It is aimed at estimating costs for biomass at it is received at the gate of the conversion/pre-treatment plant.

6) Synthesizing the results in terms of economic supply estimates (cost-supply).

The results in this report should serve as a first basis for:

1) further discussion and guidance from project partners and stakeholders as to the further elaboration of the environmentally constrained biomass potentials in 2020 and 2030.

2) As input for the model chains of PRIMES, RESOLVE and GLOBIOM to assess final bioenergy production and shares and related environmental impacts.

The biomass supplies and related cost levels are presented for the actual situation (2010), 2020 and 2030. The same applies to the cost levels.

1.2 Types of biomass potentials

Several biomass potential studies have been done in the last decades. Their approaches have been very different and their results difficult to compare and interpret. The BEE study was developed in response to this[1]. It provides a wide overview of state-of-the-art biomass resource assessments and it also proposes several generic approaches, definitions, conversions and a classification of biomass feedstock types in order to improve the accuracy and comparability of future biomass resource assessments (Retenmaier et al., 2008 and Vis et al., 2010). In the Biomass Futures project we have therefore built as much as possible on the state-of-the-art overview of biomass assessment studies provided by BEE and we use as much as possible the same biomass classification, definitions and conversions.

[1] See BEE project website: http://www.eu-bee.com/

Table 1.1 Overview of main biomass categories, their definition and criteria to spatially identify their potential

Sector	Biomass category	Biomass type detail	General definition	Specific definition
Biomass from agriculture	Energy crops	Woody/ligno-cellulosic biomass	Biomass from agricultural production activities	Solid (ligno cellulosic& woody) energy crops (for generating electricity & heat, 2nd generation biofuels)
	Energy crops	Sugar, starch, oil	Biomass from agricultural production activities	Crops for biodiesel & bioethanol (1st generation: sugar/starch & oil crops)
	Energy crops	wet biomass	Biomass from agricultural production activities	Energy maize and maize residues (for biogas)
	Agricultural primary residues	Dry manure	Biomass from agricultural production activities	dry manure (poultry, sheep & goat manure)
	Agricultural primary residues	Wet manure	Biomass from agricultural production activities	pig and cattle manure
	Agricultural primary residues	Solid agricultural residues	Biomass from agricultural cultivation, harvesting and maintenance activities	Other solid agricultural residues (prunnings, orchards residues)
	Agricultural primary residues	Solid agricultural residues	Biomass from permanent (semi-natural) grasslands	grass
	Agricultural primary residues	Solid agricultural residues	Biomass from agricultural cultivation and harvesting activities	straw/stubbles (cereals, sunflower, RAPE)
Biomass from forestry	Forestry biomass	Woody biomass	Biomass from forestry: forests and other wooded land, incl. tree plantations and short rotation forests (SRF)	Stem wood production

	Forestry biomass	Woody biomass	Biomass from forests and other wooded land incl. tree plantations)	Volume of additionally harvested wood realistically available for bioenergy
	Primary forestry residues	Woody biomass	Cultivation and harvesting / logging activities in forests and other wooded land. Biomass from trees/hedges outside forests incl. landscape elements	Available volume of felling residues (branches and roots) and woody residues from landscape maintenance activities outside forests.
	Secondary forestry residues	Woody biomass	Biomass coming from wood processing, e.g. industrial production	Bioenergy potential of wood processing residues (e.g., woodchips, sawdust, black liquor)
Biomass from waste	Primary residues	Biodegradable waste	Biomass from road side verges	Biomass residues/solid biomass resulting from maintenance activities (e.g. from grass and woody cuttings from road side verges)
	Secondary residues	Solid and wet agricultural residues	Processing of agricultural products, e.g. for food and feed	Processing residues (e.g. pits from olive pitting, shells/husks from seed/nut shelling and slaughter waste).
	Tertiary residues	Biodegradable waste	Biomass coming from private households and/or private residential gardens	Organic household waste incl. woody fractions, e.g. food leftovers, waste paper, discarded furniture,)
	Tertiary residues	Organic waste from industry and trade	Biomass from industry and trade, excl. forest industry	Organic waste from industry and trade incl.woody fractions, e.g. bulk transport packaging, recovered demolition wood (excluding wood which goes to non-energy uses),
	Waste biomass	Biodegradable waste	From industry and private households	Sewage sludge

In Table 1.1 an overview is presented of all biomass categories involved in the inventory of biomass supply in this report. As becomes clear, there are three sectors under which the biomass categories have been classified: agriculture, forestry and waste. Under these main sectors there are categories of dedicated biomass production such as biofuel crops, woody and grassy crops, stem wood production and by-products and waste categorized in primary, secondary and tertiary levels. The classification of the biomass types follows as much as possible the BEE project categories. However, from a mapping perspective it was logical to take the three sectors agriculture, forest, waste as the principle categorization as these sectors have a clear territorial component (at least for the first 2). Because of this some, differences in classification occur in the underneath table with the BEE classification. For example woody biomass and residues in BEE are all categorized under forestry and forestry residues, while in the underneath table we have decided to put residues from trees outside forest land (e.g. fruit trees, vineyards etc.) under (primary) agricultural biomass.

All categories of biomass resources in the table have been mapped in order to quantify the technical potential and the areas in which the highest concentrations are found. The mapped results are included in the report. The next steps are to translate this technical potential into on the one hand an economic potential and on the other hand an environmentally sustainable potential. The first is done by linking the potentials to a price level and the cost-supply combination of information forms the input for the demand modelling with the RESOLVE and PRIMES models. The modelling results provide the final economic potential. Specific environmental constraints are already taken into account for both present and 2020 potential estimates, but differ per biomass feedstock type. Their input together with the cost information into the demand modelling will eventually result in the economic environmental potential. The environmental constraints are addressed in scenarios.

The sustainable potential will be introduced in the next section of this paper and should result in a further adaptation of the potential estimates to environmentally constrained supplies of biomass resources. This however will be addressed in further steps in the project at various levels:

1) Supply maps taking further account of scenario specifications in relation to environmental sustainability

2) Integrated economic model (GLOBIOM) also building on sustainability criteria in different scenarios

3) Stakeholder workshops and policy briefings developed within the scope of this project

1.3 Scenarios and sustainability criteria applied in biomass potential estimates

It is not without a reason that there is large emphasis on sustainability when realizing the EU renewable targets. Firstly because reduction of GHG emissions for mitigating climate change is one of the main drivers for setting these targets. Secondly because there is still a long way to go before the targets are reached and it is clear that a tremendous increase in biomass production/collection is needed which may have important effect on EU-wide and global agricultural land demand and overall environmental quality.

This is why the biomass potential estimates in this report are made for different scenario situations taking into account different sustainability criteria. For some of these criteria it is already clear that they will constrain the near future availability of biomass as they have already been addressed in EU policy. For biomass feedstock to be used for conversion into biofuel there are already mandatory sustainability criteria formulated at EU level, while for solid and gaseous biomass feedstock there are only recommendations formulated by the Commission to be adopted at a voluntary basis by the Member States (MS).

EU policies for renewable energy and sustainability criteria

In December 2008, the European Parliament adopted the 'Directive on the promotion of energies from renewable sources' (Directive 2009/28/EC) (RES Directive) as part of the EU Climate and Energy Package. Above all, the Directive set a general binding target for the European Union to have 20 % of its final energy consumption provided by renewable sources by 2020. It also includes a specific target of having a minimum of 10 % of the total energy used in the transport sector coming from renewable energy sources.

The latter target is accompanied by a novel policy instrument: All biofuels and other bio-liquids counting towards the target must meet a set of mandatory sustainability criteria to achieve greenhouse gas reductions compared to fossil fuels[2] and to mitigate risks related to areas of high biodiversity[3] value and areas of high carbon stock[4].

The RES Directive should be implemented by Member States by December 2010. A key element of the implementation are National Renewable Energy Action Plans (NREAPs) in which Member States have to report to the European Commission how they intend to fulfil the targets set by the Renewables Directive. Based on existing data sources and information, this document will give an overview of the status quo, starting with the specific RE targets of the Member States, followed by the instruments to be applied to promote the development of renewable energies.

For solid and gaseous biomass sources the Commission has put forward recommended sustainability criteria which can be adopted by Member States, but are not binding. The following criteria for inclusion into national schemes are recommended by the Commission;

- A general prohibition on the use of biomass from land converted from primary forest, other high carbon stock areas and highly biodiverse areas.
- A common greenhouse gas calculation methodology which could be used to ensure that minimum greenhouse gas savings from biomass are at least 35 % (rising to 50 % in 2017 and 60 % in 2018 for new installations) compared to the EU's fossil energy mix.
- A differentiation of national support schemes in favour of installations that achieve high-energy conversion efficiencies.
- Monitoring of the origin of biomass.

In the framework of the Biomass Futures project detailed analyses will be provided for the way sustainability criteria may constraint the biomass feedstock availability. This will be addressed through application of three different scenarios with sustainability requirements ranging from limited to very strict criteria. These will be presented in the next section and are also used in the rest of this report to show the range in future biomass potentials.

[2] According to Art. 17.2 of the RED, biofuel production must comply with a GHG saving of 35% in 2008 compared to fossil fuels. This rate increases up to 60% in 2018.

[3] Land of high biodiversity value is address in the RED in Art. 17.3 as "primary forests and other woodlands", "nature protection areas" and "highly biodiverse grassland".

[4] The aim of Art. 17.4 of the RED is the protection of areas with high carbon stock to avoid the emission of high amounts of GHG by land conversion. Here wetlands, forested areas (tree cover above 30%) and areas with a tree cover of 10-30% are addressed. These land categories may be used for biomass production as long the status of these areas will not change. For example, a forested area can be logged, but it must be guaranteed that the forest will re-grow. In Art. 17.5 the protection of peatland is covered in a similar manner. Peatland can only be used when it is proven that cultivation and harvesting of biomass does not involve drainage of previously undrained soils.

1.4 The Biomass Futures scenarios

In this project there are 2 scenarios applied:

1) Reference scenario

2) Sustainability scenario

In Table 1.2 underneath a detailed overview is given of the sustainability criteria applied in 2020 and 2030 in every scenario. The sustainability criteria applied in the reference storyline are following the 'Directive on the promotion of energies from renewable sources' (Directive 2009/28/EC) (RES Directive) as described above and therefore only apply to biofuels and bioliquids.

Table 1.2 Criteria applied in reference and sustainability scenarios

Scenario	GHG mitigation criteria 2020	GHG mitigation criteria 2030	Other sustainability constraints 2020 and 2030
Reference	Only for biofuels and bioliquids consumed in EU a GHG mitigation of 50% as compared to fossil fuel is required. This **excludes** compensation for iLUC related GHG emissions.	Only for biofuels and bioliquids consumed in EU a GHG mitigation of 50% as compared to fossil fuel is required. This **excludes** compensation for iLUC related GHG emissions.	Only for biofuels and bioliquids consumed in EU limitations on the use of biomass from biodiverse land or land with high carbon stock.
Sustainability	For all bioenergy consumed in the EU the following mitigation requirements are set: Biofuel/bioliquids: 70% mitigation as compared to fossil fuel (comparator EU average diesel and petrol emissions 2020). Bioelectricity and heat: 70% mitigation as compared to fossil energy (comparator country specific depending on 2020 fossil mix) . This **includes** compensation for iLUC related GHG emissions.	For all bioenergy consumed in the EU the following mitigation requirements are set: Biofuel/bioliquids: 80% mitigation as compared to fossil fuel (comparator EU average diesel and petrol emission 2030) Bioelectricity and heat: 80% mitigation as compared to fossil energy (comparator country specific depending on 2030 fossil mix) This **includes** compensation for iLUC related GHG emissions.	For all bioenergy consumed in EU limitations on the use of biomass from biodiverse land or land with high carbon stock.

In the sustainability scenario stricter sustainability criteria apply and these are also applicable to solid and gaseous biomass sources. In the sustainability scenario these apply to all bioenergy sources both produced inside and outside the EU (either domestically produced or imported). A very important difference with the reference scenario is that this GHG mitigation requirement should also include compensation for emissions from indirect land use changes caused by biomass cropping in the EU. For an explanation of how this iLUC related emission is estimated and when there is an indirect land use effect applicable see next Chapter.

1.5 Outline of report

This report consists of 5 chapters including this introductory chapter. In the next 3 chapters the biomass-supply of the agricultural, forest and waste sectors is presented in the different scenario situations for 2020 and 2030 situation. For several biomass sources also present potential is presented. All biomass sources covered in the chapters are already summarized in the Table 1.1 presented in the former. This is then followed by Chapter 5 in which an overview is given of the cost levels of the different biomass feedstocks and an integration of the results is discussed.

2 Biomass from agricultural land and by-products

2.1 Actual energy cropping

It should be realised that the EU policy ambitions go far beyond current consumption of renewable energy. In 2009 the whole EU reached a total final renewable energy (RES) consumption of 70.1 MTOE which amounted to 10% of the total gross final energy consumption. Bioenergy based on waste and biomass makes up 69% of the RES consumption and is by far the most important renewable. This bioenergy makes up respectively 5.5%, 4% and 0.8% of the renewable energy share in the heat, electricity and transport sector (EC, 2010). Given the targets set in the RES-Directive and NREAPs it is clear that to reach the 2020 targets, there still needs to be a tremendous increase in RES production including bioenergy. To produce the remaining 10% RES share, particularly for the biofuels targets set for 2020, large amounts of biomass are required. This will particularly lead to further increases in cropped biomass as with present state of technology most fuels will still be based on rotational arable crops providing sugar, starch and or oils as feedstock. Second generation biofuels based on ligno-cellulosic material cannot be expected to become economically viable at large scale within the next 10 years. This implies that large land areas are needed both inside and outside Europe for biofuel feedstock production but also, although to a lesser extent, for feedstock for renewable heat and electricity production. The demand in the latter category is however less land related as it can mostly be satisfied by waste and by-products from several sources.

Although estimates of the size exhibit a large variation. The European Commission (2008) calculated that 17,5 mln hectares of land would be required to reach the 10% biofuels target, which would amount to about 10% of the total Utilised Agricultural Area (UAA) in EU27. Their starting point was that 50% of the production would come from cultivation of rotational biomass crops for 1st generation technology biofuels. The other 50% would come from ligno-cellulosic by-products and perennial biomass crops or imports from outside the EU. For conversion of these ligno-biomass feedstock they assumed 2nd generation biofuel technology to become commercially available before 2020. The OECD (2006) is less optimistic and estimates that about 45 million hectares of land are required to reach the EC-targets by 2020. Their estimates are purely based on 1st generation biofuel technologies and they assume yields to remain at the same levels as they are now.

It is clear that the pressure on land will increase strongly under a growing biomass demand. This may cause adverse effects on biodiversity as it may lead to the further intensification of existing land uses, both in agricultural and forest lands, but also the conversion of non-cropped biodiversity-rich land into cropped or forest area. The conversion of biodiversity rich grasslands for example is meant to be prevented with the sustainability scheme for biofuels to be introduced together with the approval of the biofuels target of 10%. The RES directive states that biofuels shall not be made from raw material obtained from land with recognized high biodiversity value, such as undisturbed forest, areas designated for nature protection purposes or highly biodiverse grasslands. However, the big question is how this land resource is exactly defined and identified (e.g. mapped) and whether not being accountable to the renewable energy target provides enough protection to valuable ecosystems in markets offering very high prices to biomass feedstock.

In addition there is also an increasing resistance against using existing arable land for the production of biomass at the expense of food and feed production. There are indications that this will endanger the food security situation, especially in third world countries, and that indirect land use changes may take place by bioenergy production pushing food and feed production into uncultivated areas causing loss of valuable natural habitats (e.g. tropical rain forest and savannah) and tremendous releases of green house gas (GHG) stocks in the soil.

2.1.1 Actual energy crops

Based on a compilation of a wide range of data sources it is estimated that at present there are approximately 5.5 million hectares of agricultural land on which bioenergy cropping takes place. This amounts to 3.2% of the total cropping area (and around 1% of the utilised agricultural area) in the EU-27. Practically all of this land is used for biofuel cropping, mostly oil crops (82% of the land used for biofuel production). These are processed into biodiesel; the remainder is used for the production of ethanol crops (11%), biogas (7%), and perennials go mostly into electricity and heat generation (1%).

An overview of where the present bioenergy cropping takes place is given in the underneath Table 2.1 and at regional level in Map 2.1 expressed in energy potential. The regional distribution of dedicated cropping patterns is based on the assumption that the bioenergy crops are distributed over regions in the same proportion as similar crops are used for feed and food purposes. The statistical figures on crop types and areas have therefore been used as a weighting for the distribution of biomass crops from national to regional levels (totals were derived from sources providing national estimates).

It becomes clear that 93% of the domestically grown bioenergy crops are converted into biodiesel and bioethanol. The area with fodder maize used as feedstock for biogas is also taking a large share of the biomass cropping area in Germany. This should be kept in mind when interpreting the map, but in other countries this feedstock crop is not important at all. So basically this map 2.1 reveals the present dedicated biofuel cropping situation in the EU.

It also becomes clear from the map that at present bioenergy cropping is only important in a selection of EU countries of which France and Germany are the most important. Significant areas of oil crops for biodiesel are also found in the UK, Poland and Romania. Dedicated cropping with perennials is still taking place at a very small scale. The countries that have the largest areas are Finland, Sweden, UK and Poland.

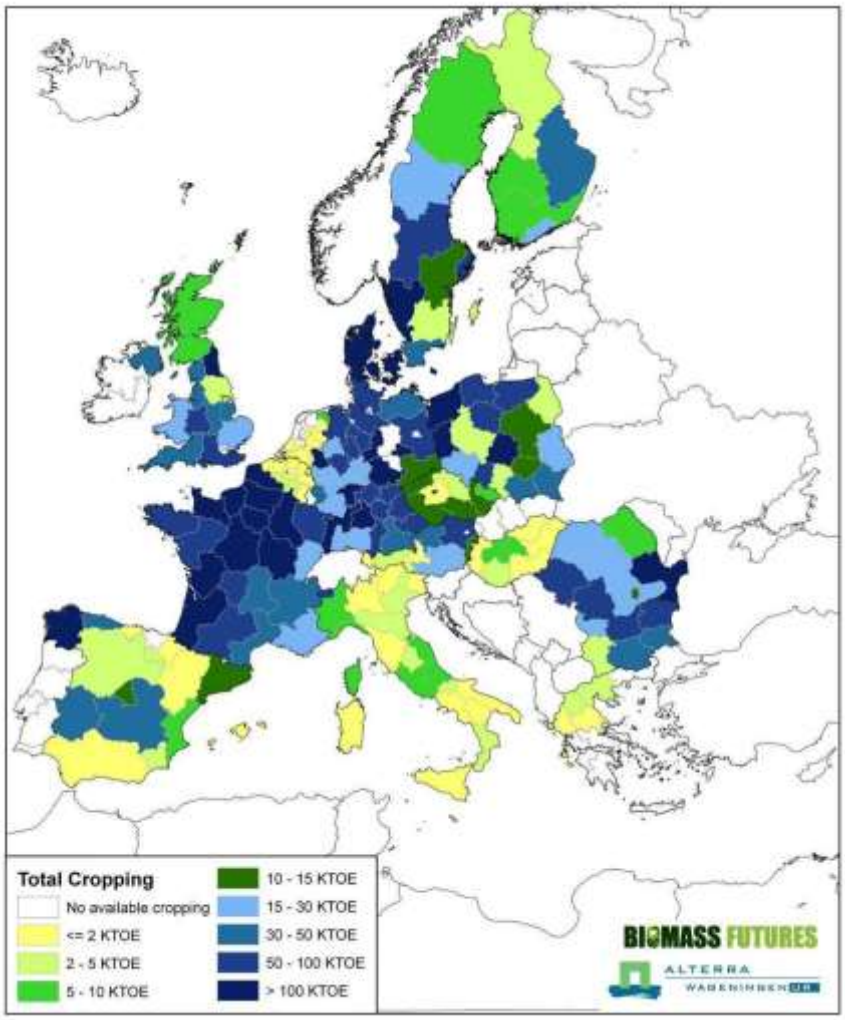
Table 2.1 Bioenergy cropping area (average 2006-2008 situation)*

	RAPE	Sunflower	Wheat	Barley	Sugar beet	Maize	Other arables (e.g. sorghum)	Reed Canary Grass (RCG)	Willow	Poplar	Miscanthus	Hemp
Belgium (only Flanders)	959		1173	191	0	660	0	0				
Bulgaria		258094	0		0	0	0	0				
Czech Republic	104000		0		0	0	0	0				
Denmark			51300	42750	0	0	0	0	2500			
Germany	1105000		78080	49920	3000	295000	0	0		500	300	
Ireland											2000	
Greece		11220	0		0	0	0					
Spain		150223	11902	21159	0	0	104			18		
France	885687	66665	225000	75000	50000	50000	0		500		1500	
Italy	5200	59800	0		0	0	0	0	0	6000	7500	
Hungary	10175	8325	0		0	0	0					
Netherlands	2500		0		0	500	0					
Austria	10200	4800	855	645	0	40000	0					300
Poland	740740		0		0	0	0		7000		13500	
Romania	22746	545912	0		0	0	0					
Finland	821		119	320	0	0	0	18700				
Sweden	50000		19600	15400	0	0	0	780	13000			390
United Kingdom	320542		10824	5093	0	0	0		5500		13500	
Total	3258571	1105038	398852	210479	53000	386160	104	19480	28500	6518	38300	690

Source: Dworak et al. (2008) and AEBIOM and own more recent up-dates with new sources. For detailed information on data sources used see Annex 1.

* Figures are only given for countries for which information was found on bioenergy cropping areas.

Map 2.1 **Energy potential from biomass cropping (average 2006-2008 situation). For sources see Table 2.1 and Annex 1.**



Energy cropping with ligno-cellulosic crops is not wide spread in most EU countries. From the data in Table 2.1 and Map 2.1 we can conclude that there are only some larger cropping areas in Sweden, Poland and the UK. In total the present EU wide perennial cropping area is estimated to be at around 93000 hectares with a total energy potential of 440 KTOE.

2.1.2 Approach to estimating future energy cropping and agro-waste potentials

Estimating the land potential for bioenergy cropping and for agro-waste potentials

In this study we build on the CAPRI model results which predict future land use changes in the EU-27 related to agricultural production including those for domestic biofuels. For 2020 the baseline scenario as run with the CAPRI model for the EC report 'Prospects for agricultural markets and income in the EU 2010-2020' was used. This outlook takes account of the most recent Health Check reform, the 2020 RES

Targets and the most recent OECD-FAO projections on agricultural prices, population and welfare developments (EC, 2010) [5]. For the 2030 CAPRI data we use the reference scenario run developed within the EC4MACS project[6]. The CAPRI model endogenously determines the changes in supply and other demand (feed, food, processing) for biofuels feedstocks. As the CAPRI market part comprises behavioural functions for oilseed and sugar and starch processing, the demand for bio-diesel and bioethanol processing can be covered either by domestically or imported processed vegetable oils, and the domestic processing may be sourced by EU produced feedstocks or by imported ones. The following technology pathways are covered:

- For total domestic ethanol production, five technology pathways are covered distinguished by usable feedstock groups: 1)Cereals - differentiated in wheat, barley, rye, oats, maize, and other cereals - , 2) sugar, 3) table wine 4) 2nd generation ethanol, and 5) non-agricultural ethanol
- For biodiesel, three technology pathways are covered distinguished by usable feedstock groups: 1) vegetable oils - differentiated in rape oil, sunflower oil, soya oil, and palm oil 2) 2nd generation biodiesel, and 3) non-agricultural biodiesel.
- There are also biofuel quantities which are produced from agricultural residues, like cereals straw or sugar beet leaves, or new energy crops i.e. perennials.

Table 2.2 Quota levels for biofuel consumption in 2020 per MS as used in CAPRI assessment

	BL	DK	DE	AT	NL	FR	PT	ES	EL	IT	IR	FI	SE
Biodiesel	6.0	6.0	9.0	7.5	6.7	8.0	5.5	7.6	5.2	8.2	6.0	5.9	6.0
Bioethanol	6.0	6.0	8.7	6.4	2.9	5.2	2.8	7.5	3.5	4.1	2.9	4.5	6.0
	UK	CZ	EE	HU	LT	LV	PL	SI	SK	CY	MT	BG	RO
Biodiesel	6.4	6.0	4.1	6.0	5.1	5.7	6.0	8.0	5.9	2.9	1.1	2.4	2.3
Bioethanol	6.4	4.1	2.8	2.4	4.2	2.0	3.4	2.6	3.0	0.8	0.6	1.7	1.6

Source: CAPRI model, calculated based on AgLink 2009 and EC (2009).

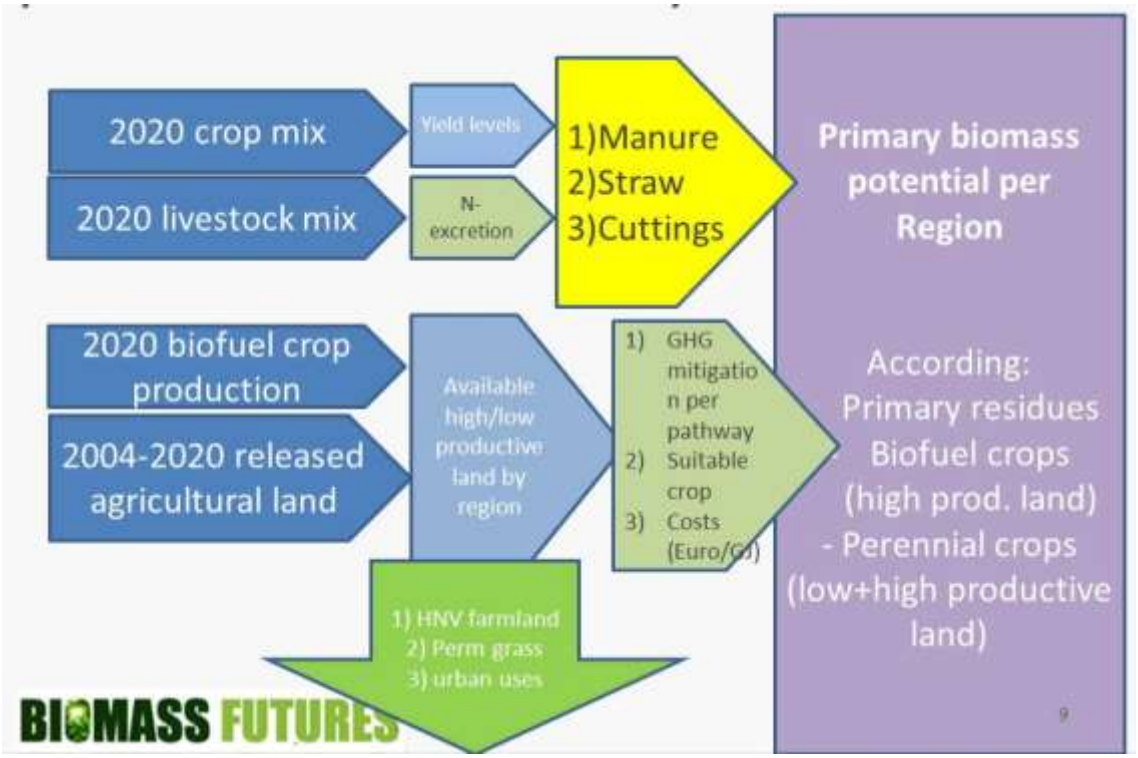
In order to incorporate domestic use and supply of biofuels in the CAPRI baseline assessment the domestic ethanol and biodiesel production is defined via a profit maximization approach as a function depending on processing margins. As for biodiesel the margins for individual vegetable oils are covered by using an average margin depending on weighted individual margins for all usable vegetable oils. The baseline assumptions leading to final 2020 mixes of RES fuels are mainly fed by expert knowledge building on different sources providing projections of domestic use and supply of biofuels for the single EU27 countries (PRIMES) and non-European countries (AgLink).

[5] EC (2010), Prospects for agricultural markets and income 2010-2020. http://ec.europa.eu/agriculture/publi/caprep/prospects2010/index_en.htm. For further information see also: http://ageconsearch.umn.edu/handle/91395
http://www.ilr1.uni-bonn.de/agpo/publ/techpap/techpap10-01.pdf

[6] see: http://www.ec4macs.eu/home/reports.html?sb=23

An important assumption is that the share of domestic biofuel demand in 2020 results from the implementation of quota obligations. These were estimated by taking the information on implemented quotas until 2009 (see in EC, 2009) and assumed that all existing quota obligations which are defined for a year before 2015 will be increased in the respective EU MS in 2020 by 1.5%. In addition, all existing quotas which are already defined for a year beyond 2015 will only exceed the existing level by 1.1%. For all EU MS where no quota exists it is assumed that a minimum quota of 6% will be introduced in 2020. For final quota levels see Table 2.2.

Figure 2.1: Estimation of bioenergy potential from agriculture in post-model analysis of Capri-2020 and 2030 for reference and sustainable scenarios.



The emphasis in the CAPRI run (Blanco Fonseca et al., 2010) is on predicting biofuel cropping response. However, in addition to this specific information is also available for the CAPRI baseline 2020 on the agricultural land use cropping and livestock patterns. This implies that in a post model process the CAPRI model output serves as an excellent basis for (see also Figure 2.1):

1. Estimating the land use implications of total domestic biofuel feedstock production
2. Estimating the potential for agricultural by-products, i.e. straw, manure, cuttings from permanent crops. These potentials can be derived from the detailed future land use patterns and livestock types and numbers combined with additional information on production levels and competing uses.
3. Estimating the future unused/released land potential (as compared to 2004) that may be used for dedicated biomass cropping with perennials taking account of additional scenario assumptions as elaborated in the baseline, sustainability and global transition scenarios of this study.

For estimating the 2030 potential the same procedure was applied as described above, the CAPRI references scenario takes the PRIMES reference scenario 2030[7] as a starting point for bioenergy demand

[7] See for description of this reference scenario: http://ec.europa.eu/energy/observatory/trends_2030/doc/trends_to_2030_update_2009.pdf

which was run within the scope of the EC4MACS project (Witzke et al., 2011). In this reference scenario it is assumed that the renewable energy targets of 2020 are maintained towards 2030. Within this context it is clear that a demand for rotational arable crops remains and that this is incorporated in the same way as in 2020 in the CAPRI model. However in the CAPRI reference scenario the area results for crops do not distinguish between output used for (traditional)biofuel production and output going to food and feed. In absence of a scenario run that excludes biofuel targets, the simple assumption was made for 2030 for estimating the biofuel crop potential land use that this shows the same amount and regional pattern as in 2020. Therefore the amount of land used for the production of biofuels in 2020 was also assumed to remain in biofuel production in 2030. The mix of biofuel feedstock will change however as priority is given to the most sustainable crop mix per region, taking account of the mitigation requirements set in both reference and sustainability scenario of this study. How this mitigation requirement is estimated is discussed in the following.

Estimating the minimum GHG mitigation requirement per scenario for the bioenergy cropping potential
In the sustainability scenario applied in this study two types of sustainability criteria are applicable (see also Table 1.2 in Chapter 1). A minimal GHG mitigation requirement for biofuels of 50% in 2020 and 2030 is assumed in the reference scenario. In the sustainability scenario this mitigation requirement is much stricter as it should include a compensation for iLUC related emissions, it should reach 70% and 80% compensation level in 2020 and 2030 respectibly and it both applies on biofuels and on cropped biomass for heat and electricity production.

For the estimation of the minimal GHG requirement we build on the approach developed in the EEA/ETC-SIA study (Elbersen, et al., 2012). An estimate of GHG payback and mitigation ability is made for all crops, including the iLUC effect and taking into account the type of feedstock and related bioenergy delivery pathway. A 20 year payback time is assumed. This is implemented by estimating the GHG mitigation efficiency factor which is build-up from 3 components:

1. Direct land use emissions from the cropping process which are strongly linked to input and output levels which differ per EU-region

2. The downstream emissions of the biomass feedstock conversion routes

3. A possible iLUC GHG emission factor if land use displacement is applicable.

Ad 1. The emissions from the land-based part of the chain, if cropping is involved, are calculated using the Miterra-Europe model[8]. This assesses the impact of measures, policies and land-use changes on environmental indicators at the NUTS-2 and Member State level in the EU27. MITERRA-Europe partly takes the input of the CAPRI and GAINS models, supplemented with an N leaching module and a measures module. MITERRA-Europe calculates all relevant GHG emissions from agriculture (CH_4 from enteric fermentation and manure management, N_2O from manure management and direct and indirect soil emissions, and CO_2 from changes in soil carbon stocks and cultivation of organic soils), according to the IPCC 2006 guidelines. GHG emissions from fertiliser production and mechanisation are also included. The emission and mitigation levels for crops depend very much on the yield at the different locations. For biofuel crops, the yield potential is taken from the 2020 and 2030 Capri baseline and reference

[8] Velthof, G.L., Oudendag, D., Witzke, H.P., Asman, W.A.H., Klimont, Z. and Oenema, O. 2009. Integrated assessment of nitrogen emissions from agriculture in EU-27 using MITERRA-Europe. Journal of Environmental Quality 38:402–417

scenarios respectively. For perennial crops, the yield potentials are derived using the GWSI crop growth model which takes soil and climate characteristics into account and predicts yield levels for high yielding systems (on good agricultural land in optimal and water limited situation) and on low yielding systems (low yielding soils). The yield and emission levels for the perennial crops were produced at these three levels per NUTS region (for further details see Elbersen et al., 2012).

Ad 2. The emissions of the downstream part of the bioenergy pathways and of the fossil comparators are based on GEMIS, which refer to full life-cycle emissions. GEMIS 10 is a life-cycle analysis program and database for energy, material, and transport systems. The GEMIS database offers information on 1) fossil fuels, renewables, nuclear, biomass and hydrogen, 2) processes for electricity and heat, 3) materials and 4) transports[9]. An overview of the average emissions of all technology pathways (including upstream land based direct emissions in case of cropping and downstream emissions) is given in the Table 1 in Annex 2. The land based up-stream emissions were calculated by the Miterra system (Veldhof et al., 2009) for every bioenergy crop grown in every EU-27 (NUTS 2) region. There are large differences between regions in soil-related climatic conditions and management and these determine the minimum and maximum emissions in the pathways based on cropped biomass.

Ad 3. A unified view on the iLUC-GHG emission factors does not exist. The major available studies regarding this issue have therefore been consulted as part of the EEA/ETC-SIA study (Elbersen et al., 2012) and an average iLUC-GHG factor is calculated to estimate the GHG payback and mitigation ability for each bioenergy pathway.

Table 2.3 GHG emission of fossil based fuel comparator in the EU in CO2 equivalents per kg/GJ fuel, assuming 100% efficiency, including upstream effects.

EU-27-mix	2010	2020	2030
Diesel	87.5	87.5	87.5
Gasoline	90.2	89.4	89.4

Source: GEMIS, 2010

Indirect land use change (iLUC) takes place when '*existing agricultural land is diverted from production into land used for the production of biofuels*' (EC, 2011). In the study of Fritsche et al. (2011) for the European Parliament it is defined much broader as '*effects through displacing current agricultural (food, feed) or forest (fibre, timber) production to other areas - e.g. grasslands or forested land – which causes direct land use changes (DLUC) at the new location in agriculture (food, feed), and forestry (fibre, wood products)'.* This latter definition shows that the iLUC effect does not only need to be related to biofuels but to bioenergy cropping in general. However, in practice it is clear the biofuels are the most important cause for iLUC as these, at least the rotational arable crops used for the first generation biofuel, need to be produced on the better agricultural lands and are therefore most likely to compete with food and feed production on existing agricultural/arable lands. The iLUC effect is however very difficult to estimate as it is caused by the introduction of a demand for bio-energy feedstock, but cannot be directly linked to the bioenergy production chain. The effects manifests itself through a change in demand for agricultural commodities at the global market. Modelling studies are therefore needed to estimate how big the effects are and how these translate into additional GHG emissions. In the mean-time several modelling studies have been published on iLUC. Their estimates of iLUC GHG effects differ strongly however (see Annex 3). This is related to the many different assumptions, input data and modelling logic. Comparison of these studies and an median iLUC-GHG emission factor per crop-conversion pathway combination was already made within the EEA/ETC-SIA study (Elbersen et al. 2012). An overview of the studies included and de median iLUC-GHG factor also used in this study is presented in Annex 2. It should be noted however that the most recent modelling results on iLUC with the IFPRI-

[9] GEMIS includes the total life-cycle in its calculation of impacts - i.e. fuel delivery, materials used for construction, waste treatment, transports/auxiliaries and includes by-product allocation (based on energy value). A further description of GEMIS and the calculated GHG emissions is given in EEA (2008) in Annex 2 and 3.

MIRAGE-BioF model referred to in the EC *Impact assessment on indirect land-use change related to biofuels* have not been taken into account in the overview presented in Annex 3.

Table 2.4 Average GHG emissions of fossil comparators 2020

	GHG emissions in (gr. CO_2eq /MJ_{out})	
Country	**Electricity**	**heat**
AT	158.1	94.0
BE	145.9	91.5
BG	249.5	169.7
CY	100.7	106.7
CZ	261.5	100.2
DE	200.0	84.5
DK	170.4	84.7
EE	267.2	87.4
ES	166.3	91.6
FI	187.1	90.4
FR	140.2	87.9
GR	214.6	105.5
HU	167.1	94.2
IE	160.5	109.3
IT	141.4	86.1
LT	110.6	105.0
LU	111.3	91.8
LV	173.4	98.3
MT	99.4	101.4
NL	158.1	73.3
PL	249.2	153.8
PT	168.4	92.1
RO	179.9	85.8
SE	119.4	93.2
SI	259.6	110.7
SK	183.6	103.2
UK	153.0	76.1

Source: GEMIS 10 and PRIMES reference scenario

To determine the final emission level for each pathway and region, the GHG emission of the whole bioenergy pathway is calculated. To come to a final mitigation potential to assess whether the feedstock conversion pathway combination per region fits with the sustainability criteria in every scenario a comparison is made with the GHG emissions of the fossil-based comparators. For the transport fuels the levels are given in Table 2.3.

The fossil fuel mix for calculating the average emission of the 2020 fossil comparators for both electricity and heat are based on the PRIMES reference scenario for 2020 (Capros et al., 2009). These emissions as presented in Table 2.4 are based on the fossil fuels only (coal, lignite, oil and natural gas), since the assumption is that these renewable energy pathways will replace fossil fuels and no other RE sources or nuclear energy.

The minimum mitigation level of 50 % against fossil comparators in the reference scenario only applies to biofuels and does not include a compensation for iLUC related emissions. For the total biofuel feedstock potential this implies that both in 2020 and 2030 biofuel crops can be grown in practically all countries where according to CAPRI predictions land is used for biofuel production.

In the sustainability storyline however, the mitigation potential is set on 70 % in 2020 and even 80% in 2030, including an iLUC GHG compensation in case of displacement effects. In the case of biofuels which are grown on existing arable land in competition with food and feed crops reaching this mitigation level is impossible. Both in 2020 and 2030. The growth of dedicated perennial crops on lands competing with food and feed crops remains possible in a few regions where yields for these crops are high and where the fossil comparator is also relatively high, because of the large share of lignite based fossil energy use. This becomes clear in the following where the final cropping potentials are presented.

Excluding high biodiverse land and land with high carbon stock
In the reference scenario biofuel crops cannot be cropped on highly biodiverse areas or areas with a high carbon stocks. In the sustainability scenario this applies to all cropped biomass. The land availability for bioenergy cropping should therefore be reduced with these type of land use categories. However, it is difficult to capture spatially all of these areas in Europe because of lack of spatially-detailed information and clear definitions. In this study both the NATURA 2000 (farmland) and the HNV farmland areas were regarded as good proxies for highly biodiverse areas and agricultural areas of high carbon stocks, and were therefore taken as no-go areas for biomass cropping. This approach was copied from the EEA/ETC-SIA-ACC study (See Elbersen et al., 2012).

2.1.3 Biofuel and biogas cropping in 2020 and 2030 in reference and sustainability scenario

In order to make an estimate of the final biofuel 2020 potential for the reference scenario the land use of biofuel crops per region as calculated in the Capri baseline 2020 run (EC, 2010) were taken. The final biofuel crop mix and related potential was determined based on the 50% minimum mitigation requirement for biofuels as specified in this reference scenario and from that selection the cheapest crop mix (Euro/GJ). For the 2030 potential the same procedure was applied, except that the calculation of the mitigation requirement was based on 2030 mitigation and cost data. The cost levels were again based on the CAPRI 2020 and 2030 cost levels.

This results in a regional distribution of biofuel crop potentials as presented in Map 2.2. The regional distribution of the 2020 and 2030 biofuel potential does not differ much. In 2020 there is a total of 11827 KTOE and in 2030 11645 KTOE available. In 2020 the use of sugarbeet for bioethanol is much higher, while towards 2030 this disappears and shifts towards a larger share for corn and cereals. This shift is caused by price shifts, which generally make sugar beet a more expensive alternative as compared to wheat and maize, and the availability of crops fitting with a 50% mitigation potential. For biodiesel both oil seed rape and sunflower remain the most important feedstock.

Map 2.2 Biofuel and biogas (energy maize) cropping potential 2020 and 2030 in the reference scenario

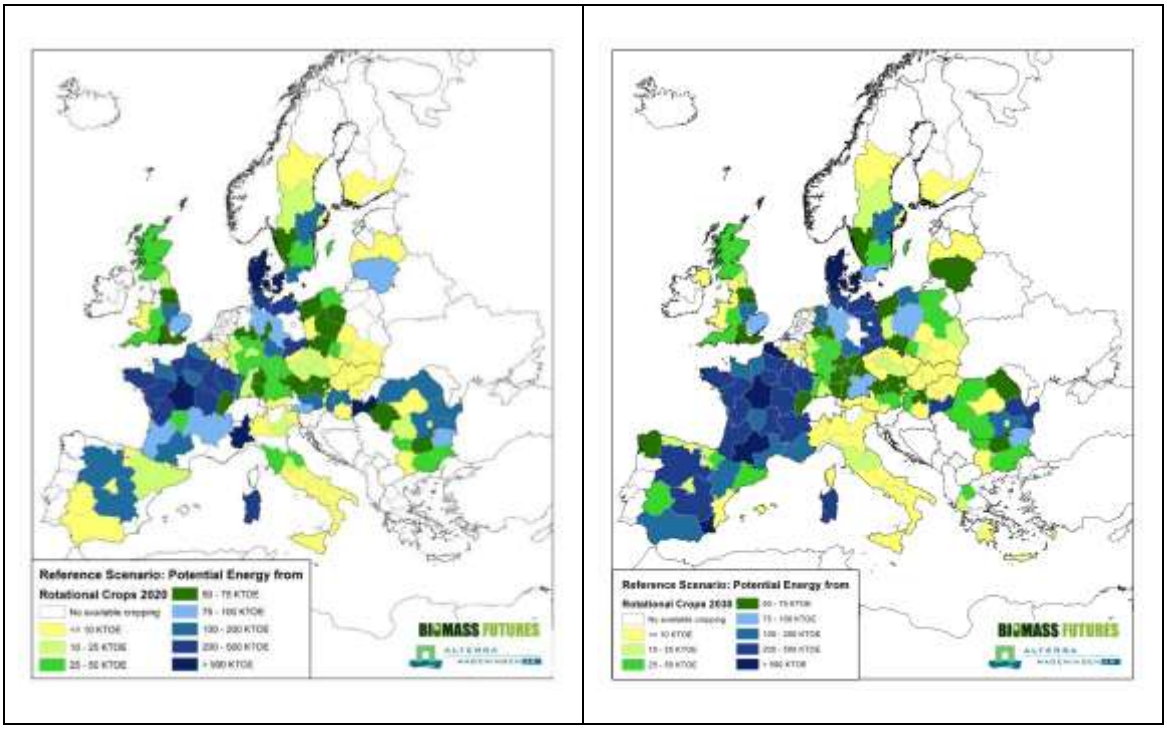


Table 2.5 Crop potential from biofuel crops and energy maize (KTOE) in 2020 and 2030 reference scenarios

KTOE	2020	2030
Austria	410	186
Bulgaria	260	264
Belgium/Luxembourg	12	13
Cyprus	0	0
Czech Rep.	63	60
Germany	2156	2904
Denmark	977	1139
Estonia	0	0
Greece	0	71
Spain	321	1974
Finland	2	2
France	5755	6526
Hungary	1863	446
Ireland	0	0
Italy	3585	321
Lithuania	85	70
Latvia	5	7
Malta	0	0
Netherlands	0	4
Poland	357	582
Portugal	0	0
Romania	649	543
Sweden	358	318
Slovenia	0	0
Slovakia	15	14
UK	462	493
EU-27	17335	15937

For biogas cropping with energy maize the potential is estimated at 5509 KTOE in 2020 and 7924 KTOE in 2030. This potential is estimated to be grown on the land that becomes released between 2004 and

2020 and 2030 respectively if biogas maize is among the cheaper feedstock crops as compared to other dedicated crops (see next). The biogas maize is assumed to be grown on land additional to the land required for biofuel cropping as estimated in the CAPRI runs. It is allocated to the released land to be used for dedicated cropping with perennials and also energy maize if fulfilling the soil and climate circumstances and the mitigation requirements per scenario and belonging to the cheaper crop mix (Euro/GJ) (see next).

2.1.4 Dedicated cropping potential 2020 and 2030

Although the actual dedicated cropping area is still very small, the future bioenergy potential from dedicated cropping with these perennials could become more important for several reasons:

1) Ligno-cellulosic material is a good feedstock for heat and power generation in increasingly efficient conversion technologies.

2) Other cheaper ligno-cellulosic waste and by-products from the waste and forest sectors will be used first. However dedicated cropping with ligno-cellulosic crops could be an attractive option to ensure that there is enough local biomass available year-round, especially when competing uses are diminishing the potential from the other sectors.

3) Ligno-cellulosic material is a feedstock for second-generation biofuel production and within the next 10 years it is expected that these types of technologies will become more economic and marketable. This certainly applies to thermochemical conversion in which biomass is gasified to syngas which is then converted to biodiesel using Fischer-Tropsch (F-T) synthesis. This Biomass to Liquids (BtL) process can be applied to woody or grass-derived biomass as well as cellulosic dry residues and wastes.

4) Ligno-cellulosic crops have generally a higher GHG efficiency then rotational arable crops since they have lower input requirements and the energy yield per hectare is much higher. At the same time most ligno-cellulosic crops have lower soil quality requirements then rotational arable crops. If they are grown on lower productive lands at which they do not compete with rotational arable crops, acceptable yields can still be reached and displacement effects are limited.

5) Because of the above reasons, second generation biofuel production is applicable for double counting for the RES-targets which could make ligno-cellulosic biomass feedstock more attractive. Res-stimulation measures can therefore also be expected to become implemented which make dedicated cropping with ligno-cellulosic crop on released or recently abandoned lands, or even in competition with rotational arable crops a plausible economic option.

In this assessment it is expected that dedicated cropping with perennials for bioenergy production is most likely to take place on land that is not needed for the production of food and feed production nor biofuel crops. In order to estimate the amount of land that can be included in this potential a post-model analysis was made of the agricultural production area as modelled in CAPRI 2020 baseline and 2030 reference scenario and 2004. By comparing the size of different types of land uses in the future years with the 2004 situation an estimate could be made of the amount of land released, but also of the type of categories of land released. Good quality land is released in the arable cropping category and low quality land is land which was used for perennial crops like vineyards, olives and fallow. The results of this comparison are shown in Table 2.5.

In addition of the land released, there is also a category of land only occurring in the sustainability scenario. It is land that according to the CAPRI 2020 baseline is used for biofuel cropping. In the reference 2020 and 2030 scenarios of this study this land indeed remains under biofuel crops. However in the sustainability scenarios 2020 and 2030 biofuel crops cannot be produced in the EU sustainably as

they do not reach the mitigation target of 70% and 80% respectively including the compensation for iLUC. This implies that in the post model assessment these lands are allocated to dedicated perennial cropping provided these crops do comply with the mitigation targets of 70% and 80% respectively for 2020 and 2030 including a compensation for iLUC. It turns out that this land category can be used in 64% of the biofuel area share in 2020 and only 55% in 2030. This decline is a result of a 10% higher mitigation target in 2030. The biofuel crop land is assumed to remain stable between 2020 and 2030 under the reference storyline as was already explained in the former section

It becomes clear from Table 2.6 that because of constraints on the use of biodiversity rich and land with high carbon stock less land is available in the sustainability then in the reference scenario. It is also clear that between 2004 and 2020 slightly more land is released then between 2004 and 2030. This is particularly caused by the larger arable land demand in 2030 as compared to 2020 as the category of released land of good quality is smaller in 2030 as compared to 2020. The total utilised agricultural area was 187 million hectares in 2004, which means that 11% of this area is expected to be released from agriculture (through market forces and policies taken into account in the CAPRI baseline run) until 2020 in the reference scenario and towards 2030 this declines towards 10%. In the sustainability scenarios this amounts to respectively 10% and 9%

Table 2.6 Land released from agricultural production (*1000 ha) between 2004 and 2020 and 2004 and 2030 in the EU-27

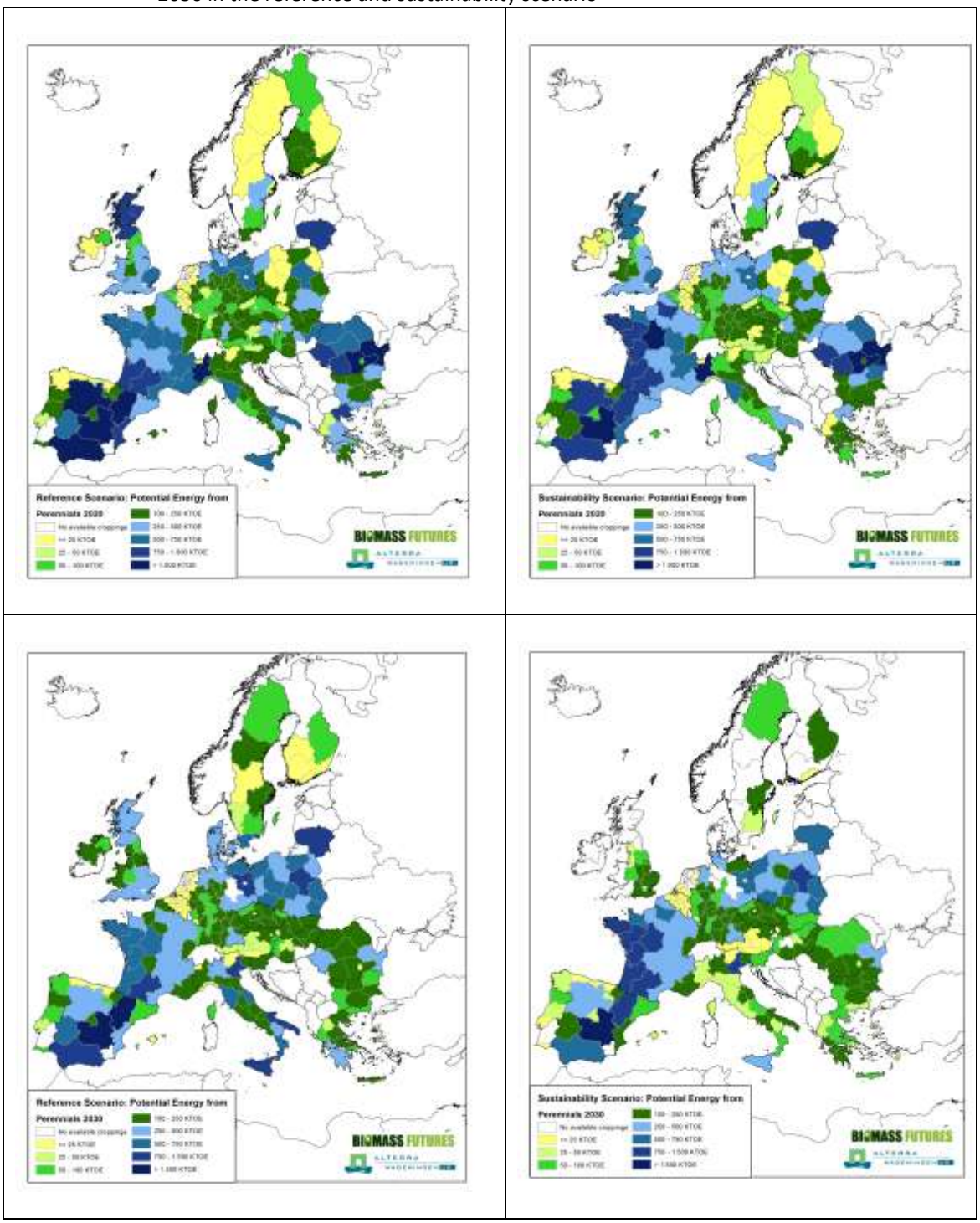
Land released between 2004 and:	Good quality released	Good quality land not fit for sustainable biofuel production	Low quality land	total
2020 reference	8200	0	13526	21726
2020 sustainability	6003	3039	9315	18357
2030 reference	5093	0	13700	18793
2030 sustainability	4016	2590	9499	16105

The land potential estimates in Table 2.5 exclude a further potential of land that has been abandoned already before 2004 and therefore not included in the total utilised agricultural area figures of 2004 used in the CAPRI modelling exercise. This abandoned land resource is expected to be considerable especially in the CEEC and the Mediterranean and could also add significantly to future potentials (Pointereau et al., 2008). This however has not been taken into account in the potential presented here. The land potential presented here should therefore be characterised as a conservative estimate.

In order to come to a total energy potential for dedicated cropping in the 2 scenarios different criteria were applied to select the final perennial crop mix. This mix firstly fits with the soil and climate characteristics per region, but to determine the final mix in the reference scenario priority was given to the cheapest crop mix per region, while in the sustainability scenario the crops with the highest mitigation potential were selected, with cost level as secondary selection criterion. Further details on the estimation of the land potential and on the crop mix with the highest mitigation potential were presented in former section 2.1.2. The results of this in terms of a final energy potential are presented in Map 2.3 for 2020 and 2030 reference and sustainability scenarios.

Differences between the final dedicated potential for the 2 scenarios occur because of the stricter sustainability criteria. Therefore in the sustainability scenario there is less land available to use for dedicated cropping and/or there are more regions where the mitigation requirement is not reached. This is the case for example in Ireland and Scotland where in the reference scenario in 2030 there is still ample potential, but in the sustainability scenario this potential disappears because there is no single perennial energy pathway in which an 80% mitigation can be reached. At the same time it can also be seen that in 2020 in the sustainability scenario there is still potential in Northern Ireland and Scotland, while towards 2030, when the mitigation requirement shifts from 70% to 80% of fossil alternative, the dedicated crop potential disappears.

Map 2.3 Dedicated cropping potential with perennials on released agricultural land in 2020 and 2030 in the reference and sustainability scenario



Differences within countries between the reference and sustainability scenario can also be large when a country has a large share of HNV farmland, which leads to a much smaller land potential in the sustainability scenario. This is for example very clear in Lithuania, Hungary, Greece and Spain (see Table 2.7). Sometimes the sustainable potential is almost as big or bigger than the potential in the reference scenario which is caused by the incorporation of biofuel land. This land would be used for biofuel cropping in the reference scenario while in the sustainability scenario (part of) it is used for dedicated cropping with a higher mitigation potential. This is clearly the case in Germany and France and also in Poland, particularly in 2020 (Table 2.7).

The large differences in woody and grassy perennials are caused by differences in cost levels (Euro/GJ). For example in the Mediterranean the cost levels of grassy crops (e.g. miscanthus and switchgrass) are much lower than the of woody crops, while towards central and western Europe this difference in price level declines and it is more the combination of type of land (soil), climate and crop type combination that determine which crop is most efficient in GHG efficiency and cost level. In Bulgaria it is for example much more difficult to reach high yields (for reaching the minimal mitigation level) at low cost levels for grassy crops. Also in the North it becomes clear that Reed Canary grass, a grassy perennial, is often delivering the same energy amount at a lower price than willow.

Table 2.7 Dedicated cropping potential (KTOE) in 2020 and 2030 in reference and sustainability scenarios

	2020				2030			
KTOE	**Reference**		**Sustainability**		**Reference**		**Sustainability**	
	woody	**grassy**	**woody**	**grassy**	**woody**	**grassy**	**woody**	**grassy**
Austria	393	362	180	285	192	239	81	112
Bulgaria	1206	184	1156	558	925	0	315	629
Belgium/Luxembourg	160	110	160	99	31	77	45	12
Cyprus	0	0	0	0	0	0	0	0
Czech Rep.	33	481	31	506	70	897	186	818
Germany	3024	2592	3881	2267	1596	5639	2390	3513
Denmark	0	0	0	0	0	278	0	0
Estonia	0	0	0	0	0	0	0	0
Greece	0	2906	0	1374	0	1752	0	792
Spain	44	10133	14	6064	73	7312	47	3949
Finland	0	374	0	229	0	102	0	102
France	5418	5008	8669	4070	4627	3231	2812	7731
Hungary	838	680	599	599	461	259	207	381
Ireland	0	16	0	12	0	98	0	0
Italy	0	5535	134	4358	0	6344	0	2550
Lithuania	272	692	382	588	314	621	448	239
Latvia	0	0	0	0	0	0	0	0
Malta	0	0	0	0	0	0	0	0
Netherlands	25	55	24	49	18	65	5	29
Poland	472	2668	392	2653	1332	6520	1782	4944
Portugal	0	489	0	252	0	358	0	165
Romania	5949	3220	5418	2660	412	1001	329	814
Sweden	304	323	277	274	505	890	151	72
Slovenia	0	96	0	38	0	121	0	77
Slovakia	63	549	42	455	51	505	87	285
UK	418	3101	383	2489	213	2114	158	560
EU-27	18619	39576	21742	29880	10821	38422	9044	27774

Overall it is clear that the stricter sustainability criteria do lead to a lower potential both in 2020, but even more so in 2030 when mitigation requirements become more strict (See Table 2.8). The biofuel potential in 2020 and 2030 in the reference scenarios are practically the same as it is assumed that between 2020 and 2030 the land used for dedicated biofuel cropping remains the same. Differences in cropping potential between the years are therefore mainly in the dedicated perennial group and are related to difference in land releases and stricter sustainability criteria.

Table 2.8 Summary of cropping potential (KTOE) in 2020 and 2030

KTOE	**Biofuel potential**	**Energy maize (biogas)**	**Dedicated woody perennial crops**	**Dedicated grassy perennial crops**	**Total**
2020 reference	11825	5509	18619	39576	70021
2020 sustainability	0	0	21742	29880	51622
2030 reference	11645	7924	10821	38422	60888
2030 sustainability	0	0	9044	27774	36818

2.2 Manure

Manure is a scarce resource in some regions, while in others there is too much of it and farmers are obliged, under the Nitrate Directive in Nitrate Vulnerable Zones, to even pay for the disposal of excess manure (above 170 kg N/Ha). In order to estimate the potential a couple of assumptions are made:

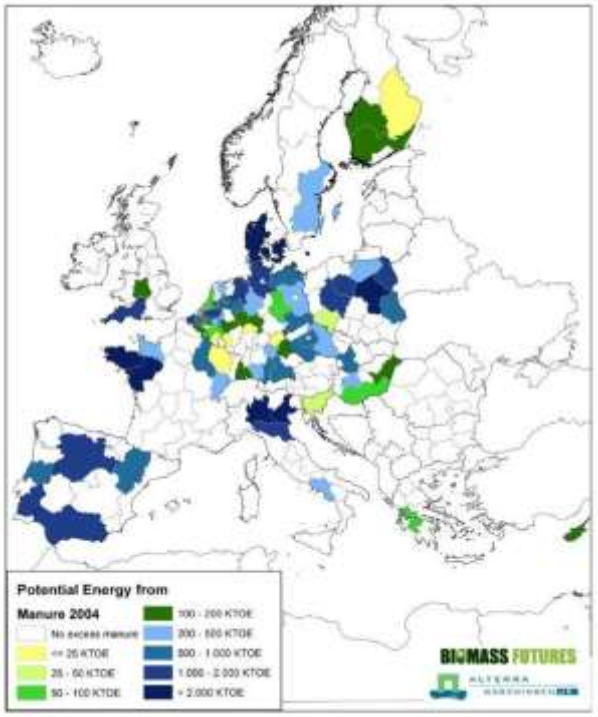
1) Farmers with excess manure are more likely to search for opportunities to produce biogas from it. This stimulus most certainly applies to farmers having higher manure production then 170 kg nitrogen per hectare as they even have to make costs to get rid of it. We set the level at which farmers start searching for alternative uses for their manure at 100 kg nitrogen per hectare. Manure in excess of this 100 kg Nitrogen per hectare of forage area (fodder crops+grazing lands) is therefore the first to be used for bioenergy generation and the cost of using it could be very low since the farmer has not make costs to get rid of it.

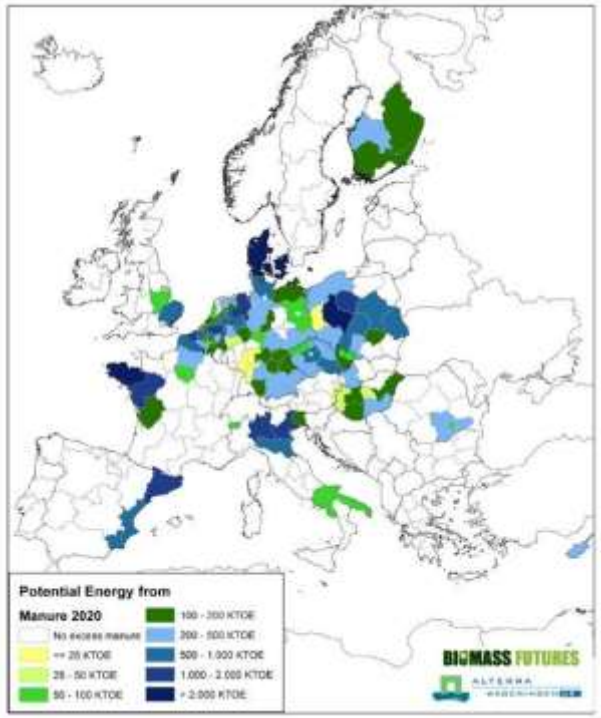
2) In areas in which there is no manure potential above the 100 kg Nitrogen per hectare it is assumed that there is not enough stimulation to put it into a biogas installation, even though the nutrients resulting from the bioenergy conversion can still be brought back to the land. The potential is assumed to be zero, although it is acknowledged that even in these regions there could be some potential to convert manure into energy.

In order to map and quantify the most accessible and cheap potential, data from CAPRI were used for the year 2004 per Nuts 2 region. For 2020 the CAPRI baseline and for 2030 the reference scenario results on livestock numbers and types per region (see Section 2.1 in this report). The CAPRI scenario results specify new 2020 and 2030 land use patterns and livestock population composition and numbers. Data from GAINS specify excretion factors per type of animal in every country. Based on these two information sources the same estimates could be made as for 2004, 2020 and 2030 regarding wet and dry manure production per hectare of forage area and on the amount of nitrogen (and related manure) production above 100 kg Nitrogen per hectare.

Results are presented in the maps (Map 2.4) and Table 2.9. They show that the manure potential towards the futures declines with declining livestock numbers as predicted in CAPRI. This also goes together with a regional shift in manure potential. Large potentials for the strong livestock concentration areas like in The Netherlands, Denmark, The Po-valley remain also towards 2030 but it is striking the manure production in large parts of Germany, Netherlands, Portugal and also Normandy will strongly decline towards 2030. While in Hungary, Poland and Romania large increases are seen.

Map 2.4 Manure potential (KTOE) in 2004, 2020 and 2030





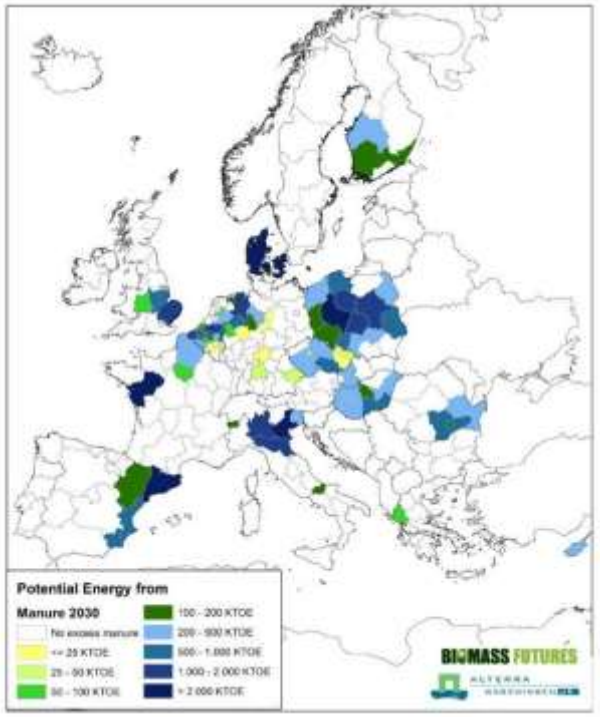


Table 2.9 Total energy potential from manure (KTOE) in EU in 2004 and 2020

KTOE	2004	2020	2030
Austria	0	10	0
Bulgaria	0	0	0
Belgium/Luxembourg	2992	3090	3078
Cyprus	101	238	298
Czech Rep.	1687	2003	1492
Germany	10769	7266	2551
Denmark	3469	3300	2377
Estonia	0	0	0
Greece	94	0	78
Spain	3933	2622	3742
Finland	316	789	423
France	9035	7309	9894
Hungary	617	721	1952
Ireland	0	0	0
Italy	7632	4834	5580
Lithuania	0	0	0
Luxembourg		0	0
Latvia	0	0	0
Malta	0	200	228
Netherlands	3916	4574	2743
Poland	7314	8177	11745
Portugal	1681	0	0
Romania	0	328	1262
Sweden	879	0	0
Slovenia	43	0	0
Slovakia	690	321	235
UK	1647	942	2174
EU-27	56817	46724	49852

2.3 Primary agricultural residues

In agriculture the main sources of primary residues come from arable crops in the form of straw and from maintenance of permanent crop plantations like fruit and berry trees, nuts, olives, vineyards, and citrus. In the underneath sections these two groups are discussed separately.

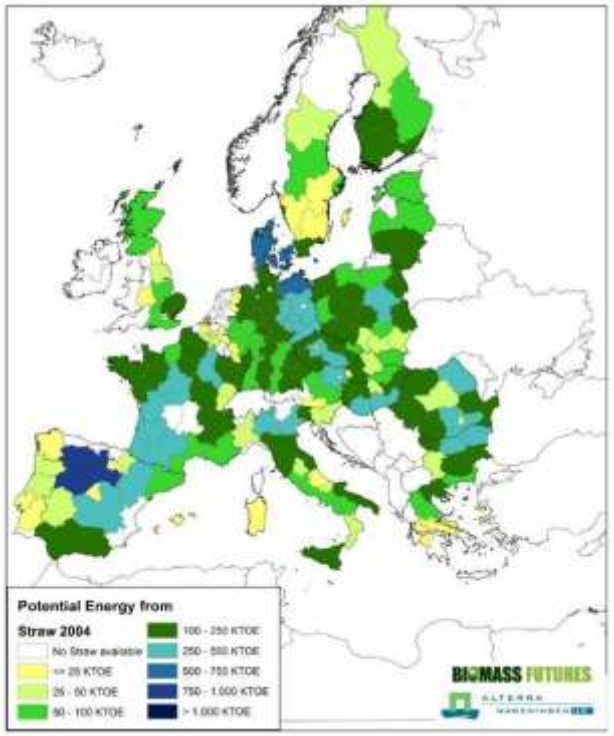
2.3.1 Straw

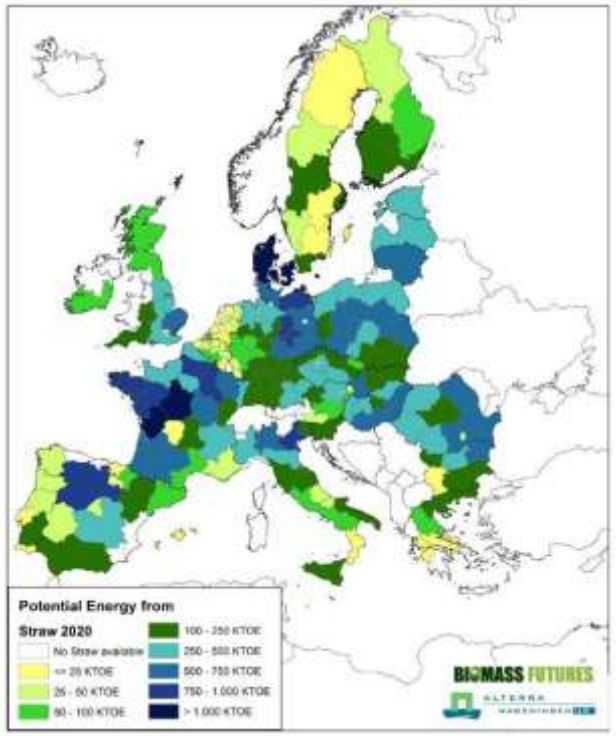
A methodology for estimating the straw potential available for bioenergy production was developed by the JRC already since 2006 (JRC and CENER, 2006 and Scarlat et al. 2009). In this work the methodology for estimating a sustainable potential applies to a wide range of crops delivering straw including all cereals, rice, and maize, sunflower and oil seed Rape. Based on a wide range of EU expertise the straw yield ratios per type of crop are provided together with sustainable harvest levels. The latter relate to harvest practices aimed at maintaining the soil carbon levels in the soil. These were estimated to be at 40% for wheat, rye, oats and barley and at 50% for the other 4 crops. Beside sustainable yield levels estimates were also made of competitive uses of straw to be subtracted from the bioenergy potential. Competitive uses are for bedding in specific livestock systems (including horses) and for mushroom production. The JRC approach was applied by us to the CAPRI 2020 baseline and 2030 reference scenario land use situations. It also takes account of competing uses of straw for livestock which is incorporated using the CAPRI 2020 and 2030 livestock numbers per region.

The results are presented in the maps (Map 2.5) and in Table 2.10 and show that the total straw potential doubles between 2004 and 2020 and then remains relatively stable towards 2030. This doubling is related to a combination of increased production of cereals and a decline in livestock numbers. The country with by far the largest increase is France where a combination of increased cereal production and declining livestock numbers leads to this growth. Large growth numbers, particularly towards 2020 are also seen in Germany, Poland, Romania, Slovakia and UK.

The straw potential is well spread over practically all of the EU, but countries like France, Germany, Poland, Italy, Hungary and in the future UK have the largest potentials. In Denmark there is the largest concentration of straw although the potential remains limited compared to the larger EU countries. Countries which show particularly large increases towards 2020 and 2030 are France, Poland, Hungary, Romania, UK and Denmark.

Map 2.5 Economic and environmentally sustainable straw potentials (KTOE) in 2004, 2020 and 2030





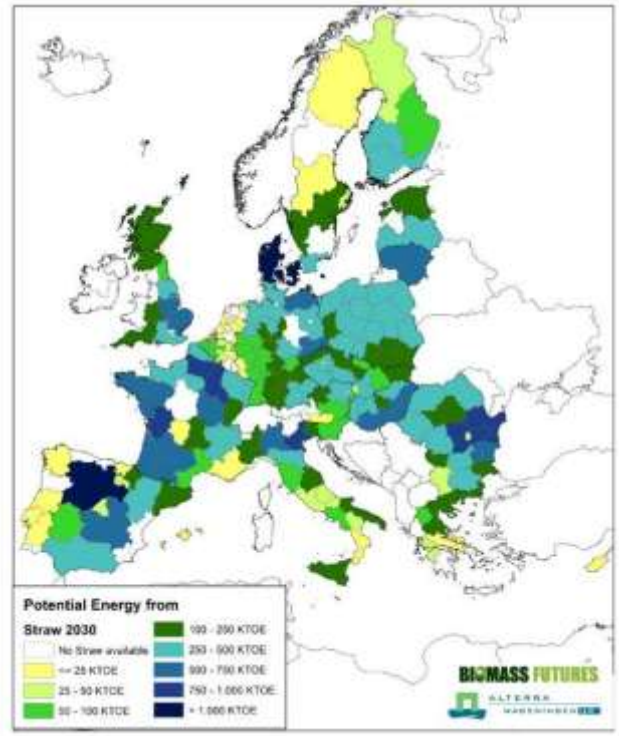


Table 2.10 Straw potential (KTOE) per country in 2004, 2020 and 2030

KTOE	2004	2020	2030
Austria	600	677	615
Bulgaria	1295	1396	1820
Belgium/Luxembourg	171	334	450
Cyprus	0	0	1
Czech Rep.	1235	1448	1478
Germany	5125	8883	7496
Denmark	575	1300	1200
Estonia	63	285	211
Greece	341	439	572
Spain	1672	2153	2850
Finland	421	576	635
France	3479	11000	10848
Hungary	1214	3182	3247
Ireland	0	55	0
Italy	2028	3205	2764
Lithuania	182	576	588
Luxembourg		0	0
Latvia	68	275	372
Malta	0	0	0
Netherlands	39	195	111
Poland	1884	6142	4447
Portugal	109	190	32
Romania	1351	3312	3601
Sweden	311	596	586
Slovenia	38	127	88
Slovakia	306	827	761
UK	430	2114	2722
EU-27	22936	49285	47493

2.3.2 Other agricultural residues

Beside the straw residues it can also be expected that woody material from prunings and cuttings in permanent crops can deliver a large potential. In certain regions of the EU, plantations with soft fruit, citrus, olives but also vineyards can cover quite a significant surface. In order to estimate the residue potential the permanent cropping areas derived from CAPRI for 2004 and the 2020 reference and 2030 baseline scenario were combined with average harvest ratios per type of permanent crop. The harvest ratios were derived from several publications[10] (See Table 2.11).

[10] Di Blasi, C., Tanzi, V. and Lanzetta (1997), M. A study on the production of agricultural residues in Italy; Biomass and Bioenergy Vol 12 No 5 pp 321-331 (1997)

Lacopo Bernetti et. Al. (2004). A methodology to analyse the potential development of biomassenergy sector: an application in Tuscany; Forest Policy and Economics 6 (2004) 415-432

Figures taken from powerpoint presentation "Bioenergy market in Greece" by Despina Vamvuka (15/12/2006): http://www.enveng.tuc.gr/Downloads/ABES_LAB/05%20Vamvuka.pdf

Siemons, R., Mc Chesney, I., Nikolaou, N., Vis, M., Berg, van den D. & Whitelye M. (2004). Bioenergy's role in the EU energy market. A view of developments until 2020. http://ec.europa.eu/environment/etap/pdfs/bio_energy.pdf

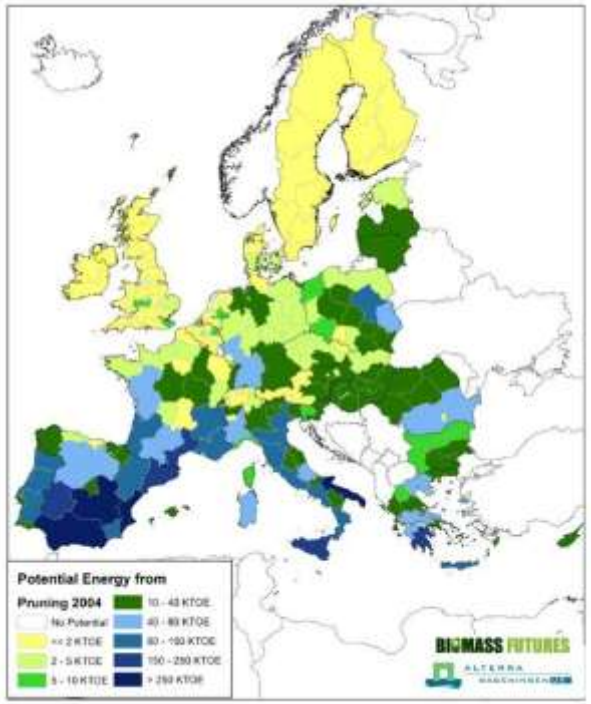
Mladen Ilic, Borislav Grubor and Milos Tesic (2004). The state of biomass energy in Serbia; BIBLID: 0354-9836, 8 (2004), 2, 5-19; http://www.doiserbia.nb.rs/ft.aspx?id=0354-983604020051

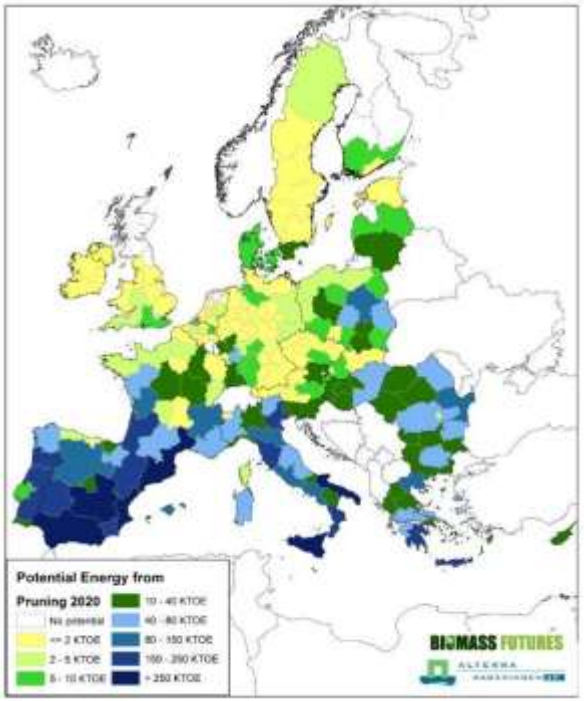
Table 2.11 Average residue harvest ratios per type of permanent crop

Land use category	Residue yields Ton DM/HA/Year
Fruit and berry plantations - total	2.15
Temperate climate fruit and berry plantations	
Subtropical climate fruit and berry plantations	
Nuts fruit and berry plantations	2.15
Citrus plantations	2.75
Olive plantations - table olives	1.77
Olive plantations - oil production	
Vineyards - quality wine	2.81
Vineyards - other wines	
Vineyards - table grapes	
Vineyards - raisins	

The results in the maps (Map 2.6) and Table 2.12 show that especially in the Mediterranean this by-product could be an important resource with Spain as the largest contributor followed by Italy, Greece and Portugal. The largest potential is delivered by vineyards and olives because of their large extent. The total EU potential is not as large as for manure nor straw but for some countries as mentioned before it is a very important resource. In comparison to the other agricultural residues this potential seems to remain relatively stable towards 2030.

Map 2.6 Potential from woody residues of fruit trees, nuts and berry plantations, olives, citrus and vineyards (KTOE) in 2004, 2020 and 2030





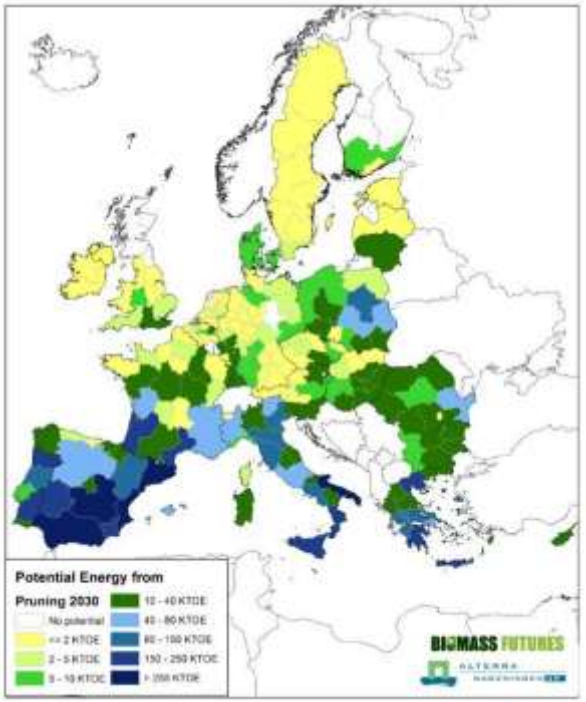


Table 2.12 Potential from woody residues of fruit trees, nuts and berry plantations, olives, citrus and vineyards (KTOE) in 2004, 2020 and 2030

KTOE	2004	2020	2030
Austria	68	48	39
Bulgaria	81	242	106
Belgium/Luxembourg	14	18	26
Cyprus	33	31	17
Czech Rep.	33	10	29
Germany	162	135	129
Denmark	6	6	6
Estonia	2	2	1
Greece	858	801	1163
Spain	3570	4164	3680
Finland	3	8	7
France	1133	996	760
Hungary	150	255	130
Ireland	1	0	2
Italy	1966	2067	1624
Lithuania	23	14	18
Luxembourg	0	0	0
Latvia	20	7	1
Malta	1	0	3
Netherlands	16	13	14
Poland	260	323	360
Portugal	564	586	512
Romania	318	314	150
Sweden	2	22	5
Slovenia	27	20	18
Slovakia	26	9	5
UK	25	15	31
EU-27	9362	10105	8833

Finally it can be mentioned that there are many other primary residues from agriculture which have not been mapped and quantified in this chapter. One important example of this group is the potential from permanent grasslands which are found in agricultural lands, but also in areas like recreational or nature conservation areas or dykes. Furthermore there are also many grasslands abandoned. Most of these permanent grasslands produce a lot of biomass. Management of these areas through cutting could often be beneficial for biodiversity as low levels of human disturbance stimulate larger diversity as it prevents one floristic species to become dominant over others and thus creates more ecological niches for a wider range of species. In the Table 2.13 a potential distribution is given of grass from abandoned grasslands. These lands were abandoned between 2004 and 2020 and 2004 and 2030 only include grasslands that were in agricultural management. So the Map 2.7 and Table 2.13 only provide a limited quantification of the biomass potential from grasslands as they exclude the potential from non-agricultural lands. The reason these recreational area potentials were excluded is related to data limitations as it was difficult to bring together these data within the scope of the project. In Chapter 4 on the waste categories an estimate of biomass potential from road side verges is also given.

The abandoned grassland cuttings potential seems to be significant in 2020, but towards 2030 it is expected in CAPRI that most of these lands will be converted in productive use again either for grazing livestock or to convert in some cropping land, which brings down the potential to almost nothing.

Table 2.13 Grassland cutting potential (KTOE) from abandoned agricultural grasslands in 2020 and 2030

KTOE	2020	2030
Austria	50	0
Bulgaria	35	0
Belgium/Luxembourg	2	0
Cyprus	0	0
Czech Rep.	62	0
Germany	687	0
Denmark	0	0
Estonia	6	0
Greece	30	7
Spain	272	6
Finland	69	49
France	946	0
Hungary	37	0
Ireland	101	0
Italy	73	8
Lithuania	27	0
Luxembourg	0	0
Latvia	21	0
Malta	0	0
Netherlands	44	1
Poland	179	0
Portugal	50	1
Romania	309	0
Sweden	55	20
Slovenia	9	0
Slovakia	50	0
UK	536	170
EU-27	3649	262

3 Biomass from forestry

3.1 Biomass from forests and other wooded land

Forests cover fast amounts of Europe and most forests are managed to supply some kind of ecosystem function (MCPFE 2007[11]). 80 % to 90% of these forests are available for wood supply (MCPFE 2007). The potential sustainable biomass production from forests consists of various streams, most markedly 1) the current production of stem wood for industry, 2) the potential stem wood that could be additionally harvested within the sustainable harvest limit, 3) primary forestry residues, e.g., logging residues, early thinnings and extracted stumps, and 4) secondary forestry residues, residues from the processing of wood in industry.

The potential availability of these four streams is interdependent; primary forestry residues will be available proportional to the amount of stem wood harvested, secondary forestry residues will be available proportional to the activity in the woodworking industry. The secondary forestry residues are already accounted for in the current round wood production, but if these are not directly converted into energy, which is mostly the case with these very expensive feedstock resources, they are more likely to be converted in a later stage of their lifecycle as secondary forestry residues.

The forestry potentials presented here are derived from the EUwood project. The present the present (2010), 2020 and 2030 situation for different forest potentials. The sustainability scenarios 2020 and 2030 are based on the medium mobilisation scenario[12] of EUwood (Mantau et al., 2010 a, Chapter 3). This scenario assumes that regulations and practices enabling or restricting forest operations will be similar to the current situation. The reference scenarios 2020 and 2030 take the EUwood results from the High mobilisation scenario. This assumes that one has been more successful in translating the different measures into practice then in the medium mobilisation scenario sothat more additional harvestable stem wood and primary forestry residues are harvested. In addition, stronger mechanisation is taking place then in the medium mobilisation scenario and possible negative environmental effects of intensified use of forest resources are considered less important than in the medium mobilisation scenario where forests with high biodiversity are excluded from harvesting and more measures are taken to prevent loss of site productivity and soil erosion[13]. The results of the EUwood project also take beside environmental also technical and social constraints into account, including requirements of workforce.

Within EUwood the potential yearly sustainable biomass extraction from forests from stem wood, additional harvestable round wood and primary forestry residues was calculated with the EFISCEN model (Verkerk et al. 2010[14]). Due to the long rotation length of many tree species sustainable harvest should be calculated for prolonged periods. The biomass potential derived through simulation with

[11] MCPFE 2007. State of Europe's forests 2007. The MCPFE report on sustainable forest management in Europe. MCPFE/UNECE/FAO. Warsaw. 263 p.

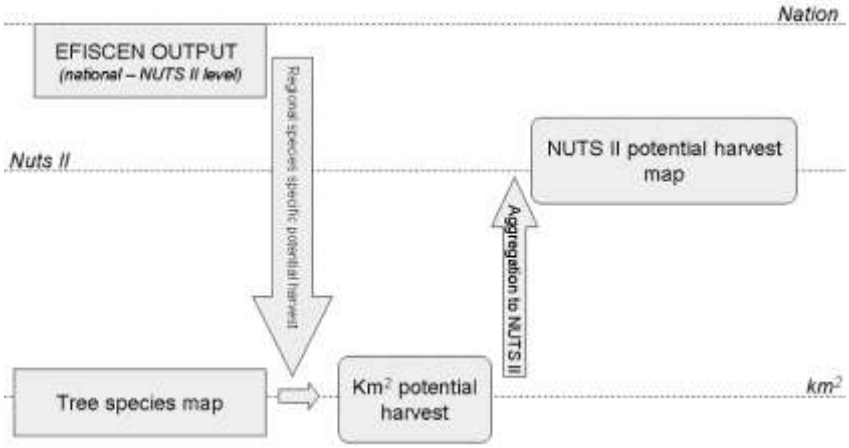
[12] Additionally EUwood provides a high and a low mobilisation scenario. Sustainability is calculated from 2010.

[13] It should be realised that the high mobilisation scenario is presented by EUwood as an extreme scenario, needed to reach the RED targets, but one that would imply large changes in the forestry sector and probably large negative environmental impacts.

[14] Verkerk, P. J., Anttila, P., Lindner, M. and Asikainen, A. (2010). The realistic supply of biomass from forests in EUwood - Real potential for changes in growth and use of EU forests. Final report. Mantau, U. et al. Hamburg, Germany, Centre of Wood Science, University of Hamburg: pp 56-79., http://ec.europa.eu/energy/renewables/studies/doc/bioenergy/euwood_final_report.pdf, http://ec.europa.eu/energy/renewables/studies/doc/bioenergy/euwood_methodology_report.pdf

EFISCEN (*Schelhaas et al., 2007*[15]). The results from the EUwood project are all provided at national level. For the purpose of this study all current (2009), 2020 and 2030 potentials are provided at this level too, but the EFISCEN figures for 2009 were also further downscaled to regions. This was only done for the 2009 results as the pattern for this year will not change very drastically towards 2020 and 2030 as large changes in the forest potential can only be expected after 50 years.

Figure 3.1 Approach followed to identify the regional specific forest potential



The downscaling to regions was done using the tree species distribution maps (Brus et al. in prep[16]), which provide results at 1 km2 grid. The EFISCEN data were first further downscaled to 1 km^2 grid and then upscaled to the desired NUTS2 regions (Figure 3.1).

The EFISCEN model takes the current stemwood production figures from the FAOSTAT database. Here production for 2009 is reported *(FAOSTAT 2011)*[17]. Additionally harvestable stem wood is calculated as the difference between the potential sustainable harvest and the current harvest. Calculated biomass potential from logging residues, stumps and other biomass are summed as primary forestry residues.

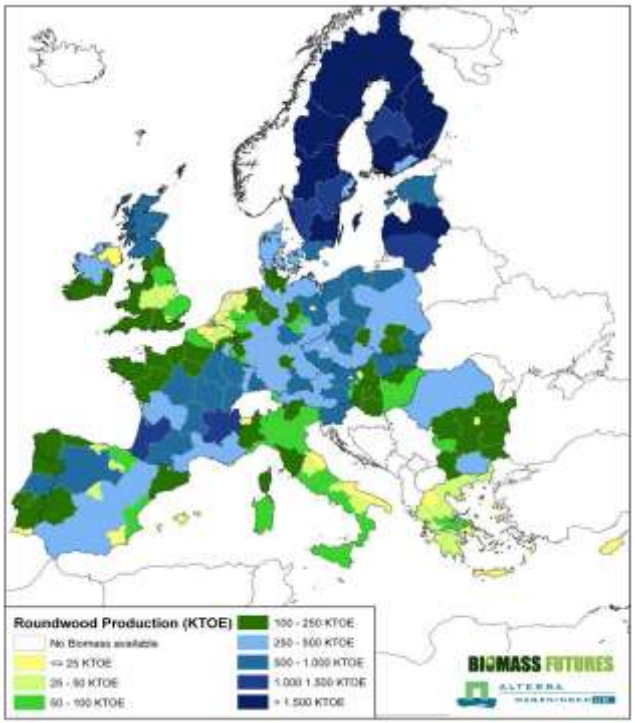
EFISCEN calculates the future state of the forest based on forest information ranging from national to NUTS II level. For some countries, most notably Spain, the information is spread over a very large area. EFISCEN output is confined to the scale of the input data. The EFISCEN output is however also split at the species-level. Although very rough, this species level output has a spatial component, as the different tree species have a heterogeneous distribution within each region. In order to further improve the potential estimates at regional level this tree-species distribution information was used in a further spatial desegregation of the EFISCEN output. For the distribution of species, the tree-species map of Europe provides a spatial distribution at 1 km resolution (Brus et al.in prep.). Keeping in mind that not in every km^2 where a species is present, the complete range in age-classes will be present proportional to the national or regional sums, we coupled the species specific potential harvest per ha to the species cover per km^2 cell. Summing the potential harvest for all species in a cell yields a potential harvest per

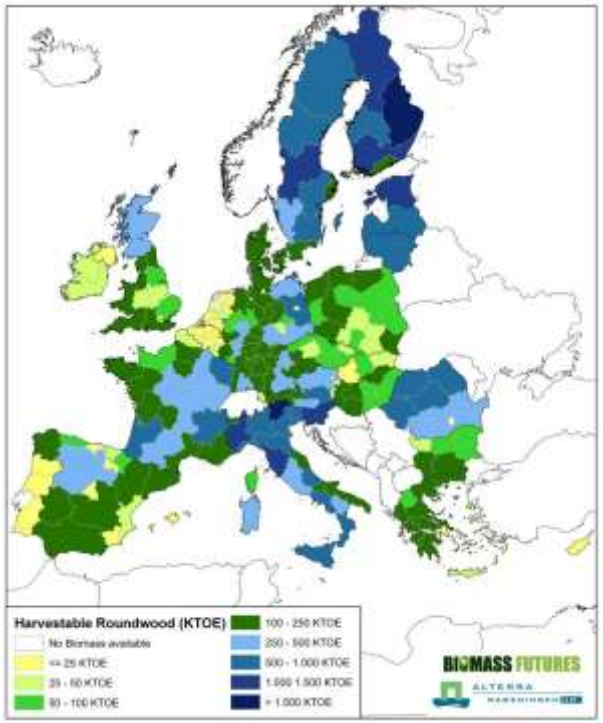
[15] Schelhaas, M.J., J. Eggers, M. Lindner, G.J. Nabuurs, A. Pussinen, R. Päivinen, A. Schuck, P.J. Verkerk, D.C. van der Werf, S. Zudin, 2007. Model documentation for the European Forest Information Scenario model (EFISCEN 3.1). Wageningen, Alterra, Alterra report 1559, EFI Technical Report 26, Joensuu, Finland. 118 p.

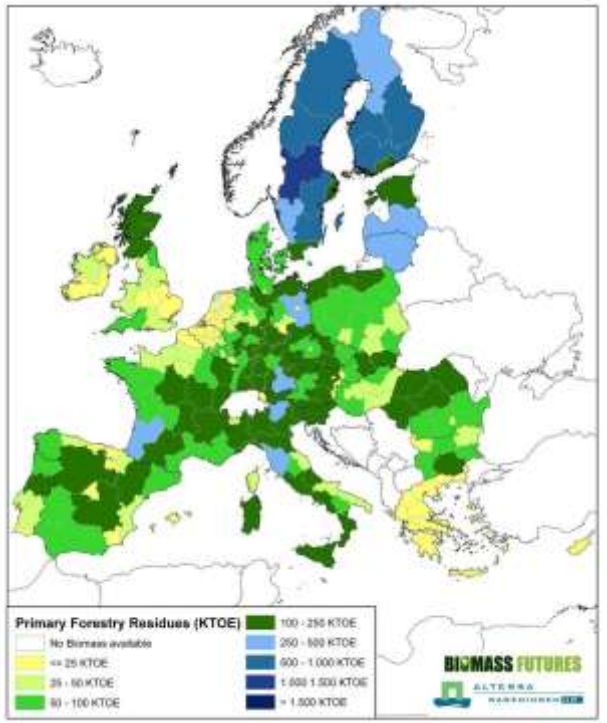
[16] Brus, D., G.M. Hengeveld, N. Heidema, D. Walvoort, P. Goedhart, G.J. Nabuurs et al. in prep, a European tree species disrtibution map.

[17] http://faostat.fao.org/site/626/DesktopDefault.aspx?PageID=626#ancor, roundwoodproduction 2009, downloaded 07 Mar 2011

Map 3.1 Current round wood, additional harvestable round wood and primary forestry residues potential (KTOE) for the current situation*







Source: EUwood project (2010) and own elaboration.

*This spatial disaggregation does not further apply the restrictions of the EU wood scenario assumptions to the regional levels, it only distributes the national EUwood figures according to species distributions per region. A further refinement of this analysis could therefore be done in the future by incorporating the spatial interaction between species distribution and restrictions on forest activities.

Table 3.1 Round wood, additional harvestable round wood and primary forestry residues potential in 2010, 2020 and 2030 reference and sustainability scenario situation

	Current 2010			2020 Reference scenario			2020 Sustainability scenario scenario			2030 Reference scenario			2030 Sustainability scenario scenario		
KTOE	Round wood	Additionally harvestable round wood	Primary forestry residues	Round wood	Additionally harvestable round wood	Primary forestry residues	Round wood	Additionally harvestable round wood	Primary forestry residues	Round wood	Additionally harvestable round wood	Primary forestry residues	Round wood	Additionally harvestable round wood	Primary forestry residues
Austria	2443.3	1554.2	653.6	2398.2	2502.7	1779.4	2398.2	2291.3	779.6	2398.2	2384.2	1739.9	2398.2	2177.9	755.5
Bulgaria	671.8	425.3	310.5	675.2	329.6	543.9	675.2	324.6	287.5	675.2	335.7	523.0	675.2	330.7	279.2
Belgium	642.0	53.9	113.1	611.5	85.8	224.5	611.5	64.6	108.6	611.5	59.5	224.4	611.5	39.1	109.8
Cyprus	1.4	10.0	0.8	1.4	3.6	1.7	1.4	3.4	0.9	1.4	3.6	1.7	1.4	3.4	0.9
Czech Rep.	2364.4	369.6	713.7	2318.2	922.7	1401.7	2318.2	881.9	663.8	2318.2	677.0	1290.4	2318.2	639.4	619.9
Germany	8272.4	5601.3	3584.8	8353.2	5014.6	6536.8	8353.2	4713.9	3096.4	8353.2	4602.0	6415.3	8353.2	4310.6	3000.6
Denmark	247.9	84.7	69.3	219.8	238.2	252.4	219.8	221.3	104.6	219.8	278.0	256.9	219.8	259.7	106.2
Estonia	709.9	950.7	164.9	699.9	1139.1	413.6	699.9	1097.5	184.7	699.9	1054.6	395.7	699.9	1014.9	176.1
Greece	254.6	874.2	74.8	254.6	416.8	171.1	254.6	389.9	87.7	254.6	358.8	156.3	254.6	334.2	80.1
Spain	2118.1	1103.6	777.1	2127.5	1170.3	1460.1	2127.5	1003.5	817.7	2127.5	1069.0	1451.5	2127.5	907.3	840.3
Finland	6084.2	3893.1	2148.1	6297.0	5083.2	6189.4	6297.0	4656.2	2486.7	6297.0	4896.7	6215.0	6297.0	4476.7	2503.4
France	7903.5	4640.0	2033.7	7788.1	3690.7	4337.3	7788.1	3239.3	1808.1	7788.1	4320.0	4644.1	7788.1	3843.9	1874.4
Hungary	766.0	751.6	282.1	771.9	665.0	573.7	771.9	632.5	285.4	771.9	632.4	555.0	771.9	600.6	280.7
Ireland	343.1	57.6	49.3	401.3	135.2	147.9	401.3	123.7	56.4	401.3	230.0	167.3	401.3	216.5	65.8
Italy	1107.3	9960.1	2292.6	983.6	2574.9	970.6	983.6	2394.4	568.2	983.6	2454.8	940.4	983.6	2280.5	558.8
Lithuania	797.5	520.1	247.2	734.5	543.4	631.8	734.5	521.7	245.3	734.5	664.7	684.0	734.5	641.0	272.9
Luxembourg	40.0	113.5	24.1	37.4	92.5	39.3	37.4	88.7	20.2	37.4	84.7	38.3	37.4	81.1	19.7
Latvia	1520.4	598.2	350.7	1363.5	851.7	944.7	1363.5	796.6	410.2	1363.5	1309.9	1114.9	1363.5	1243.4	488.2
Malta	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Netherlands	148.4	69.0	22.0	137.8	60.8	71.5	137.8	55.7	22.0	137.8	78.9	77.4	137.8	73.3	25.1
Poland	5058.2	951.4	997.4	4938.8	2595.6	3219.8	4938.8	2520.8	1262.2	4938.8	2354.0	3111.3	4938.8	2281.5	1223.1
Portugal	900.7	0.0	264.9	856.7	484.1	582.6	856.7	416.8	313.6	856.7	653.7	643.6	856.7	577.8	345.8
Romania	1834.1	2711.4	730.7	1783.3	2573.2	1364.2	1783.3	2519.9	767.8	1783.3	2518.2	1307.9	1783.3	2465.5	749.5
Sweden	9509.0	3752.5	3522.8	9560.5	4788.7	7965.4	9560.5	4250.2	3560.8	9560.5	5571.0	8476.4	9560.5	5003.1	3570.2
Slovenia	428.0	922.0	120.0	398.7	825.0	228.5	398.7	772.9	131.8	398.7	781.3	228.2	398.7	731.0	130.3
Slovakia	1327.3	77.9	337.9	1274.5	121.3	548.0	1274.5	95.3	309.7	1274.5	142.0	592.3	1274.5	115.6	322.0
UK	1241.2	1000.2	399.0	1127.9	962.6	586.5	1127.9	896.3	358.2	1127.9	1014.1	590.8	1127.9	946.2	370.3
EU-27	56735	41046	20285	56115	37871	41186	56115	34973	18738	56115	38529	41842	56115	35595	18769

Source: EUwood project (2010)

km^2. From these 1 km^2 we up-scaled back to the NUTS II level. This resulted in a finer spatial resolution result than the original EFISCEN output for the current situation (see Map 3.1). The overall results of the EFISEN-EUwood assessment are presented at national level in the Table 3.1 for round wood, additional harvestable round wood and primary forestry residues in the current, 2020 and 2030 reference and sustainability scenario situation.

Map 3.1 further emphasises the importance of regions in Scandinavia and Central France in the total round wood production. They also show that small countries of the Baltic States, especially Latvia, have a significant production of round wood. The additional harvestable potential shows that in spite of the already high present stem wood production in most regions of Scandinavia and France they also contribute significantly to the still harvestable resource for bioenergy production. Other regions with a large contribution to the bioenergy feedstock potential are found in Italy, Romania and Slovenia. This latter group has clearly a large un-harvested potential in comparison to the present stem wood production.

Table 3.2 Country summary of 2010 additional harvestable round wood potential expressed relative to round wood potential and present round wood production

Country	% additional harvestable round wood from round wood potential	% additional harvestable round wood from current production
Austria	0.39	0.64
Belgium	0.08	0.08
Bulgaria	0.39	0.63
Cyprus	0.87	6.95
Czech	0.14	0.16
Denmark	0.25	0.34
Estonia	0.57	1.34
Finland	0.39	0.64
France	0.37	0.59
Germany	0.40	0.68
Greece	0.77	3.43
Hungary	0.50	0.98
Ireland	0.14	0.17
Italy	0.90	8.99
Latvia	0.28	0.39
Lithuania	0.39	0.65
Luxembourg	0.74	2.84
Malta	0.00	0.00
Netherlands	0.32	0.46
Poland	0.16	0.19
Portugal	0.00	0.00
Romania	0.60	1.48
Slovakia	0.06	0.06
Slovenia	0.68	2.15
Spain	0.34	0.52
Sweden	0.28	0.39
United Kingdom	0.45	0.81

Source: EUwood project: Mantau et al. (2010)

The regions with a relatively large contribution to the primary forest residues are again concentrated in Scandinavia, France, Italy, Germany and Austria. This potential is generally much smaller than the additional harvestable potential although still a significant contribution in absolute terms. The results at national level show that the largest current round wood production is in Sweden, France, Germany and Finland, but also smaller countries like Austria, Czech Republic and Latvia have a large current production. It should be realised that most of this production is going to wood-industry for non-energetic use, but part of it is also already going to bioenergy production (see Table 3.3).

In general there is a relationship between present stem wood production and additionally harvestable stem wood and primary forestry residues. This becomes clear when also looking at the figures in Table 3.2. If the round wood production is already highly efficient i.e. removing a large part of the total potential (see before last column), the additional harvestable potential (last column) is relatively smaller. From a bioenergy perspective the largest potential should then be searched in the primary and secondary residues categories.

The additional harvestable and the primary residues categories are the main sources for feedstock for bioenergy. Presently these resources are practically not harvested, but they could become mobilised if enough economic and other stimulation measures become applied. Generally, there is an economic relationship between round wood production and additional harvestable potential. If there is a large demand for round wood more will be harvested and the immobilised additional harvestable category becomes smaller. This is clearly the case in Belgium, Czech Republic and Ireland. Italy seems to be a strange out layer as the present production of round wood is considerably smaller than the additional harvestable potential. Possibly the FAO figures are underestimating the present Italian production and/or the EFISCEN calculations are over-estimating the additional harvestable potential. For Portugal the EFISCEN projections do not account for the potentials from Eucalypt plantations. Hence a smaller sustainable yield for Portugal than the reported yield (6.2 mill. m^3 vs. 9.6 mill. m^3), one could however question the sustainability of Eucalypt biomass production (Lindner et al. 2005).

Table 3.3 Landscape care wood potential

KTOE	2010	2020	2030
Austria	171	264	254
Bulgaria	202	261	251
Belgium	93	139	134
Cyprus	16	23	22
Czech Rep.	202	306	294
Germany	763	1144	1103
Denmark	109	175	169
Estonia	62	102	98
Greece	233	79	76
Spain	980	409	394
Finland	327	510	492
France	1961	2877	2772
Hungary	218	301	290
Ireland	171	259	250
Italy	514	230	222
Lithuania	140	212	205
Luxembourg	0	6	6
Latvia	125	194	187
Malta	0	2	2
Netherlands	109	149	144
Poland	763	1100	1060
Portugal	218	159	154
Romania	436	595	573
Sweden	560	859	828
Slovenia	31	38	37
Slovakia	109	173	167
UK	560	851	820
EU-27	9073	11419	11002

Source: EUwood project and own elaboration

Estimates of the wood potential from landscape care wood were also made by the EUwood project. However, these estimates included the potentials from cuttings from permanent crops in agriculture. Since the estimates of this group were already presented in Chapter 2 and were based on more detailed land use information from CAPRI, then what was used in the EUwood estimates an elaboration was applied to the EUwood figures. The EUwood potential on cuttings from fruit trees and vineyards was separated from the rest of the landscape care potential. The remaining potentials of EU wood now only refer to landscape care potentials outside agricultural permanent crop land. The results are given in the

table 3.3. France has by far the largest potential, but countries like Poland and Germany also contribute significantly.

3.2 Secondary and tertiary forestry residues

As already pointed out in the beginning of this Chapter, the potential of secondary residues is interdependent of the rest of the forestry round wood production and immobilised categories, unless there is a large industry based on imported wood like in The Netherlands, Denmark and Portugal. Furthermore, it should also be emphasised that the secondary forestry residues are already accounted for in the current round wood production, as presented in the former. This implies that they cannot be summed up, but their quantification is very interesting as their use as feedstock for bioenergy is more likely than the use of round wood as feedstock. These secondary residues have a lower price and are therefore more competitive.

In the EUwood project estimates and projections were also made for the secondary residues from forest derived from the wood processing industry in the EU. The categories quantified in the EU wood project include residues from sawmills, Black liquor and other industrial residues, post-consumer wood and landscape care wood. In the following we will present the potentials for all these categories making use of the EUwood data, but also using additional sources as will be explained.

3.2.1 Black liquor, sawmill and other industrial residues

The pulp industry creates a huge amount of secondary residues in the form of black liquor. This by-product of pulp mills is almost completely used for energy production in the pulp and paper industry.[18] Efficiency of pulp making is low and thus a huge source of biomass for bioenergy is created. However, sources estimating the production of black liquor provide very different volumes (see Table 3.4).

As becomes clear the estimates by IIASA and certainly the EUwood are generally underestimating the total black liquor production as estimated in the FAOSTAT (table 3.4). For the IIASA figures this is probably explained by the fact that not all processing industries have been covered in the IIASA database. The final potential production of black liquor was therefore estimated as an average between IIASA and FAO data. To come to an estimate of the future production in 2020 and 2030 the growth figures of BL production between 2010 and 2020 and 2030 respectively from the EUwood project were applied to the total average of BL based on FAOSTAT and IIASA sources.

The most important BL producers are by far Finland and Sweden, but countries like Germany, Austria, France, Poland, Portugal and Spain also deliver a significant contribution.

[18] Saal 2010, http://ec.europa.eu/energy/renewables/studies/doc/bioenergy/euwood_methodology_report.pdf, Fonseca, 2009 http://timber.unece.org/fileadmin/DAM/publications/DP-49.pdf .

Table 3.4 Black liquor (BL) production as reported to FAOSTAT and derived from IIASA database and EUwood and final estimate of BL potential in 2020 and 2030 (Countries that are missing do not produce BL)

Table	1) FAOSTAT [t]	2) IIASA [t]	3) Average BL production (tons) based on FAOSTAT & IIASA	4) 2010	5) 2020	6) 2030	7) 2010-2020 growth rate	8) 2010-2030 growth rate	total BL production 2020 (Kton)	total BL production 2030 (Kton)
Austria	3324	3109	3217	1032	1032	1118	0.0	0.1	3217	3485
Belgium	1677	490	1084	602	516	731	0.2	0.4	1264	1535
Bulgaria	267	200	233	258	172	344	0.5	1.0	350	467
Czech Republic	1350	798	1074	1075	688	1634	0.6	1.4	1678	2550
Denmark	5	215	110	0	0	0	0.0	0.0	110	110
Estonia	262	208	235	129	86	172	0.5	1.0	352	469
Finland	18783	13506	16145	7998	6966	8944	0.1	0.3	18536	20729
France	3639	4958	4299	1677	1634	1978	0.0	0.2	4412	5204
Germany	4435	4335	4385	2107	1548	2709	0.4	0.8	5968	7673
Hungary a)	n.a.	n.a.	42	0	0	0	0.0	0.0	42	42
Italy	488	938	713	86	43	43	1.0	0.0	1425	713
Netherlands	142	239	190	0	0	0	0.0	0.0	190	190
Poland	1861	1552	1707	1290	860	1935	0.5	1.3	2560	3840
Portugal	4044	2025	3034	2408	1978	2752	0.2	0.4	3694	4222
Romania	68	420	244	172	172	258	0.0	0.5	244	366
Slovakia	1275	363	819	903	559	1333	0.6	1.4	1323	1953
Slovenia	175	360	267	172	86	215	1.0	1.5	535	668
Spain	3892	4594	4243	1935	1677	2279	0.2	0.4	4895	5766
Sweden	20464	14209	17336	8858	7869	9933	0.1	0.3	19515	21884
United Kingdom	277	2223	1250	43	43	43	0.0	0.0	1250	1250
a) Eurobserv'ER 2008										

The other industrial wood residues were derived from the EUwood project (see Table 3.5) as were the sawmills by-products. They show that the main producers of secondary wood products like Sweden, Finland, Germany and France have large potentials for all four categories. The largest potentials are in the saw-mill by products and the BL. The latter is however expected to decline towards 2030.

Table 3.5 Potentials (KTOE) for sawmill by-products, other industrial wood residues and black liquor

	Current 2010				2020				2030			
KTOE	Sawmill by-products (excl saw dust) a)	saw-dust a)	Other industrial wood residues a)	black liquor b)	Sawmill by-products (excl saw dust) a)	saw-dust a)	Other industrial wood residues a)	black liquor b)	Sawmill by-products (excl saw dust) a)	saw-dust a)	Other industrial wood residues a)	black liquor b)
Austria	757	378	311	248	840	420	342	772	913	456	389	268
Bulgaria	51	27	47	41	61	33	62	84	71	38	93	83
Belgium	95	45	109	124	106	50	140	303	127	60	156	175
Cyprus	0	0	0	0	0	0	0	0	0	0	0	0
Czech Rep.	322	161	140	165	353	176	156	403	405	202	202	392
Germany	1467	680	1074	372	1723	798	1183	1432	2063	956	1369	650
Denmark	20	11	16	0	20	11	16	26	30	16	16	0
Estonia	180	85	47	21	222	105	62	84	264	125	78	41
Greece	20	11	31	0	20	11	47	0	20	11	62	0
Spain	285	135	296	403	296	140	373	1175	306	145	451	547
Finland	1299	708	389	1672	1400	763	467	4449	1501	817	529	2147
France	755	350	420	392	787	365	467	1059	829	384	529	475
Hungary	21	10	31	0	31	15	47	10	42	21	62	0
Ireland	101	55	47	0	111	60	62	0	131	71	78	0
Italy	115	56	311	10	125	62	373	46	136	67	436	10
Lithuania	121	65	47	0	141	76	62	0	162	87	62	0
Luxembourg	10	5	31	0	10	5	31	0	10	5	31	0
Latvia	344	185	93	0	424	229	124	0	515	278	156	0
Malta	0	0	0	0	0	0	0	0	0	0	0	0
Netherlands	21	10	0	0	31	0	0	46	21	10	16	0
Poland	264	125	405	206	275	130	498	614	317	150	638	464
Portugal	84	40	78	475	106	50	109	887	127	60	140	661
Romania	327	155	140	41	412	195	187	59	507	240	249	62
Sweden	1909	954	280	1889	2044	1021	311	4684	2179	1089	342	2384
Slovenia	43	20	47	21	53	25	62	0	74	34	78	52
Slovakia	159	74	78	134	181	84	93	317	234	108	124	320
UK	301	150	171	10	322	161	187	300	332	166	202	10
EU-27	9072	4496	4637	6223	10093	4984	5461	16751	11316	5597	6488	8742

a)Source EUwood

b)Based on calculation in Table 3.4

3.2.2 Post-consumer wood potentials

In the EUwood project estimates were also made of the post-consumer wood potential a category that can be categorised as tertiary wood potential as it is derived after its use by a consumer like a household as part of waste or from building sites (see Table 3.6). Results show that the total potential is in the same size category as the sawmill by-products and the BL. The potentials are generally linked to population size of the EU-countries explaining the largest potentials in countries like Germany, France, Poland, Italy and UK.

Table 3.6 Post-consumer wood potential (KTOE) 2010, 2020 and 2030

KTOE	2010	2020	2030
Austria	161	190	204
Bulgaria	15	44	29
Belgium	307	336	394
Cyprus	15	15	15
Czech Rep.	102	146	190
Germany	1270	1373	1475
Denmark	190	190	204
Estonia	29	234	44
Greece	131	146	175
Spain	613	701	789
Finland	175	190	219
France	920	964	1022
Hungary	73	102	131
Ireland	88	88	102
Italy	905	1051	1241
Lithuania	44	58	73
Luxembourg	0	0	0
Latvia	44	44	58
Malta	0	0	0
Netherlands	365	380	409
Poland	511	686	978
Portugal	102	131	146
Romania	248	336	438
Sweden	146	161	175
Slovenia	15	29	29
Slovakia	29	44	58
UK	1095	1154	1241
EU-27	7593	8791	9842

Source: EUwood project

3.3 Conclusions on potentials from the forestry sector

In order to make an estimate of how much of the forestry potential is already used for renewable energy production in Table 3.7 the total primary and secondary forestry potential for 2010, excluding landscape care wood, is compared to the present renewable energy production from wood and wood residues as reported in the Eurostat Waste Statistics for 2007. The first column illustrates to which extend countries are already using forest products for renewable energy production and the relative comparison (last column) illustrates to which extent the potential is already used. The comparison is not straight forward, since the potentials in the second column only refer to the domestic potential, while the first column also includes the use of wood and wood residues which can be imported. This also explains why countries like Denmark, The Netherlands and Portugal end up with more than 100% of their potential used. Clearly these countries have a large wood processing industry largely based on imported wood. What becomes clear however that the production of bioenergy from wood resources is already very large, particularly in countries like Germany, France, Finland, Austria, Sweden, but to which extent this is based on domestic of imported resources is not clear.

Table 3.7 Total and relative 2007 production of renewable energy from wood and wood residues and total forest potential

	Production of Renewable energy from forestry (2007) (KTOE) 1)	Total forest potential 2010 2)	% renewable production from total potential
Austria	3930	6346	62%
Belgium	649	1182	55%
Bulgaria	709	1573	45%
Cyprus	n.a.	12	n.a.
Czech Republic	1948	4235	46%
Denmark	1441	449	321%
Estonia	731	2157	34%
Finland	7149	16193	44%
France	9234	16494	56%
Germany	10578	21051	50%
Greece	1005	1266	79%
Hungary	1146	1862	62%
Ireland	169	652	26%
Italy	1707	13853	12%
Latvia	1532	3092	50%
Lithuania	732	1798	41%
Luxembourg	16	224	7%
Malta	0	0	0%
Netherlands	524	271	194%
Poland	4550	8007	57%
Portugal	2808	1843	152%
Romania	3304	5940	56%
Slovakia	484	2189	22%
Slovenia	429	1600	27%
Spain	4206	5117	82%
Sweden	8441	21816	39%
United Kingdom	784	3273	24%
Total	68206	142494	48%

1) Eurostat Waste statistics: Overview of Renewable energy from Forestry (Wood and wood residues) (2007)
2) EUwood project: Mantau et al. (2010) = total present round wood production + additional harvestable wood potential+primary forestry residues

To make a final estimation of the present potential of biomass from forestry we need to clarify which categories presented in the former are included in the total potential. For this we need to involve the cost levels of the different wood, wood residues and secondary residues into account. A possibility is to build on the economic analysis done by Lindner et al. (2005). In this study it is indicated that the cheapest resource from forestry is from industrial wood residues especially because they are already available at the site of the energy plant, so no extraction nor transport cost involved. The cost for extraction of forest residues and especially complementary fellings are significantly higher making their mobilisation for bioenergy production already complicated. This also explains their very limited use under the present market conditions (Table 3.3). According to Lindner et al. (2005), the competition for wood fibre starts to be turned in favour of bioenergy conversion as from 70 Euro per m3 (343 Euro/toe) onwards. This can only be paid if carbon credit prices move towards 60 Euro per ton of CO_2 and/or fossil energy prices increase significantly. At this moment the EU Carbon credit level is around 14 Euro/ton CO_2 and it is therefore logical to exclude the present stem wood production from the total forest biomass potential.

The cheapest forest resource will then come from secondary residues such as other forestry residues, sawmill by-products and post-consumer wood and BL. Most of the latter category is already re-used as an energy feedstock in the paper and other wood processing industry. So it is not really available on the open market, although used for bioenergy production inside the wood/paper industry. Further details on the cost-supply relations can be found in Chapter 5.

4 Biomass from waste sector

Since the regulation on waste statistics is implemented (EC 2150/2002) EU member States are obliged to report data on waste amounts to Eurostat. A distinction is made in 48 categories of waste and a distinction is also made in hazardous and non-hazardous. The data reported for 2008 to Eurostat together with the biodegradable waste estimates from the EEA (2006) study were used as the main basis for our potential assessment. For road side verges potential and used fats and oils other sources and own elaborations were used. A selection was made from the waste categories reported in Eurostat and EEA (2006) that were fitting to the waste categories selected and defined by the BEE project. For missing waste categories additional sources of information were searched like for verge grass and used fats and oils. The categories reported are all seen as major feedstock for bionergy generation.

In the following an overview at national level is given of the main waste categories of biodegradable waste. Data were also recorded on the present treatment and recovery rates for different types of waste. These data were analysed in order to get a better understanding of the present use of waste for bioenergy generation, their competitive use and thus their real technical and economic availability for bioenergy generation. It should also be mentioned that doubts exist about the consistency of the data reported by the different Member States. In some countries amounts of waste are very large, while this is not reflected by the number of inhabitants. It is therefore expected that the amount of waste reported depends not only on the total production of waste, but also on whether there is a collection system enabling the registration of the total amount of waste. In this study we assume that if waste amounts are well reported, they are also collected and this implies that they will also be more likely to become used as feedstock for bioenergy generation.

4.1 Primary residues

The first waste category to be mapped and quantified is verge grass (see Map 4.1). Data on this resource do not come from Eurostat, but its amount is assumed to be directly linked to length and density of roads. These were estimated using an EU-wide road network map[19] combined with a more precise road network map for The Netherlands (TOP10, Kadaster). Since the EU-wide data source only contains the main roads, the more detailed information from The Netherlands could be used and extrapolated EU wide using road density relations between the 2 data sources to the EU-wide data layer. A 10 meter boundary was estimated along the total road length in every region for which an average grassland potential was calculated. This estimation was made based on an analysis of aerial photographs (AEROGRID) and Google Maps. For the estimation of the grassland yield we build on Smit et al. (2008) who estimated an average grassland productivity factor for different types of grassland per environmental zone in Europe. The type of grassland used in this map was the extensive grassland type. The environmental zonation ensures that grassland productivity is directly linked to climatic factors such as rainfall, evapotranspiration and length of growing season.

[19] ESRI® Data & Maps 2008 Update. Europe Roads represents the roads (European Highway System, national, and secondary roads) in Europe. Europe Roads provides a base map layer of roads for Europe. Largest scale when displaying the data: 1:10,000.

Map 4.1 Actual biomass potential (KTOE) from roadside verges assuming grassland cover at 10 meters of either sides

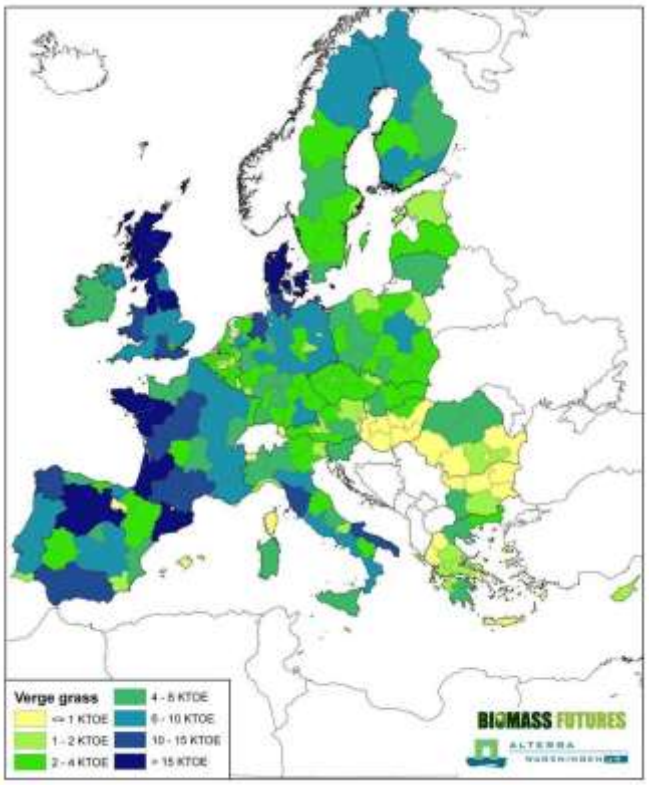


Table 4.1 Potential of grassland cuttings (KTOE) in road side verges 2010, 2020 and 2030

KTOE	2010	2020	2030
Austria	18	19	19
Bulgaria	8	8	7
Belgium	22	23	23
Cyprus	1	1	1
Czech Rep.	21	21	20
Germany	190	199	204
Denmark	29	31	31
Estonia	2	2	2
Greece	20	21	21
Spain	107	112	115
Finland	27	28	29
France	202	212	217
Hungary	7	7	7
Ireland	10	10	10
Italy	100	105	108
Lithuania	4	4	4
Luxembourg	0	0	0
Latvia	4	4	3
Malta	0	0	0
Netherlands	38	40	41
Poland	58	57	56
Portugal	36	37	38
Romania	17	17	17
Sweden	30	31	32
Slovenia	4	4	4
Slovakia	14	14	13
UK	129	135	138
EU-27	1097	1143	1162

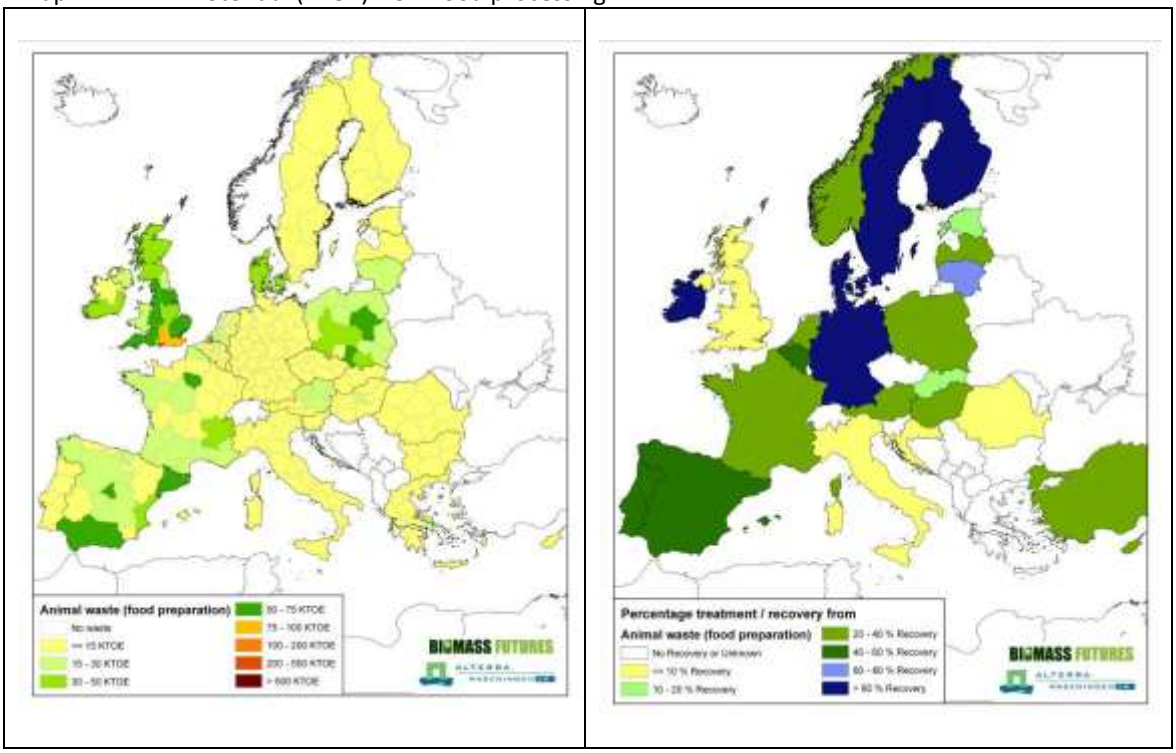
The results show that verge grass could amount to almost 1100 KTOE per year which is not amongst the highest potential, but could be an interesting resource to top-up the woody-feedstock amount in regions where large biomass conversion installations are based. The potential towards 2020 and 2030 is assumed to develop according to population and income growth predictions for these 2 periods as based on FAO-OECD projections (EC, 2010).

4.2 Secondary residues from the food processing industry

The residues mapped in this category are defined by Eurostat as 'Animal waste of food preparation and products'. It is waist consisting of animal tissue coming from preparation and processing of meat, fish and other foods of animal origin and of sludges from washing and cleaning of these products. The total current potential EU wide is estimates at 2763 KTOE.

However, whether this potential is really completely available for bioenergy generation is very much the question as becomes clear from the map on the right hand (Map 4.2). In many EU countries, particularly Germany, Sweden, Finland and Ireland, this type of waste is already recovered, but not only for energy conversion.

Map 4.2 Potential (KTOE) from food processing



Source; Eurostat waste statistics, 2008

The potential is particularly important in countries with a large food-processing industry like for example Spain, Poland, UK, France, Belgium, The Netherlands and Austria. The potential towards 2020 and 2030 is assumed to develop according to population and income growth predictions as based on FAO-OECD projections (EC, 2010).

Table 4.2 Animal waste potential (KTOE) 2010, 2020 and 2030

KTOE	2008	2020	2030
Austria	105	110	112
Bulgaria	2	2	2
Belgium	105	111	113
Cyprus	5	5	5
Czech Rep.	16	16	15
Germany	80	84	86
Denmark	40	42	43
Estonia	14	14	14
Greece	69	73	74
Spain	392	412	421
Finland	37	39	39
France	384	403	412
Hungary	61	60	59
Ireland	53	56	57
Italy	34	36	36
Lithuania	21	20	20
Luxembourg	0	9	0
Latvia	9	9	8
Malta	2	2	2
Netherlands	122	128	131
Poland	516	509	499
Portugal	34	35	36
Romania	12	12	11
Sweden	35	37	38
Slovenia	8	8	8
Slovakia	10	10	10
UK	609	639	653
EU-27	2777	2880	2907

Source; Eurostat waste statistics, 2008 and own extrapolations

4.3 Tertiary residues

Within the tertiary waste we distinguished 5 categories of waste. The first three are derived from households and industries and refer to the municipal household waste and the common sludges. The other 2 categories are the used fats and oils and the paper cardboard wastes collected from households and industries.

4.3.1 Biodegradable waste from private households and industry

The total potential of this category was estimated as part of the EEA (2006) study for 2010, 2020 and 2030. Two categories of municipal solid waste are distinguished: municipal solid waste not going to landfill, composting and recycling and the category going to land fill. It is clear that the first category was estimated to identify the part of the MSW potentially available to be converted to bioenergy. The second category also provides a potential as it can be expected that in the future less of this MSW is going to landfill and more is used as bioenergy feedstock.

In order to make the estimates for the biodegradable component of municipal solid waste projections by the European Topic Centre on Resource and Waste Management (Skovgaard et al., 2005 and EEA, 2005). However, it is assumed that waste generation can be reduced by 25 % by 2030, due to household waste prevention measures (based on data from Gewiese et al., 1988). The fraction of waste that is biodegradable is assumed to remain constant into the future.

Table 4.3 Municipal solid waste potential (KTOE) 2010, 2020 and 2030

	MSW not used for landfill, composting and recycling			MSW landfill		
KTOE	2010	2020	2030	2010	2020	2030
Austria	0	0	0	180	180	180
Bulgaria	258	258	73	360	360	360
Belgium	0	86	146	900	540	360
Cyprus	0	0	0	180	0	0
Czech Rep.	86	258	220	720	360	360
Germany	948	1120	879	3240	2160	1800
Denmark	172	172	146	0	0	0
Estonia	0	0	0	180	0	0
Greece	258	431	366	540	180	180
Spain	861	861	659	1080	900	720
Finland	86	172	146	180	180	180
France	861	1034	879	2340	1800	1440
Hungary	0	86	146	1080	540	360
Ireland	86	172	146	180	180	180
Italy	689	689	586	1080	900	720
Lithuania	0	0	73	360	180	180
Luxembourg	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Latvia	0	0	0	180	0	0
Malta	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Netherlands	258	258	146	180	180	180
Poland	258	431	366	2160	1080	900
Portugal	172	345	293	720	360	360
Romania	172	345	220	1620	720	720
Sweden	86	86	73	360	360	180
Slovenia	0	0	0	180	180	180
Slovakia	0	86	73	360	180	180
UK	1120	1981	1611	3780	1800	1440
EU-27	6375	8873	7249	22140	13320	11160

Source: EEA, 2006

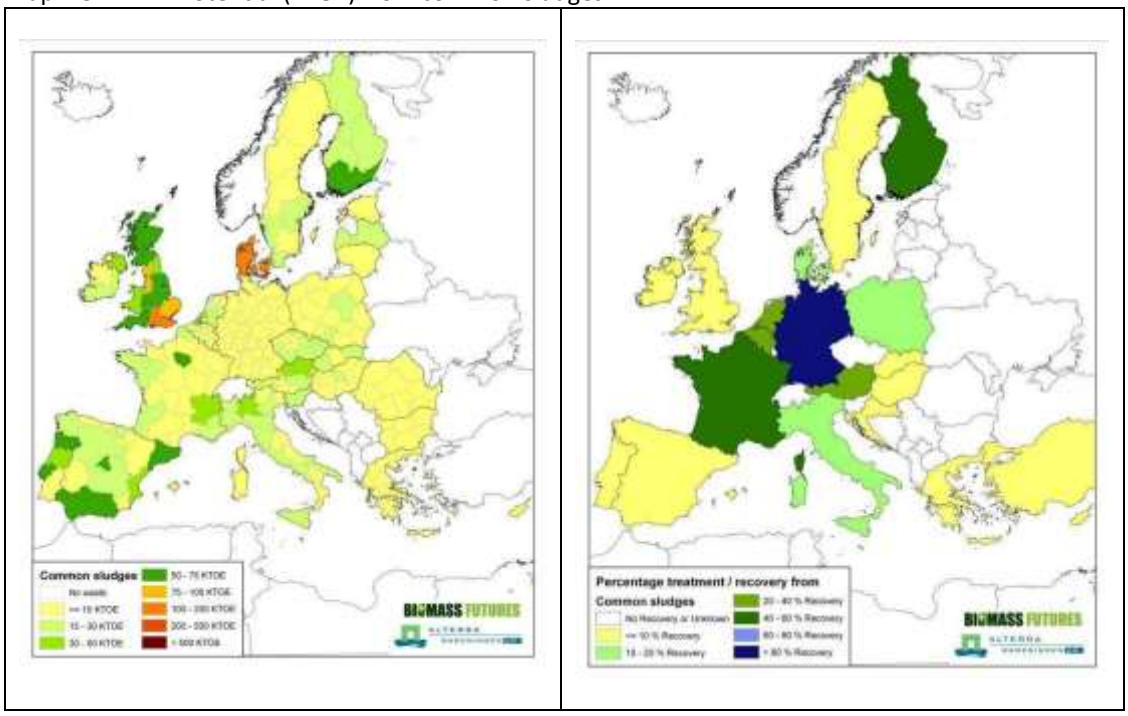
The estimates show that the extent of the MSW potential is generally strongly related to population size. However, some countries have a larger share of their MSW in the first category while others show the largest potential going to landfill. In Austria for example all MSW is collected and going to landfill, but in Spain the MSW is relatively evenly distributed over landfill and no-landfill composting and recycling. Overall it becomes clear that the first category is estimated to be smaller, but expected to remain relatively stable into the future and the landfill part is expected to decline by half. This is the result of the assumption made in the EEA study that MSW production will decrease by 25% by 2030.

4.3.2 Common sludges

The category common sludge which is defined by Eurostat as 'Industrial effluent sludges' includes all kinds of sludges originating from wastes from waste water treatment and water preparation. The total current potential is estimated to be at 3700 KTOE and is particularly concentrated in the UK, France, Italy and Spain (Map 4.3 and Table 4.4).

Present recovery rate of this category is still very low in most EU countries which is related to the limited possibilities to recover this waste other than into energy. At this moment most of the sludges are incinerated and/or deposited into land and only a small part is already used for energy recovery, like in Germany, France and Finland.

Map 4.3 Potential (KTOE) from common sludges



Source; Eurostat waste statistics, 2008

Table 4.4 Potential (KTOE) from common sludges for 2010, 2020 and 2030

KTOE	2008	2020	2030
Austria	427	448	458
Bulgaria	77	76	75
Belgium	250	263	269
Cyprus	5	4	4
Czech Rep.	276	273	267
Germany	267	280	286
Denmark	273	286	293
Estonia	23	22	22
Greece	71	74	76
Spain	813	853	872
Finland	412	432	442
France	668	701	717
Hungary	113	111	109
Ireland	49	51	52
Italy	594	623	637
Lithuania	29	28	28
Luxembourg	0	0	0
Latvia	48	48	47
Malta	0	0	0
Netherlands	337	354	362
Poland	288	284	279
Portugal	399	419	428
Romania	115	113	111
Sweden	300	315	322
Slovenia	45	45	44
Slovakia	110	109	107
UK	1779	1866	1907
EU-27	7768	8080	8213

Source; Eurostat waste statistics, 2008 and own extrapolations

4.3.3 Used fats and oils

Data on this category of waste were derived from the REFUEL report (Deurwaarder et al., 2007). It provides estimates for the potential in 2020. In this study the potentials for 2020 were back-casted to 2010 and forecasted to 2030 by using country specific population and income growth projections towards 2020 and 2030. This results in a potential for used fats and oils as presented in Table 4.5. As expected, the large countries have large potentials.

Table 4.5 Potential for used fats and oils for 2010, 2020 and 2030

KTOE	2010	2020	2030
Austria	34	35	36
Bulgaria	45	44	44
Belgium	33	34	35
Cyprus	32	31	30
Czech Rep.	45	44	43
Germany	345	356	364
Denmark	22	23	24
Estonia	6	6	6
Greece	46	47	48
Spain	172	177	181
Finland	22	22	23
France	251	259	264
Hungary	44	43	42
Ireland	16	17	17
Italy	241	248	254
Lithuania	15	15	15
Luxembourg	0	2	0
Latvia	10	10	10
Malta	71	69	68
Netherlands	19	20	20
Poland	170	167	164
Portugal	42	43	44
Romania	99	97	95
Sweden	37	38	39
Slovenia	9	9	8
Slovakia	24	23	23
UK	249	256	262
EU-27	2099	2137	2159

Source; REFUEL (Dearwaarder et al., 2007) and own extrapolations

4.3.4 Paper cardboard

The final potential presented is for paper card board. Eurostat reports on this potential but it takes no account of recycling. It includes 'fibre, filler and coating rejects from pulp, paper and cardboard production. It's origin is either from pulp and paper industry and separate collection of waste (e.g. from households, industry, offices). According to CEPI (2009) the level of recycling in this commodity is very large. Most of the used paper and card board is re-used as feedstock for the paper industry again. The potential available for bioenergy generation is therefore much lower than the Eurostat figures indicate. To make a more realistic assumption the Eurostat figures have been corrected with the CEPI (2009) recovery rate (see Table 4.6).

Table 4.6 Recycling rates of used paper card board wastes

	CEPI recycling rate paper wastes
Austria	0.8
Bulgaria	0.56
Belgium	0.5
Cyprus	0.5
Czech Rep.	0.55
Germany	0.83
Denmark	0.8
Estonia	0.5
Greece	0.5
Spain	0.75
Finland	0.71
France	0.72
Hungary	0.54
Ireland	0.7
Italy	0.63
Lithuania	0.5
Luxembourg	n.a.
Latvia	0.5
Malta	0.8
Netherlands	0.82
Poland	0.37
Portugal	0.73
Romania	0.52
Sweden	0.74
Slovenia	0.5
Slovakia	0.5
UK	0.78

Source: CEPI, 2009. Annual statistics

Table 4.7 Potential (KTOE) from paper card board for 2008, 2020 and 2030

KTOE	2008	2020	2030
Austria	183	191	177
Bulgaria	42	41	37
Belgium	617	644	595
Cyprus	34	34	30
Czech Rep.	185	182	161
Germany	1713	1788	1653
Denmark	70	73	67
Estonia	44	43	38
Greece	162	169	156
Spain	643	671	621
Finland	163	170	158
France	1373	1433	1324
Hungary	153	152	134
Ireland	153	159	147
Italy	2222	2319	2144
Lithuania	25	25	22
Luxembourg	0	3	0
Latvia	3	3	3
Malta	1	1	1
Netherlands	291	304	281
Poland	2643	2611	2300
Portugal	355	371	343
Romania	126	125	110
Sweden	532	555	513
Slovenia	55	55	48
Slovakia	85	84	74
UK	2001	2089	1931
EU-27	13876	14297	13067

Source; Eurostat waste statistics, 2008 and own elaboration with CEPI (2009) paper recovery data and extrapolations toward 2020 and 2030 according to population growth figures.

After applying the recycling rate to the Eurostat paper card board waste collection figures an estimate of the actual potential could be made. This potential was the extrapolated towards 2020 and 2030 using country specific population growth figures as an extrapolation factor like was done with the other waste potentials based on Eurostat figures. While doing this is was also assumed that the recovery rate increased by 10% between now and 2020. This results in the potentials as presented in Table 4.7.

Particularly large potentials are found in the larger countries especially when this goed together with relatively low recovery rates such as in Italy, Poland, Belgium and Spain.

The contribution to the renewable energy potential of paper card board waste could be significant. Also after applying the recovery rate correction figures it cannot be expected that this resource will be exploited fully as prices are still relatively high for this resource as will be discussed in the next chapter.

5 Total potentials and cost-supply relations of different biomass sources

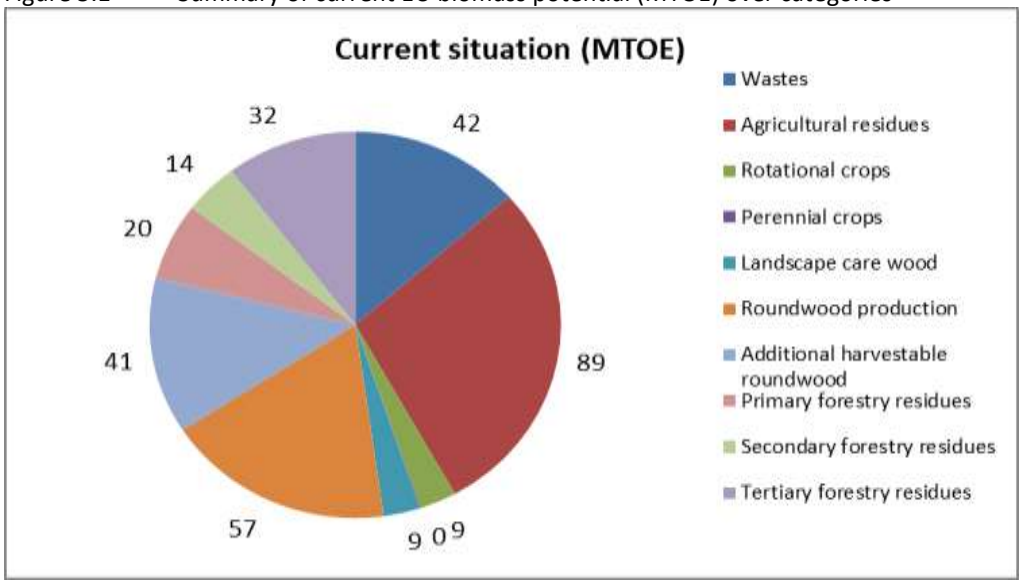
From the former it has become clear that all three sectors can make significant contributions to bioenergy generation especially when by-products and wastes are efficiently used for this purpose. First a summary is given of the presented potentials in terms of relative contribution to the total. After this the potentials are linked to cost levels to get an understanding of their likely use for conversion into bioenergy. These are presented per country and for the total EU.

5.1 Summary of potentials

In the figures underneath a summary is made of the relative contribution every category can make to the total potential, per year and scenario situation and the contribution of every country to the whole EU potential.

From a summary overview of the current potentials in aggregated groups it becomes clear that the largest potential is in the agricultural residues class (Figure 5.1). This class consists of manure, straw and cutting and prunings from permanent crops. The second largest contribution comes from round wood potential, although one can doubt whether this feedstock should really be included as the price of it is far above levels at which bioenergy can compete with competing uses of wood. The third place is covered by the waste group and the additional harvestable round wood potential. The contribution of tertiary forestry residues should not be underestimated as price levels of these potentials are generally more likely to be in the limits of commercial bioenergy production.

Figure 5.1 Summary of current EU biomass potential (MTOE) over categories



When a comparison is made with future potentials (Table 5.1) it becomes clear that potentials are expected to increase significantly, especially towards 2020 in the reference scenario. Between 2020 and 2030 the potentials will stabilise. An important contribution to the growth in potential comes from the expectation that bioenergy cropping will increase significantly both for biofuels on existing agricultural land and on released agricultural land with perennials crops. This growth is however expected to be much larger in the reference scenario then in the sustainability scenario where biofuel cropping on existing agricultural lands is not going to happen and also the perennial cropping potential is more

limited because of constraints on access to land. Towards 2030 the overall cropping potential will be smaller than in 2020 while the agricultural residues potential remains stable. Similar levels of agricultural residues are purely a coincidence as the contribution of the separate residues categories to these totals ranges between 2020 and 2030. In 2020 the contribution by manure is 2 MTOE lower than in 2030 while there is a larger contribution of straw and prunings.

Table 5.1 Potentials (MTOE) per aggregate class compared over period and scenario

MTOE	Current	2020 reference	2020 sustainability	2030 reference	2030 sustainability
Wastes	42	36	36	33	33
Agricultural residues	89	106	106	106	106
Rotational crops	9	17	0	20	0
Perennial crops	0	58	52	49	37
Landscape care wood	9	15	11	12	11
Roundwood production	57	56	56	56	56
Additional harvestable roundwood	41	38	35	39	36
Primary forestry residues	20	41	19	42	19
Secondary forestry residues	14	15	15	17	17
Tertiary forestry residues	32	45	45	38	38
total	314	429	375	411	353

Another reason for growth in biomass potentials is caused by increases in the primary and tertiary forestry residues. This is however only expected in the reference scenarios, while in the sustainability scenarios this will be lower as primary forestry residues are expected to remain at the same level as for the current situation.

Waste potentials are expected to decline towards the future and landscape care wood potential is expected to increase.

In summary it becomes clear that the contribution of the waste sector will further decline to the total potential, the forest sector contribution currently contributing with 52% will also decline to a 47% contribution. The growth in contribution to the overall potential is expected to come from agriculture. Currently it contributes with 31% tot the total potential, but this is expected to increase to above 40% in both reference and sustainability scenario in 2020 and 2030. Within the agricultural group the largest contribution may come from manure, straw and dedicated cropping.

Countries with the largest potential are not only the biggest countries, e.g. Germany, UK, France, Poland, but also the ones with a large forest area, population and/or agricultural sector (see Figure 5.2). However towards the future the contribution of countries to the potential may shift (see Figure 5.3). Overall there will be a decline in the contribution of big countries like Germany and Italy to the EU potential, while increase can be expected in France, Spain, Poland and Romania. Particularly in the sustainability scenario the contribution of Poland could increase significantly. Overall there are however very small changes expected in relative country contribution between the scenarios.

Figure 5.2 Overview of current EU potential per country

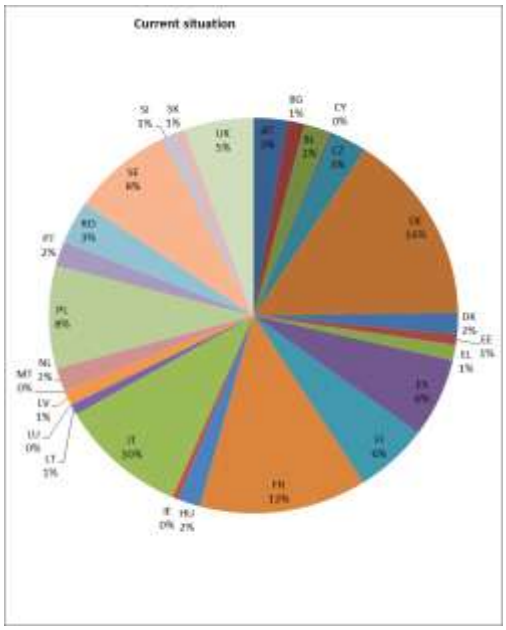
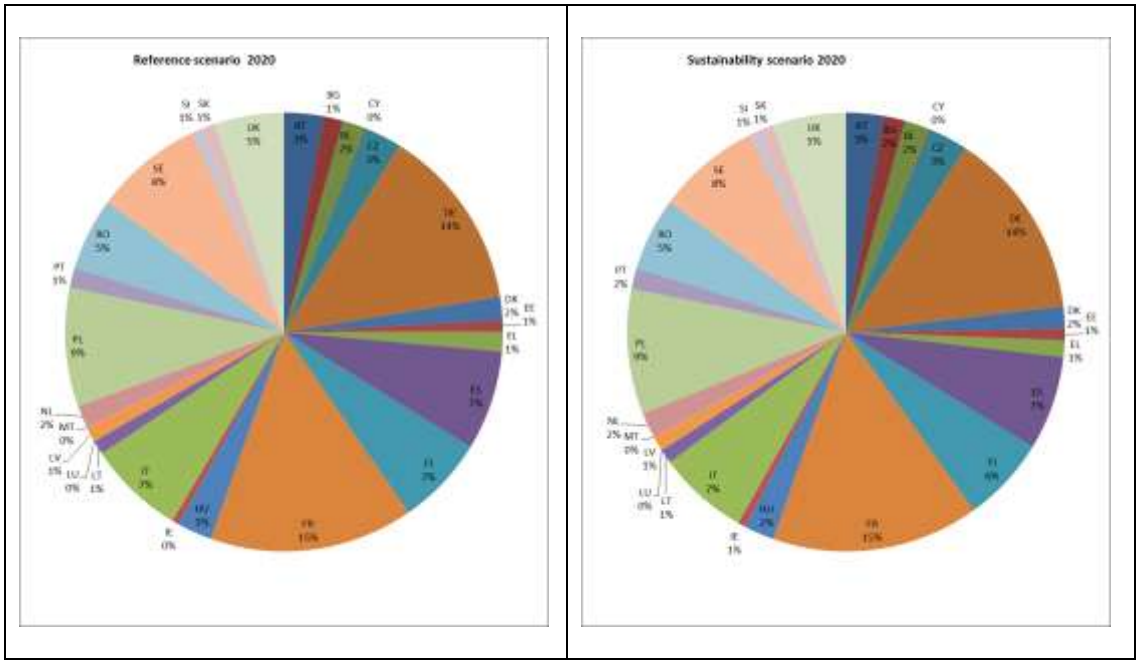
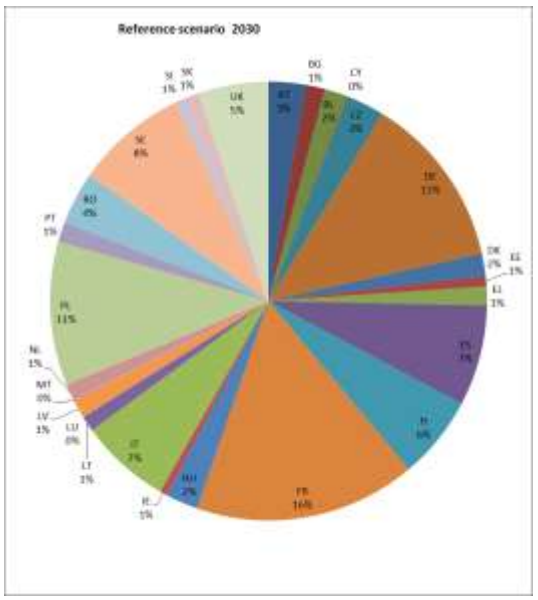
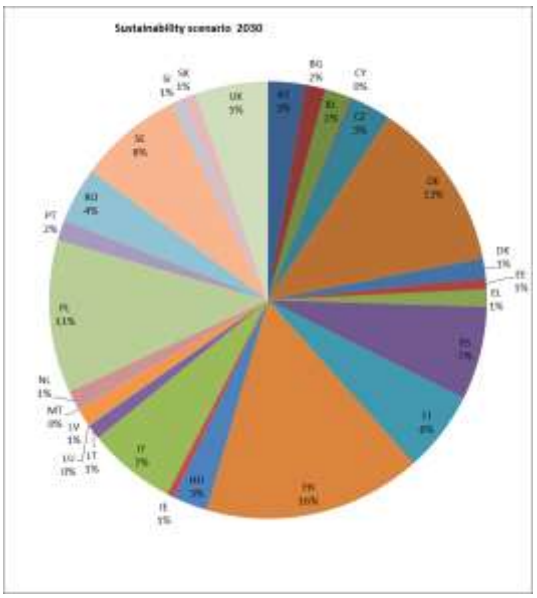


Figure 5.3 Contribution to potential 2020 and 2030 per country







5.2 Cost-supply relations

Now that the current potentials are known it is interesting to combine that with cost levels at which these are expected to be available. After all, the price supply combination will eventually determine what the most interesting potential categories are when they are fed in the next demand modelling steps with RESoLve and PRIMES in other parts of the Biomass Futures project.

A large number of sources were used for the estimation of the and the approach to price levels of the different biomass feedstocktypes. Since price levels range for many feedstocks between countries. For most potentials national price levels were estimated, but for most agricultural potentials even regional estimates were made. Most of the information collected refered to the actual situation. In order to extrapolate to the future price levels were expected to remain stable and only an inflation correction was applied. An exception was made for crops and agricultural residues where price changes towards the future were incorporated from other studies.

An overview of the sources of information used and the approach to determine the price levels of the different biomass feedstock is given in Annex 4 and 6. Since price levels range for many feedstocks between countries this information is also presented at regional level (for perennial crops in Annex 6) and national level in Annex 8 together with the total potential per type of biomass.

The cost supply relations have been determined for the 2020 and 2030 scenarios as these are used as input in the further demand modelling in other parts of the Biomass Futures project. When starting at the EU level (see Figure 5.4 and Table 5.2) it becomes clear that the largest potential of 66% in the reference and 68% in the sustainability scenario for 2020 is available at a price below 200 Euro/Toe. In 2030 this drops to 53% and 51% respectively. This potential consists mostly of potentials from the waste sector, practically all are in, primary residues and some dedicated cropping potential from agriculture and secondary and tertiary residues from the forest sector. In the class between 200 and 400 Euro/TOE the additional potential is still significant in all scenarios. This range mostly consists of primary and

secondary forestry residues, dedicated perennial crops and the rotational crops for biogas and biofuels start to enter.

Table 5.2 Overview of biomass potential (MTOE) per price class for 2020 and 2030 in the reference and sustainability scenarios for EU-27

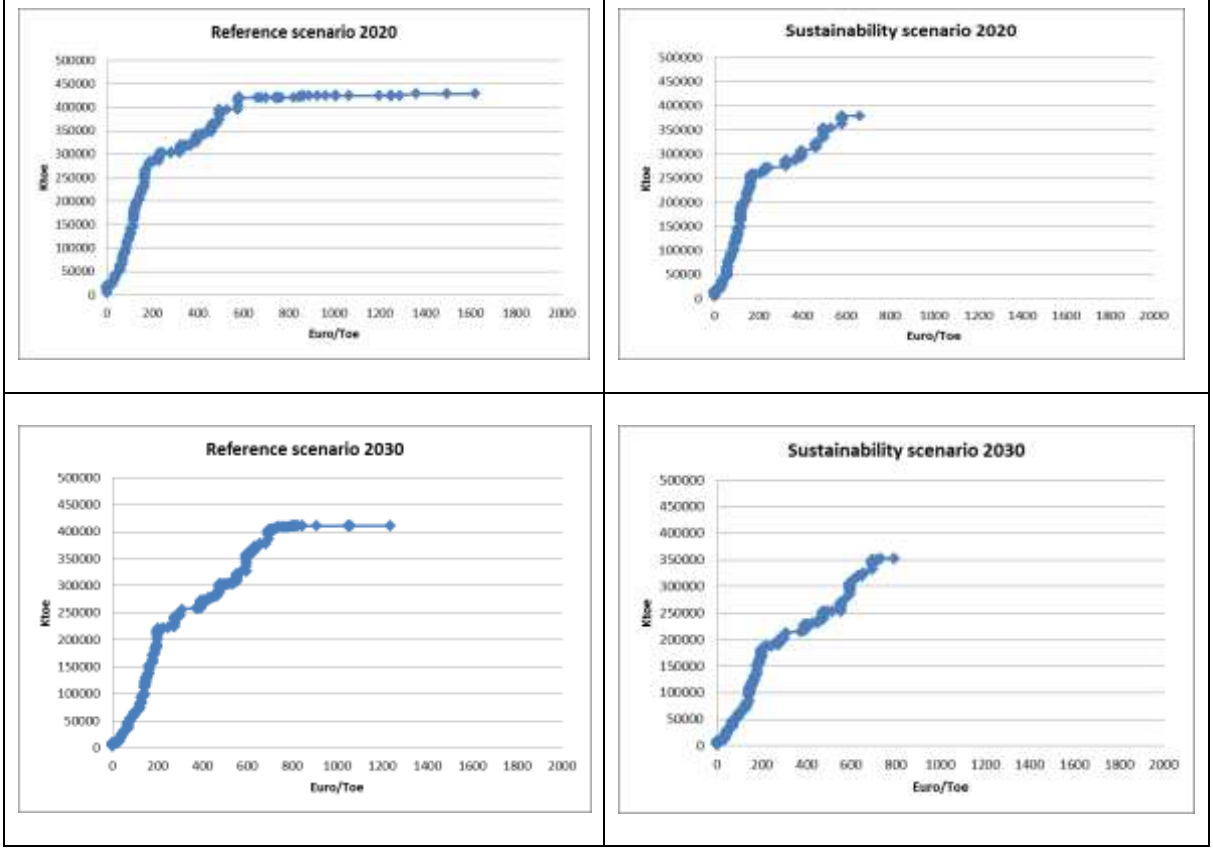
MTOE	2020		2030	
	Reference	Sustainability	Reference	Sustainability
0-199 Euro/Toe	284	259	217	179
200-399 Euro/Toe	58	49	55	49
400-599 Euro/Toe	79	72	87	79
600-999 Euro/Toe	4	0	51	46
>=1000 Euro/Toe	5	0	1	0
Total	**429**	**379**	**411**	**353**

In the range between 400 and 600 Euro/Toe still a significant potential is available. This is where the additional harvestable round wood and the round wood production itself starts to enter at large scale in the 2020 situation. For the reference scenarios this is also the range at which more biofuel crops are included, in the sustainability scenarios these are not produced.

In the range above 600 Euro/Toe practically no additional potential is found in the 2020 sustainability scenario, while in the 2030 sustainability scenario there is still potential in the round wood and manure categories included. Higher prices because of inflation correction explain these differences between the 2020 and 2030 sustainability scenarios. In the reference scenario 2020 the potential in the price above 600 Euro/Toe consists mainly of biofuel crops and some manure of regions where there are some small amounts of manure available. While in the same scenario in 2030 it contains both biofuel crops, round wood potential and manure.

The above description applies to the average EU situation. However between countries large changes occur in price levels and types of feedstock available. This is reflected in the country specific cost-supply curves provided in Annex 8 for the two scenarios in 2020. In Austria for example the biofuel potentials are cheaper than the additionally harvestable roundwood, roundwood and primary residues potentials from forestry, while in most other countries this order is less straight forward as most biofuel crops are the most expensive. In most countries the perennials are somewhere in the middle cost group, but it depends strongly per country whether the woody or the grassy perennials are cheaper. In some countries the forest potential is by far the largest, e.g. in Austria, Finland, Sweden, while in other countries the potential from agriculture is more important. In the large countries like Germany, France and Poland the potentials are more evenly distributed over all sectors.

Figure 5.4 Cost-supply of biomass potentials at EU-27 level for 2020 and 2030 in reference and sustainability scenarios



When interpreting these country results it should also be kept in mind that prices are still a national average, while between regions large difference may also occur. This particularly applies to manure and straw prices whose level is very much determined by local scarcity. In regions where it is a scarce product prices go up. While in regions where there is much excess of these feedstocks the prices are very low. In a country like e.g. Italy the average national price is still relatively high, while huge excess manure production exists in the Po-valley. Especially for large countries like France, Germany, Poland etc., the national totals and averages can provide a very misleading picture. However, for all agricultural potentials regional data are also available in this project.

References

Bauer, A., Leonhartsberger C, Bo¨sch, P., Amon, B., Friedl, A. & Amon T. (2010), Analysis of methane yields from energy crops and agricultural by-products and estimation of energy potential from sustainable crop rotation systems in EU-27. Clean Techn Environ Policy (2010) 12:153–161

Beurskens. M. Hekkenberg. P. Vethman. (2011). Renewable Energy Projections as Published in the National Renewable Energy Action Plans of the European Member. ECN-E--10-069. 28 November 2011

Blanco Fonseca, M. et al. (2010), Impacts of the EU biofuel target on agricultural markets and land use: a comparative modelling assessment. JRC 58484

Brus, D., G.M. Hengeveld, N. Heidema, D. Walvoort, P. Goedhart, G.J. Nabuurs et al. in prep, a European tree species distribution map.

Christian, D. & Riche (A. (1999). Establishing fuel specifications of non-wood biomass crops. ETSU B/U1?00612/00/00. IACR-Rothamsted.

Dworak, T., B.Elbersen, K. van Diepen, I. Staritsky, D. van Kraalingen, I. Suppit , M. Berglund, T. Kaphengst, C. Laaser, & M. Ribeiro (2008), Assessment of inter-linkages between bioenergy development and water availability, Ecologic, Tender report: ENV.D.2/SER/2008/0003r

DBFZ et al, Leipzig, Service- und Begleitvorhaben des Förderprogramms „Energetische Biomassenutzung“, METHODEN zur stoffstromorientierten Beurteilung für Vorhaben im Rahmen des BMU-Förderprogramms „Energetische Biomassenutzung“, TEIL I: TECHNOLOGIEKENNWERTE, GESTEHUNGSKOSTEN, TREIBHAUSGASBILANZEN, Stand: 17.09.2010. 2)

Deurwaarder, E.P., S.M. Lensink, H.M. Londo (2007). BioTrans biofuels data. Annex to ‘Use of BioTrans in Refuel’; functional and technical description. Refuel deliverable D10b. IEE programme.

EEA, 2005. The European Environment — State and Outlook 2005. Copenhagen.

EEA, 2006. How much bioenergy can Europe produce without harming the environment? EEA Report No 7/2006. Copenhagen.

Ekbom, T., Berglin, N., Lodgeberg , S (2005). High efficient motorfuel production from biomass via black liquor gasification. Presented at ISAF XV international symposium on alcohol fuels. San Diego. September 2005.

Elbersen et al. (2012). Review of the EU bioenergy potential from a resource efficiency perspective. An update of EEA report No 7/2006. ETC-SIA technical report (forthcoming).

Eurostat (2004). Definition and explanation of relevant EWCSTat categories. Annex to the manual on waste statistics.

Eppler, U., Petersen, J.-E & C. Couturier, 2007. Short Rotation Coppice and Perennial Energy Grasses in the European Union: Agro-environmental aspects, present use and perspectives. Background paper for a joint expert workshop of JRC/EEA/Rothamsted Research in Rothamsted, October 2007. http://re.jrc. ec.europa.eu/biof/.

ESRI (2008). Includes the roads (European Highway System, national, and secondary roads) in Europe. Largest scale when displaying the data: 1:10,000.

Gewiese A. et al., 1988. Abfallvermeidung — Ein Modellversuch in Hamburg-Harburg im Jahre 1987. Berlin: INECTUS 1988 quoted in Abfallwirtschaft — Handbuch fur Praxis und Lehre, by: Bernd Bilitewski, Georg Hardtle, Klaus Marek.

JRC-IE. (2010) Indirect Land Use Change from increased biofuels demand: Comparison of models and results for marginal biofuels production from different feedstocks, European Commission, Joint Research Centre, Institute for Energy, Ispra, Italy.

JRC-IPTS (2010), Agro-economic Modelling Platform, (AGRITRADE action), Biofuel Modelling (AGLINK, ESIM, CAPRI), Final report. Sevilla, 2010.

Kuipers, N., Linders, M. & Zeevalking, J. (2010), Vraag en aanbod van biomassa in de Energy Valey Regio. TNO 2010.

Lensink, S., Wassenaar, J., Mozaffarian, M., Pfeiffer, A., Luxembourg, S. & Cleijne, J. (2010. Conceptadvies basisbedragen 2011 voor electriciteit en groen gas in het kader van de SDE regeling. ECN-E-10-053.

Lindner, M., Meyer, J., Eggers, T., Moiseyev, A. (2005), Environmentally compatible bioenergy potential from European forests. A report commissioned by the European Environment Agency through the European Topic Centre on Biodiversity, Paris. European Forest Institute. Joensuu, Finland.

Mantau, U. et al. 2010: EUwood - Real potential for changes in growth and use of EU forests. Final report. Hamburg/Germany, June 2010. 160 p.

Mantau, U. et al. 2010: EUwood - Real potential for changes in growth and use of EU forests. Methodology report. Hamburg/Germany, June 2010. 165 p.

Montia,A., Fazioa S., Lychnarasb V., Soldatosb P.,Venturia G. (2007). A full economic analysis of switchgrass under different scenarios in Italy estimated by BEE model. Biomass and Bioenergy 31 (2007) 177–185.

Monti A., Bezzi G., Pritoni G.,Venturi G. (2008), Long-term productivity of lowland and upland switchgrass cytotypes as affected by cutting frequency. Bioresource Technology 99 (2008) 7425–7432

Noord, De M., Beurskens, L., Vries, De H. (2004). Potentials and costs for renewable Electricity generation. A data overview. ECN-C-03-006.

OECD (2006), SCOPE Biofuel Report.

Pointereau, P., F. Coulon, P.Girard, M.Lambotte, T. Stuczynski, V.Sanchez Ortega, A. Del Rio; Editors : E.Anguiano, C.Bamps, J-M.Terres (2008): Analysis of Farmland Abandonment and the Extent and Location of Agricultural Areas that are Actually Abandoned or are in Risk to be Abandoned

Rettenmaier, N., Reinhardt, G., Schorb, A., Köppen, S., Von Falkenstein, E. (2008) and other BEE partners. Status of Biomass Resource Assessments, BEE project deliverable 3.6.

Scarlat, N., Martinov, M., Dallemand J.F. (2010), Assessment of the availability of agricultural crop residues in the European Union: Potential and limitations for bioenergy use. Waste Management 30 (2010) 1889–1897.

Schelhaas, M.J., J. Eggers, M. Lindner, G.J. Nabuurs, A. Pussinen, R. Päivinen, A. Schuck, P.J. Verkerk, D.C. van der Werf, S. Zudin, 2007. Model documentation for the European Forest Information Scenario model (EFISCEN 3.1). Wageningen, Alterra, Alterra report 1559, EFI Technical Report 26, Joensuu, Finland. 118 p.

Siemons, R., Mc Chesney, I., Nikolaou, N., Vis, M., Berg, van den D. & Whitelye M. (2004). Bioenergy's role in the EU energy market. A view of developments until 2020.

Skovgaard, M., Moll, S., Møller Andersen, F., Larsen. H., 2005. Outlook for waste and material flows: Baseline and alternative scenarios, ETC/RWM working paper 2005/1, EEA.

Smit H.J., M.J. Metzger, F. Ewert (2008), Spatial distribution of grassland productivity and land use in Europe. Agricultural Systems 98 (2008) 208–219

Vis et al. (2010). Harmonisation of biomass resource assessment. Volume 1. Best practices and methods handbook. Bee project deliverable 5.3.

Wit De, M. & Faaij A. (2008). Biomass resources potential and related costs. Assessment of the EU 27, Switserland, Norway and Ukraine. REFUEL work package 3 final report. Copernicus Institute – Utrecht University, The Netherlands.

Annex 1 Dedicated cropping 2008 main data sources used

In 2007 the European Commission tendered a study with the specific objective to analyse the different water needs and distribution of bioenergy crops grown or potentially grown in the next decades in the EU. The resulting report (Dworak et al., 2008) contains an overview of the dedicated bioenergy cropping area which has been used for this study and which has been updated with additional (more recent) sources from AEBIOM (2009). The reference year for the data ranges between 2006 and 2008. The main sources used per country are listed below:

Austria

Bioenergy production in 2006 (Brainbows Informationsmanagement GmbH (2007) and Raab (2007)):

- SRC (Miscanthus und others): some 100 ha
- Cereals for heating: more than 1,500 ha
- Biogas (Silage Maize and fodder: around 40,000 ha
- Bioethanol: no production ha
- Rape seed (biodiesel): about 15,000 ha

Belgium (only Flanders)

Information was received from Linda Meiresonne working for the Linda Research Institute for Nature and Forest. The underneath figures were derived from the Ministry of Agriculture. Arable crops: inventory based on applications for energy subsidy (45 €/ha) or set aside subsidy.

Energy – Situation 2007:

- Rapeseed: 507 ha
- Wheat: 200 ha
- Mais: 521 ha

Energy – Situation 2008:

- Rapeseed: 116 ha
- Mais: 508 ha

Set aside – Situation 2007:

- Rapeseed: 452 ha
- Wheat: 1,164 ha
- Mais: 139 ha
- Tricale: 2 ha

The Flemish region had 622,133 ha of agricultural land in 2007 (normal arable land and set-aside). So 0.45% of the agricultural area was occupied with targeted energy crops.

Bulgaria

A rough indication on oil cropping area for biodiesel purposes were derived from a European Biodiesel Board (EEB) report.

In this report it is stated biodiesel production first started in Bulgaria as early as 2001, and was mainly based on used cooking oils collected from restaurants, as developed by the company SAMPO in Brussartzi (North-Western Bulgaria). However, there has been a rapid increase in production of sunflower and Rapeseed-based biodiesel. Today indeed, the energy crops used as raw material for biodiesel are mainly Rapeseed and sunflower, although it should be noted that some climatic restrictions exist for Rapeseed cultivation' (Garofalo, 2007).

Based on this statement the present area of Rape and oil seeds was taken from the FSS 2007 and then it was assumed that 1/3 of the production coming from this area was used for biodiesel production.

This leads to the following cropping area:

- Oil seed Rape: 335 ha
- Sunflower: 257,759 ha
- Total: 258,094 ha

Cyprus

The hectares in agriculture used for bioenergy cropping in Cyprus is zero. In general the main reasons for not having such a RES in Cyprus is a) the requirements in high level technological knowledge (planning of installation, treatment of raw material). b) Lack of previous experience, c) Increased water requirement of energy crops in relation to the water stressed agriculture (Personal communication Ayis I. Iacovides).

Denmark:

Information on the cropping area was derived the Danish Ministry of Food, Agriculture and Fisheries, which specifies a total area of 95,000 hectares of oil seed Rape. Leppiman (2005) also specifies that in Denmark biomass (mainly straw, wood and manure) accounts for nearly 10 % of the total energy production.

Estonia

Today energy crops (mainly Rapeseed) are grown within an area that does not exceed 50 thousand hectares. The harvest is about 70 – 80 thousand tonnes, which is not sufficient to produce biodiesel. Cereal production (approximately 600-760 thousand tonnes) does not currently cover domestic demand for fodder, foodstuff, seed and industrial needs. Therefore additional cereal is being imported to cover demand (not for conversion into ethanol)(Barz and Ahlhaus, 2005).

France

Until 2005 bioethanol in France was produced primarily from sugarbeet and secondarily from wheat: most bioethanol production is likely to be derived from wheat in 2008, at the expense of sugarbeet. According to the French Ministry of Agriculture, 300,000 hectares of wheat, 50,000 hectares of corn and 50,000 hectares of sugar beet are expected to produce bioethanol by 2008. For wheat and corn, this will represent less than 5% of the total grain acreage (Hénard and Audran, 2007).

France: Situation 2007/2008:

- RAPE: 872,352 ha
- Sunflower: 80,000 ha
- Corn maize: 50,000 ha
- Starch (cereals): 300,000 ha
- Sugerbeet: 50,000 ha
- Total: 1,352,352 ha

Germany

There is significant increase in biomass cultivation for bioenergy purpose in Germany. The biggest production is focused on biodiesel. The oil seed crop cover already over 1,100,000 hectares, which is almost 10% of the arable land (Figure 3). Germany as a large central European country has 11.8 mill. hectares of arable land. Future biomass potentials in Germany for energy crops are stipulated to be even up to 2 mill. hectares or 17% of the arable land on medium to long terms.

Rapid growth in interest in biogas has been noticed recently in Germany. Between 2004 and 2005 the area dedicated for biogas energy crops increased over six times. Around 80% of the applied crops is maize, harvested for maize silage. Further growth is expected. In 2007 Germany had the highest number

of biogas plants in Europe (around 3000). Biogas is produced from manure, industrial organic waste but especially from cultivated energy crops. Energy crops state for over 46% of the substrates. Share of animal manure is around 24% of feedstock applied for biogas in Germany. The biogas potential in Germany was calculated as 24 bill. m^3 biogas per year. The amount will increase rapidly and boost the number of biogas plants.

Figure 1: Cultivation of non-food crops in Germany in 2006

Table VIII: Cultivation of non-food crops in Germany in 2006 [28]

Raw materials	Surface area in ha			Total
	Base areas*		Set aside	
	without energy crop premium	with energy crop premium		
Rapeseed	610,000	172,000	318,000	1,100,000
Oilseed lin	3,000			3,000
Sunflower	4,000		1,000	5,000
Other energy crops(incl.maize)	30,000	188,000	77,000	295,000
Starch	128,000			128,000
Sugar	18,000			18,000
Fibres	2,000			2,000
Pharmaceutical crops	10,000			10,000
Total	805,000	360,000	396,000	1,561,000

*estimate, as no detailed data and trade statistics available

Source: http://websrv5.sdu.dk/bio/JHN_paper_07.pdf

Energy Maize production Germany 2008-2009 :

Maisanbaufläche Deutschland in ha, 2008 und 2009 (vorläufig)
nach Bundesländern und Nutzungsrichtung in ha

Bundesland	Körnermais inkl. CCM		Silomais		Futter- oder Energiemais*		Energiemais		Anbaufläche gesamt	
	2008	2009	2008	2009	2008	2009	2008	2009	2008	2009
Baden-Württemberg	71.625	67.325	63.236	69.989	1.464	0	30.076	25.707	166.401	163.021
Bayern	95.821	64.822	322.767	319.860	5.737	10.780	49.474	53.029	473.799	448.490
Brandenburg	9.279	9.109	142.892	145.055	0	0	4.327	4.839	156.498	159.003
Hessen	4.413	2.180	26.476	27.481	1.503	1.181	4.609	4.655	37.001	35.497
Mecklenburg-Vorpommern	7.404	1.033	91.980	66.463	413	10.228	16.702	21.867	116.499	99.591
Niedersachsen	110.136	102.575	243.977	242.651	82.962	93.432	37.724	50.672	474.799	489.329
Nordrhein-Westfalen	111.823	111.326	126.325	121.201	4.731	9.347	16.521	16.374	259.400	258.249
Rheinland-Pfalz	741	741	19.995	21.264	2.645	2.645	11.320	11.320	34.700	35.970
Saarland	.	.	.	.	.	.	.	.	.	.
Sachsen	39.203	40.585	40.563	38.722	0	0			83.200	82.741
Sachsen-Anhalt	21.817	22.752	69.828	65.599	0	0	9.355	15.941	101.000	104.293
Schleswig-Holstein	0	0	106.753	118.676	4.323	3.603	24.346	34.742	131.099	153.418
Thüringen	120	120	44.456	47.074	3.434	3.434	600	985	49.500	51.782
Bundesgebiet	472.382	422.568	1.299.249	1.284.035	107.212	134.648	205.053	240.132	2.083.896	2.081.382

Quelle: DMK, Kleffmann, Stand Mai 2009
. = kein Wert vorhanden
* = Verwendung als Energie- oder Futtermais noch offen, großenteils Energiemais

DMK

Greece

The Hellenic Ministry of Rural Development and Food has outlined that during 2007 (Panoutsou, 2008):

- Approximately 73,000 tonnes of indigenous oil seeds (mainly comprising of 69,000 tonnes cotton seeds) would be used for biodiesel production,
- In addition, 11,200 hectares of agricultural land would be cultivated with energy crops, under contractual schemes, for biodiesel production.
- Hellenic Sugar Industry announced in 2006 that two sugar mills in north (Xanthi) and central (Larisa) Greece will be converted to bioethanol plants. This fact is expected to provide robust incentives for energy farming, since the annual resource requirements of the two plants are expected to be in the range of 600,000 tonnes of sugar beets and 600,000 tonnes of cereals (since these were estimates and no confirmation was found for the plants already being in production these areas were not taken into account in this study).

Situation 2004:

- Maize crop: 10,628 ha
- Total crop cultivation for biogas: 13,603 ha

Situation 2005:

- Maize crop: 66,988 ha

- Total crop cultivation for biogas: 86,912 ha

Hungary

In Hungary on 18,500 hectares energy crops were grown in 2008 (Doran, 2008).

Ireland

At present, biomass provides over half of Ireland's renewable energy - mainly through wood used for heating in the domestic and wood processing industry sectors (Bruton and McDermott, 2006).

Italy

Biodiesel in Italy is mainly produced from Rapeseed oil (about 70% of the total) and soybean oil (20%), with the remainder coming from both sun and palm oils. Rapeseed oil is imported from other EU countries, while soybean oil is either imported from the EU or domestically produced from imported beans (oil from domestic beans, being GM free, is used for food consumption). According to industry sources, this year (2007) some 65,000 hectares have been or will be planted to oilseeds (50,000 hectares to sunflower seeds and 15,000 hectares to Rapeseeds) under cultivation contracts between growers and the processing industry for the production of biodiesel. In 2006 bioethanol production rose to 1,280,000 hectoliters, obtained from alcohol produced from both the distillation of wine surpluses and molasses (Perini, 2007).

Poland

With plantations of about 2,000 hectares (2006) willows are mostly used as energy crop. Secondly, straw is becoming more popular for energy use, but it is currently only marginal in relation to overall production. Poland has set a target for expanding the area used for energy crops up to 160-200 thousand hectares in 2010 representing 1.2 – 1.4% of whole arable land in Poland. It may be an alternative sources of income for farmers. Now cultivation area of energetic willow is only 5.4 thousand hectares (Wesolowski, 2005).

Portugal

9,000 hectares area under energy crops in 2008 (Doran, 2008)

Romania

Romania has a significant potential for production of bioethanol from sweet sorghum and biodiesel from Rape oil and sunflower oil. It also has very good prospects as a net exporter within the EU. In Romania, in 2004, almost all of 100,000 tonnes of Rapeseed, 70,000 tonnes of sunflower and 408,000 tonnes of sunflower seeds were exported possibly for bioenergy production (Kondilia and Kaldellis, 2007).

UK

Final data used were derived from www.nnfcc.co.uk (National non-food crops website). The data on this website specify the following (in hectares):

England:

- SRC-willow: 3,083 ha

- SRC-poplar: 5 ha
- Miscanthus: 5,772 ha

Wales:

- SRC-willow: 7 ha

Scotland:

- SRC-willow: 289 ha

N-Ireland:

- SRC-willow: 289 ha

UK (region unknown):

- SRC-willow: 2,486 ha
- Miscanthus: 1,960 ha

Total:

- RAPE: 320,542 ha
- Wheat; 14,614 ha
- Barley: 1,303 ha
- SRC-willow: 5,865 ha
- SRC-poplar: 5 ha
- Miscanthus: 7,732 ha

In addition other information was also provided on: http://www.rcep.org.uk/biomass/chapter2.pdf

It specified that willow (Salix spp.) has already been used in commercial or near commercial operations in the UK. Investment in developing new varieties with increased yield stability and improved crop management has made willow increasingly competitive as an energy source. Willow chips are a reliable source of fuel of a consistent quality, suitable for firing in CHP and district heating plants. Willow has been grown extensively in Scandinavia for fuel, and in Sweden some 15,000 hectares of land are dedicated to its production for renewable energy. Consequently, much more information about cultivation, harvesting and yields is available for willow than for the other potential energy crops.

The grass miscanthus (Miscanthus spp.) is attracting an increasing amount of interest but it is still largely at trial stage in the UK. Among other potential candidate species, poplar (Populus spp.) is closest to providing an alternative source of fuel. Poplar is being trialled in short rotation coppice (SRC) plantations, as well as being tried in silvoarable agro-forestry where it is intercropped with arable species.

There are currently 1,795 hectares of land under cultivation of commercial willow SRC and miscanthus in the UK; at least 1,500 hectares of this is willow. The land dedicated to energy crops totals less than 0.01% of the total arable land in the UK. The Defra Non- Food Crops Strategy states that domestically grown crops should meet a significant part of the demand for energy and raw materials in the UK. The National Farmers' Union suggests that up to 20% of crops grown in the UK could be made available for non-food uses (i.e. for fuels or industrial materials), by 2020; hence, there is scope for a significant expansion of energy crop production in the UK. Planning crops in order to achieve the maximum environmental benefits and yields in areas close to demand is the challenge to be met by the farmers and energy generating companies.

http://www.defra.gov.uk/farm/crops/industrial/research/reports/biofuels_prospects.pdf

In 2001, over 23,000 hectares of oilseed Rape was grown on UK farms for biodiesel production, though virtually all was processed in mainland Europe on an .equivalence trade basis.. Until recently UK biodiesel production was limited to 200 tonnes. The reduction in duty from April 2002 is likely to increase this significantly. However, currently no crops are registered for bioethanol production on set-aside and no bioethanol is currently being produced.

BIOMASS FUTURES

Annex 2 Upstream and downstream GHG emissions

data from GEMIS 4.8 (Dec. 2011), background energy system: EU 27 (PRIMES baseline), year:						
data use direct land use GHG emissions from feedstock cultivation as based on Miterra						
Feedstock	Pre-treatment	conversion			Net GHG emissions 2020	Net GHG emissions 2030
heat output:			comment 1	comment 2	CO_2eq g/MJ_{out}	CO_2eq g/MJ_{out}
primary forest residues	Pelletisation	Residential pellet boilers(small)	no upstream according to RED	10 kWth	9.6	8.6
Other industrial wood residues	Pelletisation	Residential pellet boilers(small)	no upstream according to RED	10 kWth	6.9	6.1
Woody Perennials	Pelletisation	Residential pellet boilers(small)	assumed poplar SRC	10 kWth	38.3	37.3
primary forest residues	Pelletisation	Residential pellet boilers(medium)	no upstream according to RED	50 kWth	9.2	8.2
Other industrial wood residues	Pelletisation	Residential pellet boilers(medium)	no upstream according to RED	50 kWth	6.6	5.8
Woody Perennials	Pelletisation	Residential pellet boilers(medium)	assumed poplar SRC	50 kWth	37.3	36.2
primary forest residues	Pelletisation	Residential pellet boilers(large))	no upstream according to RED	0.5 MWth	8.6	7.7
Other industrial wood residues	Pelletisation	Residential pellet boilers(large))	no upstream according to RED	0.5 MWth	6.1	5.3
Woody Perennials	Pelletisation	Residential pellet boilers(large))	assumed poplar SRC	0.5 MWth	36.4	35.4
Primary forest residue	chipping	local heating plant-small scale (1MW	no upstream according to RED	excludes heat distribution	4.5	4.1
Perennials (woody)	chipping	local heating plant-small scale (1MW	assumed poplar SRC	excludes heat distribution	33.3	32.8
Primary forest residue	chipping	local heating plant-large scale (5MW	no upstream according to RED	excludes heat distribution	4.3	3.9
Perennials (woody)	chipping	local heating plant-large scale (5MW	assumed poplar SRC	excludes heat distribution	32.4	31.9
black liquor		liquid combustion-heat	no upstream according to RED	onsite combustion for process heat assumed, with FGD	0.7	0.7
used fat/oil	used fat/oil	liquid combustion(heat only)	no upstream according to RED	boiler with 1 MWth assumed	0.9	0.9
electricity/CHP output, allocation to electricity only						
sawdust + sawmill by-products	pelletisation	direct co-firing (coal process	no upstream according to RED	coal PP 800 MWel, 15% cofiring, only biomass share shown	12.9	11.9
sawdust	pelletisation	co-firing in a coal fired CHP plant	no upstream according to RED	coal SE 100 MWel, 15% cofiring, only biomass share shown	36.9	36.0
sawdust	pelletisation	CHP plant	no upstream according to RED	20 MWel cogen steam-turbine backpressure plant	12.2	11.2
straw	TOP straw	direct co-firing	no upstream according to RED	coal PP 800 MWel, 10% cofiring, only biomass share shown	36.9	33.0
straw	TOP straw	o-firin in a coal fired CHP	no upstream according to RED	coal SE 100 MWel, 10% cofiring, only biomass share shown	51.7	48.0
straw	TOP straw	CHP plant	no upstream according to RED	20 MWel cogen steam-turbine backpressure plant	31.5	27.9
Primary forest residue	chipping	direct co-firing coal process	no upstream according to RED	coal PP 800 MWel, 15% cofiring, only biomass share shown	9.0	8.6
Perennials (woody)	chipping	direct co-firing coal process	assumed willow SRC	coal PP 800 MWel, 15% cofiring, only biomass share shown	62.1	61.6
Perennials (woody)	chipping	direct co-firing coal process	assumed miscanthus	coal PP 800 MWel, 15% cofiring, only biomass share shown		49.8
Perennials (woody)	chipping	direct co-firing coal process	assumed RCG	coal PP 800 MWel, 15% cofiring, only biomass share shown		91.3
Perennials (woody)	chipping	direct co-firing coal process	assumed Switchgrass	coal PP 800 MWel, 15% cofiring, only biomass share shown		53.2
Primary forest residue	chipping	co-firing in a coal fired CHP plant	no upstream according to RED	coal SE 100 MWel, 10% cofiring, only biomass share shown	34.0	33.6
Perennials (woody)	chipping	co-firing in a coal fired CHP plant	assumed willow SRC	coal SE 100 MWel, 10% cofiring, only biomass share shown	83.3	82.8
Perennials (woody)	chipping	co-firing in a coal fired CHP plant	assumed miscanthus	coal SE 100 MWel, 10% cofiring, only biomass share shown		64.8
Perennials (woody)	chipping	co-firing in a coal fired CHP plant	assumed RCG	coal SE 100 MWel, 10% cofiring, only biomass share shown		102.6
Perennials (woody)	chipping	co-firing in a coal fired CHP plant	assumed Switchgrass	coal SE 100 MWel, 10% cofiring, only biomass share shown		69.8
Primary forest residue	chipping	CHP plant	no upstream according to RED	20 MWel cogen steam-turbine backpressure plant	8.3	6.8
Perennials (woody)	chipping	CHP plant	assumed willow SRC	20 MWel cogen steam-turbine backpressure plant	60.1	59.7
Perennials (woody)	chipping	CHP plant	assumed miscanthus	20 MWel cogen steam-turbine backpressure plant		46.7
Perennials (woody)	chipping	CHP plant	assumed RCG	20 MWel cogen steam-turbine backpressure plant		74.0
Perennials (woody)	chipping	CHP plant	assumed Switchgrass	20 MWel cogen steam-turbine backpressure plant		50.3
Rapeseed	oil extraction	liquid combustion(electricity only)	SVO assumed	ICE dieselmotor 0.5 MWel	85.9	79.0
soya import from XX	oil extraction	liquid combustion(electricity only)	SVO from AR assumed	ICE dieselmotor 0.5 MWel	35.3	34.9
Sunflower	oil extraction	liquid combustion(electricity only)	SVO assumed	ICE dieselmotor 0.5 MWel	94.5	88.9
Rapeseed	oil extraction	CHP-liquid	SVO assumed	ICE cogen with dieselmotor 0.5 MWel	47.7	43.9
soya import from XX	oil extraction	CHP-liquid	SVO from AR assumed	ICE cogen with dieselmotor 0.5 MWel	19.6	19.4
Sunflower	oil extraction	CHP-liquid	SVO assumed	ICE cogen with dieselmotor 0.5 MWel	52.5	49.4
MSW (not landfill, composting)		Combustion(electricity only)	50% biogenic share	incineration assumed	276.8	276.8
MSW (not landfill, composting)		CHP	50% biogenic share	incineration assumed	153.8	153.8
verge grass	biogas	CHP	grass cuttings assumed	ICE 500 kWel lean enginge	15.9	15.3
animal waste	biogas	CHP	liquid manure assumed	ICE 500 kWel lean enginge	5.6	5.1
dry manure	biogas	CHP	no upstream according to RED	ICE 500 kWel lean enginge	5.8	5.2
liquid biofuel output, with allocation for byproducts based on RED						
straw	TOP straw	FT production	no upstream according to RED	BtL plant, synthetic diesel	31.6	27.8
grassy perennials	TOP	cellulosic ethaonl	assumed switchgrass		10.5	10.2
grassy perennials	TOP	FT production	assumed switchgrass	BtL plant, synthetic diesel	41.4	36.9
primary forest residues	TOP	FT production	no upstream according to RED	BtL plant, synthetic diesel	35.2	30.9
woody perennials	TOP	FT production	assumed willow SRC	BtL plant, synthetic diesel	91.4	86.9
used fat/oil		transestrification of used fats/oils	no upstream according to RED		6.8	6.6
rapeseedoil		transesterification			38.3	35.2
palm oil		transesterification	Import from ID assumed		26.1	23.8
soybean oil		transesterification	Import from AR assumed		18.2	17.5
sunflower oil		transesterification			41.7	39.0
Cereals		ETOH	wheat, DDGS as feed		46.5	41.1
maize corn		ETOH	maize, DDGS as feed		42.2	38.8
sugar beet		ETOH	beet chips as feed		44.1	41.8
upgraded gaseous fuels, no transport/use included						
primary forest residues	Pelletisation	LT-gasification (SNG production)	no upstream according to RED		10.1	9.0
verge grass	biogas	upgarding of biogass	grass cuttings assumed	output > 2 MWth, BACT	12.2	11.4
animal waste	biogas	upgrading of biogas	no upstream according to RED	output > 2 MWth, BACT	6.2	5.5
dry manure	biogas	upgrading of biogas	no upstream according to RED	output > 2 MWth, BACT	6.3	5.6
Primary forest residue	chipping	LT gasification (SNG production)	no upstream according to RED			3.3
Perennials (woody)	chipping	LT gasification (SNG production)	assumed willow SRC			41.7
perennail (grassy)	torrefaction	LT gasification (SNG production)	assumed switchgrass			22.6
other industrial wood	chipping	LT gasification (SNG production)	no upstream according to RED	same as primary forest residues		3.3

Annex 3 Estimation of iLUC factor

In the EEA/ETC-SIA-ACC study (Elbersen et al. , 2012, forthcoming) an inventory was made of the following studies:

1. IFPRI study: 'Global trade and environmental impact study of the EU biofuels mandate' (Al-Riffai et al., 2010).
2. ADEME study: 'Analyses de Cycle de Vie appliquées aux biocarburants de première génération consommés en France' (ADEME, 2010).
3. E4tech study: 'A causal descriptive approach to modelling the GHG emissions associated with the indirect land use impacts of biofuels' (E4tech, 2010).
4. PBL study: Identifying the indirect effects of bio-energy production (PBL, 2010a) and
    a. 'The contribution of by-products to the sustainability of biofuels' (PBL, 2010b).
    b. 'Indirect land-use change emissions related to biofuel consumption in the EU based on historical data' (Overmars et al., submitted).
5. CARB study: Proposed Regulation to Implement the Low Carbon Fuel Standard Volume I. Staff Report: Initial Statement of Reasons (CARB, 2009a; CARB, 2009b).
6. JRC study: Indirect Land Use Change from increased biofuels demand (JRC-IE, 2010).
7. Oeko-Institut: The 'iLUC Factor' as a means to hedge risks of GHG emissions from indirect land use change (OEKO, 2010).

The main characteristics of these studies are summarized in Table 1.

Table 1: Main model features in relation to biofuel modelling

Study	Model/method	Includes by-products	intensification	conversion emissions data
CARB	GTAP: General equilibrium model	yes	yes	Woods Hole[20]
E4tech	Causal-descriptive approach	yes	yes	Winrock[21]
ADEME	Life cycle assessment (LCA), with sensitivity analysis for ILUC	yes	yes	Guide for biofuels LCAt 2008, IPCC
IFPRI	MIRAGE: General equilibrium model	yes	yes	IPCC
PBL	Historic analysis of FAO	yes	yes	IPCC

[20] As used in Searchinger (2008). Figures in http://www.arb.ca.gov/fuels/lcfs/ef_tables.xls

[21] US Environmental Protection Agency (2010a) "Renewable Fuel Standard Program (RFS2) – Regulatory Impact Analysis"

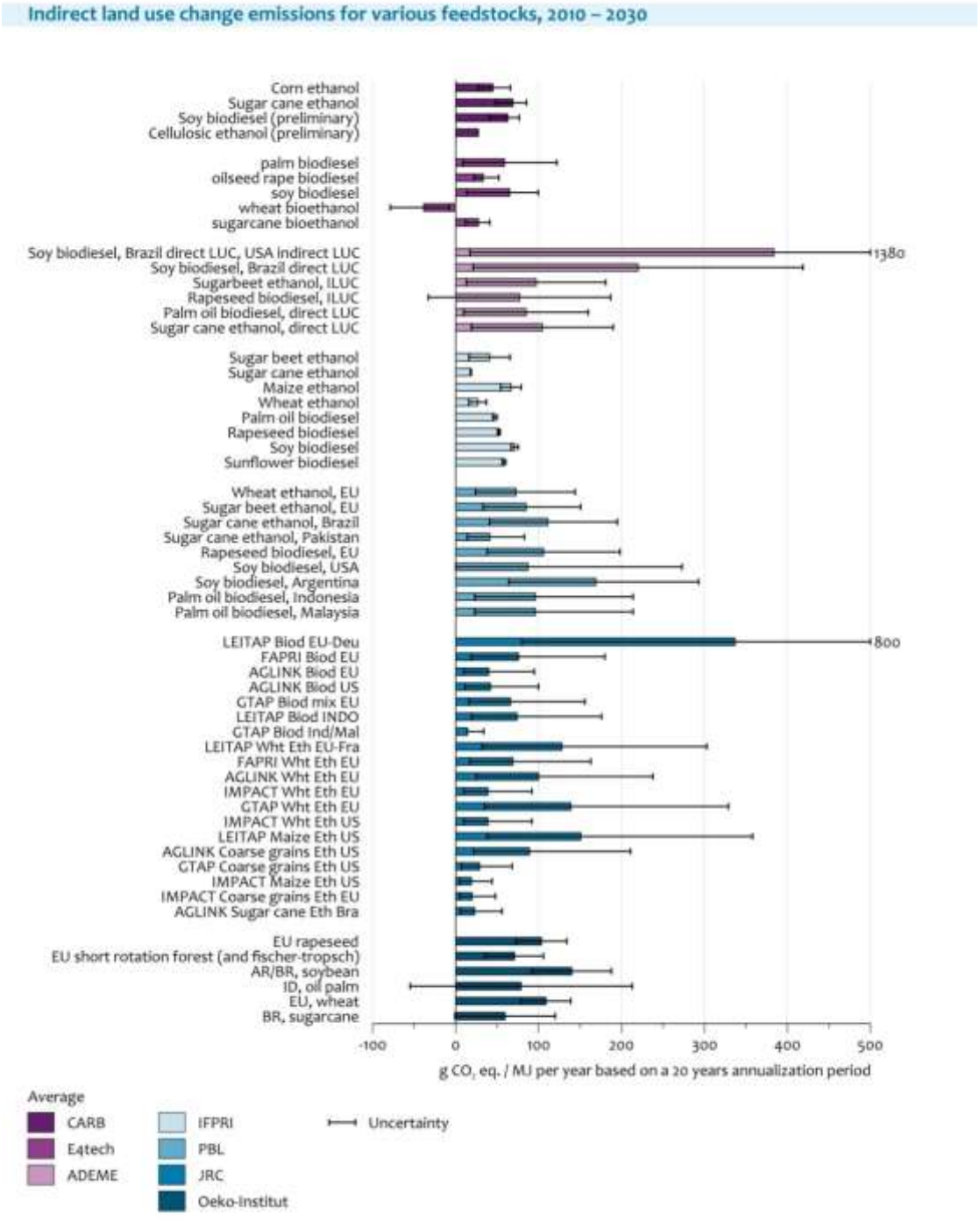
	data			
JRC	LEITAP: General equilibrium model	yes	yes	40 tC/ha for soil C emissions was used (Based on IPCC). The error bars represent the maximum range using 95 tC/ha (Searchinger et al, 2008), and the minimum derived from an emission factor of 10 tC/ha[22]
	FAPRI: Partial equilibrium model	yes	yes	
	AGLINK: Partial equilibrium model	yes	yes	
	GTAP: General equilibrium model	yes	yes	
	IMPACT: Partial equilibrium model	no	yes	
Oeko-Institut	LCA-approach based on trade patterns and land-use change due to displacement	yes	yes[23]	IPCC

The range in iLUC related GHG emissions per study is given in Figure 1.

[22] used in FAPRI-CARD calculations with GREEN-AGSIM reported to the JRC

[23] Expressed as a bandwidth of the „risk level“ from 25 to 50%, see OEKO (2010) for details.

Figure 1: ILUC emissions based on a 20 years annualization period



A summary of extremes and median values is presented in Table 2 in order to summarise the model outcomes to provide one generalised number for each biofuel type that can be used as a baseline in the storyline assessments presented in this report. The median value was used instead of the average as this indicator is less susceptible to outliers.

Table 2 Summary of review

Type of biofuel	Minimum ILUC emission factor (in g CO_2eq/MJ biofuel)	Maximum ILUC emission factor (in g CO_2eq/MJ biofuel)	Median from average values (in g CO_2eq/MJ biofuel)*
Biodiesel based on rapeseed from Europe	-33 - 80	80-800	77
Ethanol based on wheat from Europe	-79-79	-8-329	73
Ethanol based on sugar beet from Europe	13-33	65-181	85
Biodiesel based on palm oil from South-East Asia	-55-45	34-214	77
Biodiesel based on soy from Latin America	13-67	75-1 380	140
Biodiesel based on soy from US	0-11	100-273	65
Ethanol based on sugar cane from Latin America	-1-48	19-95	60
Bioelectricity based on perennial on arable land from Europe	32	75	56

* Where studies report only a minimum and maximum, the average between these was taken. Most studies report the average and a range.

Annex 4 Approach and sources used for estimating price levels of different biomass sources 2020 and 2030

	Source/Reference(s) used	Elaboration/assumptions
dry manure wet manure	1) Lensink et al., 2010, Conceptadvies basisbedragen 2011 voor electriciteit en groen gas in het kader van de SDE regeling. ECN-E--10-053	According to the Nitrate Directive farmers in Nitrogen Vulnerable Zones (NVZ) need to take care of their own manure disposal. This will lead to costs unless the manure is used in a biogas installation on farm. Therefore it is assumed that all manure exceeding the 170 kg Nitrogen/ha per region is available for bioenergy production for free. The price of manure in regions with a large manure excess is therefore at 0 Euro/ton. In regions with limited or no excess (so not between 170-100 kg N/ha) the price is 40 Euro/ton as bioenergy use does not compensate for the cost made to get rid of the excess manure. This price is kept stable over 2020 and 2030, except that an inflation correction factor is applied per year as specified in Annex 5.
straw	1) DBFZ et al, Leipzig, Service- und Begleitvorhaben des Förderprogramms „Energetische Biomassenutzung“, METHODEN zur stoffstromorientierten Beurteilung für Vorhaben im Rahmen des BMU-Förderprogramms „Energetische Biomassenutzung“, TEIL I: TECHNOLOGIEKENNWERTE, GESTEHUNGSKOSTEN, TREIBHAUSGASBILANZEN, Stand: 17.09.2010.	The price level of 50 Euro/ton was taken from DBFZ 1) for regions with an excess straw potential (total straw production - straw use for competing uses) and this level was assumed to be 80 Euro/ton in regions where there is limited straw available. This price is kept stable over 2020 and 2030, except that an inflation correction factor is applied per year as specified in Annex 5.
Road side verge grass		Cost estimates: No costs are connected to the grassland cuttings, as these are costs connected to normal management. However, transport and drying of the grass is assumed to be around 10 Euro/ton DM. This price is kept stable over 2020 and 2030, except that an inflation correction factor is applied per year as specified in Annex 5.
Prunings from fruit trees, nuts, vineyards, olives and citrus.	National cost levels for supply costs for agricultural residues derived from Siemons et al. (2004).	To estimate the 2020 and 2030 cost the 2004 costs are kept constant, but an inflation rate is applied per year as specified in Annex 5 underneath
Wood-waste	National cost levels for supply costs for agricultural residues derived from Siemons et al. (2004).	To estimate the 2020 and 2030 cost the 2004 costs are kept constant, but an inflation rate is applied per year as specified in Annex 4 underneath
Animal waste	Cost levels derived from Lensink et al. (2010).	Cost levels were only found for the Netherlands, but the same cost levels was applied to rest of EU countries. To estimate the 2020 and 2030 cost the 2004 costs are kept constant, but an inflation rate is applied per year as specified in Annex 4 underneath
Biodegradable municipal waste		Cost levels were only found for the Netherlands, but the same cost levels was applied to rest of EU countries. To estimate the 2020 and 2030 cost the 2004 costs are kept constant, but an inflation rate is applied per year as specified in Annex 5 underneath
paper cardboard		The present paper card board price of low quality paper card board is set at about 115 Euro/ton. This translates itself at a LHV of 18.58 into a price of 6.3 Euro per GJ. To estimate the 2020 and 2030 cost the 2004 costs are kept constant, but an inflation rate is applied per year as specified in Annex 5 underneath
common sludges		Price was assumed to be 0 except for transport and pre-treatment cost which were estimated at 10 Euro/ton DM. To estimate the 2020 and 2030 cost the 2004 costs are kept constant, but an inflation rate is applied per year as specified in Annex 5 underneath
Maize	CAPRI 2020 baseline and 2030 reference scenario cost levels.	Prices for these crops at regional level were taken from the CAPRI runs for the 2020 and 2030 years. To come to

	These are specific per region	an average national price for the feedstock a weighted average price was calculated according to the contribution of every region to the total national feedstock.
Rape	CAPRI 2020 baseline and 2030 reference scenario cost levels. These are specific per region	Prices for these crops at regional level were taken from the CAPRI runs for the 2020 and 2030 years. To come to an average national price for the feedstock a weighted average price was calculated according to the contribution of every region to the total national feedstock.
Sugarbeet	CAPRI 2020 baseline and 2030 reference scenario cost levels. These are specific per region	Prices for these crops at regional level were taken from the CAPRI runs for the 2020 and 2030 years. To come to an average national price for the feedstock a weighted average price was calculated according to the contribution of every region to the total national feedstock.
Sunflower 2008	CAPRI 2020 baseline and 2030 reference scenario cost levels. These are specific per region	Prices for these crops at regional level were taken from the CAPRI runs for the 2020 and 2030 years. To come to an average national price for the feedstock a weighted average price was calculated according to the contribution of every region to the total national feedstock.
Cereals	CAPRI 2020 baseline and 2030 reference scenario cost levels. These are specific per region	Prices for these crops at regional level were taken from the CAPRI runs for the 2020 and 2030 years. To come to an average national price for the feedstock a weighted average price was calculated according to the contribution of every region to the total national feedstock.
Perennials	1) Carrasco & Sixto, expert consultation Rothamsted (2007). In: Eppler et al., (2007). and Mitchell (1999) 2) Schweinle (2007), Erricson (2006) and Dudly and Riche et al. (2007). 3) Christian and Riche (1999), Monti et al. (2007), Knanna et al. (2008). 4) Dudly and Riche et al. (2007). 5) Dworak et al., 2008.	Production cost estimates were made for poplar 1), Willow 2), Miscanthus and Switchgrass 3) and Reed Canary grass 4) based on different publications who provided detailed overview of types of costs -yield level combinations. In order to extrapolate the cost to other EU regions the yield level in every region was used as a distribution factor 5). A distinction is made in cost for high yield, medium yield and low yield situations. Calculations of yield level estimates per region were made with a crop growth model (see Annex 6). Detailed yield and cost levels per region are provided in Annex 7.
Round wood	Price estimates derived from Lindner et al., 2005	See for detailed price levels the per country overview of the cost-supply combination in Annex 8.
Additional Harvestable Round wood	Price estimates derived from Lindner et al., 2005	See for detailed price levels the per country overview of the cost-supply combination in Annex 8.
Primary forestry residues	Price estimates derived from Lindner et al., 2005	See for detailed price levels the per country overview of the cost-supply combination in Annex 8.
Black liquor	Price assumed to be at 0 Euro/ton as BL is used immediately in pulp industry.	

Annex 5 Macro-economic figures used for extrapolation of potentials and inflation corrections of prices

	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Population growth (% per year)												
EU 27	0.4	0.4	0.4	0.4	0.3	0.3	0.3	0.3	0.3	0.2	0.2	0.2
of which EI15	0.5	0.5	0.5	0.5	0.4	0.4	0.4	0.4	0.3	0.3	0.3	0.3
of which EU12	0.0	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.2
GDP growth (EU 27)	-4.2	1.8	1.7	1.8	2.1	2.2	2.3	2.2	2.1	2.0	2.0	2.0
Inflation rate (EU-27)	1.0	1.8	1.7	1.8	1.9	2.0	2.0	2.0	2.0	2.0	2.0	2.0
	2021	**2022**	**2023**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**		
Population growth (% per year)												
EU 27	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1		
of which EI15	0.3	0.3	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2		
of which EU12	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2		
GDP growth (EU 27)	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0		
Inflation rate (EU-27)	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0		
Macro-economic factors 2009-2020 derived from DG-Agri report Prospects for agricultural markets and income in the EU.												
Used source data are from Eurostat, DG-AGRI, ECFIN												

Annex 6 Estimation of yield levels for perennials per region in different management systems

The FAO method of modelling crop water use and crop yield response allows taking account of the effects of suboptimum temperatures and water deficits on the length of the growing season, on crop water use and crop yield (Doorenbos and Kassam, 1979; Allen et al., 1998). In the present study, two separate assessments have been made, one for a typical C_3 grass and another one for a typical C_4 grass. For each reference crop the water use and crop yield have been assessed for two theoretical reference situations, one for completely irrigated conditions (potential production situation) and one for purely rainfed conditions (water-limited production situation). This allows to identify a zonation across Europe for the climatic suitability for each crop type and to analyze the differences in climatic suitability between C_3 and C_4.

Table 1: Examples of C_3 and C_4 perennial biomass grasses already grown in EU countries and to which the presented modelling approach is assumed to be applicable

Crop	**C_3**	**C_4**
Reed Canary Grass *(Phalaris arundinacea L.)*	X	
Miscanthus (*Miscanthus spp.)*		X
Switchgrass (*Panicum virgatum L)*		X
Giant Reed (*Arundo donax L.)*	X	

The basic assumption on the length of growing season for grass is that grass starts growing (in fact, using water) as soon as the average temperature (over 10 days) exceeds a threshold, and that it stops growing as soon as the temperature drops below this threshold. However, the full evapotranspiration rate (and therefore full growth rate) are reached when the temperature exceeds a second threshold at a higher temperature, above which the temperature is optimum for realizing the full potential in terms of evapotranspiration and related crop growth. In between the lower and higher threshold temperatures the evapotranspiration is reduced linearly from a complete reduction (reduction factor = 1) at the lower threshold to no reduction (reduction factor = 0) at the second threshold.

The values for the two temperature thresholds chosen in the present study are:

C_3 grass, 5 and 10 degree Celcius

C_4 grass, 10 and 18 degrees Celcius

A full green cover is assumed all year, so Kc = 1 during 36 dekades (10 days periods) of the year.

The progress index increases from 0 to 100 over 1 + 35 periods of 10-days

The rooting depth of the grass is 1,000 mm, so it is rather deep and this makes that the soil-limited maximum rooting depth plays a role in the modelling of the soil water balance in all soils shallower than 100 cm.

Climate and soil data

The results have been obtained by applying the FAO-method within the European part of the **Global Water Satisfaction Index system (GWSI)** JRC-Agri4Cast for the two grass crops. The GWSI system contains the soil and climatic data and calculation modules. The two grasses have been defined especially for the present study. The climatic data are a time series of 10 years (1996-2005) 10-daily weather data of the ECMWF model on a 1x1 degree long-lat grid. The European land area counts 1,088 of such major grid cells. The European soil map has been overlaid with a much finer grid at 0.1x0.1 degree resolution. This full grid counts 79,607 cells over Europe. Within each fine grid the soil map distinguishes up to 50 different Soil Typological Units. The soil data used in GWSI is a result of grouping

these soil typological units (STUs) of the European soil map into physical soil types with identical agro-hydrological properties (rooting depth, water holding capacity). These physical soil types have been mapped on a 0.1x0.1 degree grid, where each grid cell has one or more soil types, known by their percentage are occupation, of which the total amounts to 100%. Combining climate grid and soil types resulted in 17,912 calculation units (unique combinations of climate and soil type). For each calculation unit five water related output variables have been stored for each crop assessment, namely the cumulative values over the growing season of the water surplus, the water deficit, the maximum evapotranspiration ETm (irrigated situation), the actual evapotranspiration ETa (rainfed situation) and precipitation. Each variable is expressed in mm/season.

Spatial aggregation

The calculations have been made for the complete European area (all calculation units), without considering a land use mask. The water output variables are mapped to the full grid of 0.1x0.1 degree by assignment of the values from the calculation units. Several aggregation options are possible to assign a value from the calculation units to a fine grid cell. In the present study we have chosen for the area weighted average value over each grid cell as the area weighted value is the most unbiased value in a post processing, e.g, in order to combine these results with a land use mask and to distinguish regional yield patterns in relation with current agricultural areas.

The fine grid data are a basis for aggregation to NUTS-2 or national values, or to values at River Basin.

Translating water use into biomass potential yields

The maximum grass yield has been calculated by assuming characteristic water use efficiency values from the literature for C_3 and C_4 grasses. WUE is expressed in gram dry matter per kg used water. The range in WUE values is from 1 to 5 gr DM per kg ET. (Note that its inverse value is used as well. The corresponding range is expressed as between 1,000 and 200 kg ET per kg DM). As the basic conversion is 1 mm water = 10 m^3 per hectare = 10,000 kg/ha, another expression of the range in WUE is from 10 to 50 kg DM per mm ET. We have chosen the WUE values as follows:

- WUE for C_3 crops 30 kg DM per mm water use
- WUE for C_4 crops 40 kg DM per mm water use
- nil (zero percent) harvest losses

With a range in ETa values of 200 to 1,300 mm over Europe , this leads to a range in theoretically maximum yield levels of 6,000 – 40,000 kg/hectare for C_3 grasses, and 7000- 50,000 kg/hectare for C_4 grasses. Note that on the same location the modelled ETa value for C_3 and C_4 are often different, due to differences in temperature response. These maximum yield levels are not attainable in reality for reasons explained earlier. A fraction of 70% of these maximum yields is a more realistic biomass yield ceiling, and 50% of these maximum yields is probably a fair target under intensive management.

When the choice is for extensive management and low external inputs, the value of the maximum yield maps is that it shows zones with relatively high and relatively low potential, and the differences in potential between typical C_3 and C_4 crops.

Results of the biomass assessment for C3 and C4 grasses

The assessment of the maximum yields leads to a set of four European yield maps, of which as an example only the two for C3 are displayed, showing a regional yield pattern for the reference situations in Table 2.

Next, difference between C_3 and C_4 and potential and water limited yield levels can be analyzed, which results in four additional yield difference maps as displayed in Table 3 and the Maps underneath.

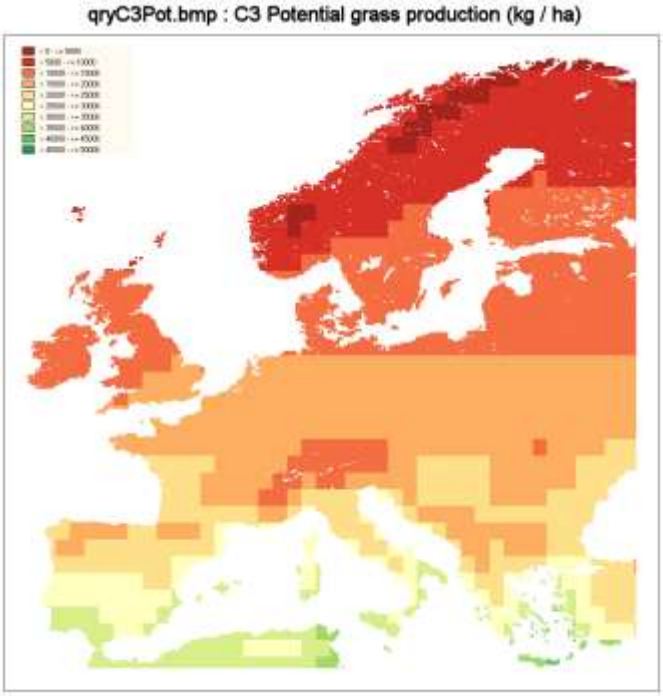
Table 2: Assessment of the maximum yields

Grass yield kg/ha	C_3	C_4
Potential	<5,000 to 40,000 *(see Map 1)*	<5,000 to >45,000
Water limited	<5000 to 20000 *(See Map 2)*	<5,000 to 20,000

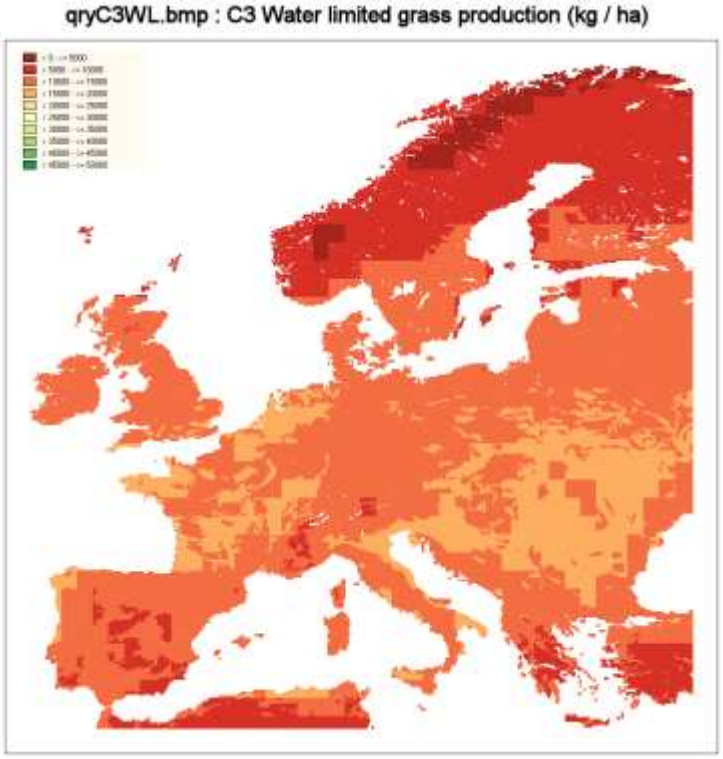
Table 3: Difference between C_3 and C_4 and potential and water limited yield levels

Grass yield kg/ha	C_3 grass	C_3-C_4 difference	C_4 grass
Potential grass yield	<5,000 to 40,000	Up to + 8,000 in northwest Europe Up to – 8,000 in south Europe *(Map 3)*	<5,000 to >45,000
Pot – watlim difference	0-30,000		0-35,000
Water limited grass yield	<5,000 to 20,000	Up to + 8,000 in northwest Europe Up to – 4,000 in south and SE Europe *(Map 4)*	<5,000 to 20,000

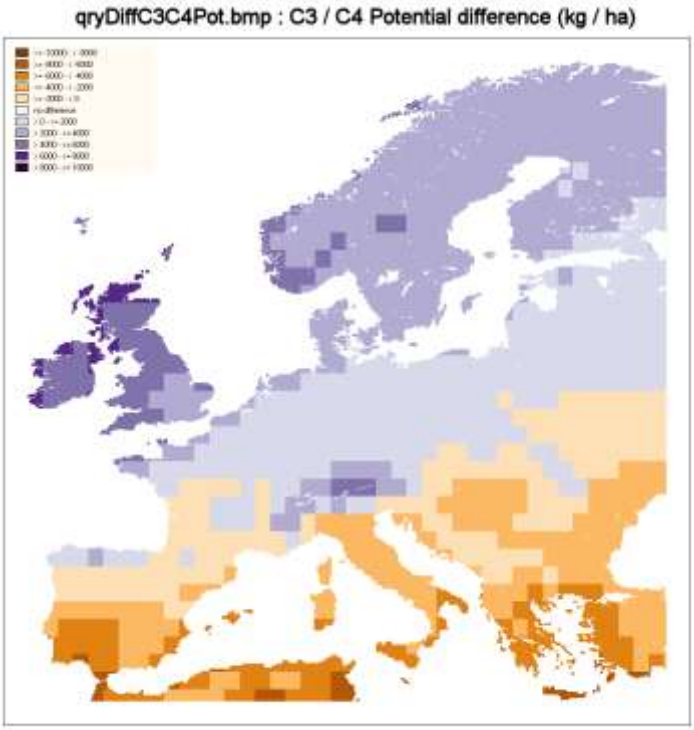
Map 1: C_3 Potential grass production



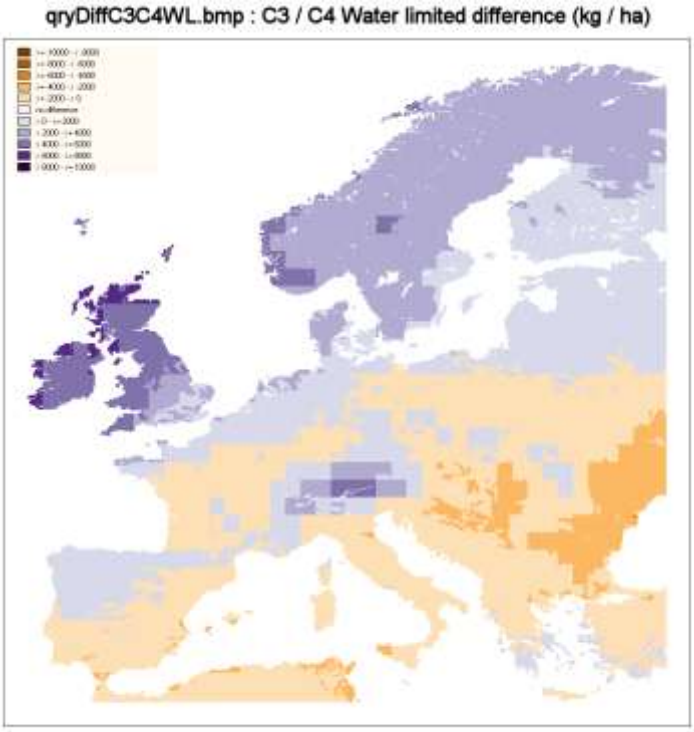
Map 2: C_3 water limited grass production



Map 3: C_3/C_4 Potential difference in yield



Map 4: C_3/C_4 water limited difference in yields



It shows that the potential yield of C_3 grass exceeds the C_4 yields in the northern half of Europe and in the mountainous areas. The dividing line runs in the direction SWW-NEE. Just north of the dividing line are the Spanish Atlantic north coast, Brittany, Massif central and Alps in France, the Alpine and Carpathian regions, Poland and Bielorussia. The highest differences are in the areas with long cool summers, especially in Ireland and western UK, in the Alps and in a few areas in Norway. A second area with somewhat smaller differences in yield are Scandinavia and Finland, which have a short cool summer, and in the coastal strips along the Channel and in the North Sea region, which have slightly warmer summer than western UK. In the whole area of the northern European low plain from Brussels to Moscow the C_3 grass has a slight advantage above the C_4 grass. However, under water limited conditions the C_4 grass has a relative advantage in those parts of the European plain where drought periods occur regularly in the summer, especially in soil regions with a lot of sandy soils, such as eastern Germany and Poland, except the coastal strip along the Baltic Sea. In northern European regions without water stress in the growing season the C_3 grass maintains its advantage over C_4 crops.

In the southern half of Europe potential grass yields of both C_3 and C_4 grasses increase southwards, but the C_4 grass out-yields the C_3 grass. The difference is largest under potential (irrigated) conditions in the far south

Also under water-limited conditions C_4 grass out-yields the C_3 grass. But the difference is levelled of by the increasing drought stress in the most southern regions.

The highest water-limited grass yields for C_4 grasses are southern France, but not in the Mediterranean part, in northern Italy and all the area east of Warschau-Vienna, down to the west coast of the Black Sea, and West-Ukraine. The highest water-limited yields of C_3 grass are slightly below the C_4 yields and occur in the same regions as for C_4 grass and for the Benelux countries. In all these areas a rather favourable balance exists between rainfall and evapotranspiration, during the entire growing season between winter and late autumn.

Conclusions for mapping of yield and water need levels:

From the above exercise we have derived a database specifying per 0.1x0.1 degree grid the total attainable and water limited yield for C_3 and C_4 crops and the related water need to arrive to this potential. From the latter we can also specify the irrigation water need per grid by subtracting the water need under total attainable yield with the under a water limited yield. The yields per location also provide the basis for the calculation of the potential energy yield from the C_3 and C_4 crops per location.

Translating biomass yield of perennial grasses over Europe into obtainable under current farming conditions fort the scenario assessments

The estimated theoretical maximum biomass productions for C_3 and C_4 perennial grasses assume optimum management conditions, which implies that a full green cover is reached soon after the start of the growing season, and sufficient nutrient availability from soil and fertilizer during the entire growing period. Two separate yield estimates have been made, one under full irrigation and one under purely rainfed conditions. These two production levels, potential and water-limited, should correspond with the highest yields observed in field experiments. In reality, the yields are often lower, due to suboptimum conditions such as nutrient shortage or incomplete plant cover.

For example, Consentino et al. (2007) report for a Miscanthus (a C_4 grass) field experiment in Catania, Sicily, where the combined irrigation and nitrogen effects were studied, a maximum biomass yield of 27 T/ha dry matter under full nitrogen and water supply, 19 T/ha under full irrigation and low nitrogen level, and 17 T/ha under limited irrigation and nitrogen conditions. For the second year in the same multi-year experiment, the biomass yields were 18, 14.6 and 14.5 T/ha dry matter respectively. Note that there was not a purely rainfed situation in this trial. This shows inter-year yield variability, and important yield reductions due to limiting soil and water conditions. The annual biomass production for the same region (Sicily, NUTS itg1) according to to GWSI model was 35 tonnes DM ha^{-1} $year^{-1}$ under potential conditions and 15 tonnes under water-limited conditions. In this case the maximum observed yield is 77% of the potential yield.

Lewandowski et al. (2003) provide a review of perennial grasses for use as energy crops in the US and Europe, and provide reported yield ranges from literature, per grass species and country. It is not clear under what kind of conditions these grasses were grown but we may assume that the highest yields are related to intensive crop management and our estimates of potential and water-limited biomass yields should correspond to these observed yields, while the lowest yields are probably below our water-limited yields, unless the reported yields are from irrigated fields.

The following table provides the comparison of yields reported by Lewandowski et al. (2003) and our yield estimates, per crop type and country.

Table 4: Comparison of observed and simulated biomass yield in some European countries for C3 and C4 grasses

Country	**C_4 grass observed**	**C_3 grass observed**	**C_4 simulated**	**C_3 simulated**
Biomass yields are in tonnes DM ha^{-1} $year^{-1}$	sw switch gra mi Miscanthus	gr giant reed rcg reed canary	pot potential wl water-lim	Pot potential wl water- lim
Finland		5-12 (rcg)		8-11 (pot)
				8-10 (wl)
Sweden		5-12 (rcg)		7-12 (pot)
				7-12 (wl)
Denmark	5-15 (mi)		10-11 (pot)	
			10-10 (wl)	
Britain	11 (sw)	6-12 (rcg)	5-13 (pot)	10-16 (pot)

	10-15 (mi)		5-13 (wl)	10-14 (wl)
Germany	4-30 (mi)	15-20 (gr)	12-16 (pot)	14-16 (pot)
			12-15 (wl)	13-14 (wl)
Switzerland	13-19 (mi)		9-13 (pot)	
			9-12 (wl)	
Austria	22 (mi)		7-20 (pot)	
			7-17 (wl)	
North Italy		3-32 (gr)		11-22 (pot)
				10-15 (wl)
Italy	30-32 (mi)		8-35 (pot)	
			8-16 (wl)	
South Italy		15-34 (gr)		25-30 (pot)
				12-15 (wl)
North Greece		5-17 (gr)		21-26 (pot)
				10-12 (wl)
Greece	26-44 (mi)		28-46 (pot)	
			9-14 (wl)	
South Greece		7-31 (gr)		29-38 (pot)
				9-11 (wl)
Turkey	28 (mi)		23-33	
			8-16	
Spain	14-34 (mi)	8-37 (gr)	16-40 (pot)	18-34 (pot)
			10-14 (wl)	10-14 (wl)

* For both observed and simulated data the range in yields is given: for the observed data the range within a country, for the simulated data the lowest and highest regional yields (NUTS-2) within a country.

Analysis of the figures in the table 4 shows that in general:

- The highest simulated biomass yields for perennial C_4 grasses vary from 11 tonnes DM ha^{-1} $year^{-1}$ in Denmark, 13 in Britain to 40 in Spain and even 46 in Greece.
- For C_4 grasses the picture over all countries with data the highest observed biomass yields are between 80 and 120% of the highest potential yields. For C_3 grasses the highest observed yields are between 85 and 95% of the highest potential yields.

This leads to the assumption that under modern intensive fully irrigated farming all (C_3 and C_4) perennial grasses could reach 90% of the potential biomass yield. Comment: this assumption (introduction of modern intensive farming practices, including irrigation when drought occurs), may not be very realistic as a scenario that will be applicable over large regions, because water availability will be a real constraint in southern Europe.

In situations where no irrigation water is available, but otherwise under modern farming practices and good soils, the biomass yields will show more regional variation. For regions in southern Europe it may yet be necessary to apply one irrigation at the start of the growth cycle to ensure crop establishment. A good crop cover is necessary for reaching the full water-limited yield. The assumption of modern farm technology is justified, as large scale biomass cropping will be organized as a new agricultural business which will not evolve from traditional farming.

The basic set of assumptions for the yield level is that the production ceiling is set at 90% of the potential production, but will not exceed the water-limited production. This can be assessed easily by

taking the lowest value of the following two yield levels: (90% of potential yield) and (100% of the water-limited yield).

Comment 1: This assumption means that in areas with sufficient rainfall (northern half of Europe) the attainable production will be close to 90% of the potential yield, but especially in southern Europe the rainfed biomass production will be down to 50 or 25% of the maximum irrigated production.

Comment 2: It appears from the observed yield data however, that for the southern European countries the lowest observed yields are above the water-limited yields, implying that the bio-energy grasses are irrigated at least partly. Yet the assumption that the large scale introduction of new biomass crops should rely largely on rainfed cropping seems justified.

In situation of extensive arable farming and low soil quality (shallow, or otherwise marginally productive soils) the biomass production will be lower than under intensive farming. We assume that the range of yield levels for these situations can be found in the tail of the observed yields, e.g. below the midyield in the range of observed yields per region (or per country). In reality the variability in biomass yields under extensive farming systems will be high. A reasonable first guess would be to assess the obtainable biomass yield under extensive arable farming as the lowest yield of the following two: (50% of the potential yield) and (80% of the water limited yield).

Summary on assessment in three yield levels (High, medium and low yield)

In summary we can derive from the set of regional mean potential and water-limited yields for C_3 and C_4 perennial grasses three attainable biomass yield levels, according to the type of cropping:

- High yield: Modern fully irrigated cropping: all grasses could reach 90% of the potential biomass yield.
- Medium yield: Modern rainfed cropping (apart from crop establishment irrigation): attainable grass yield equals the lowest value of the following two yield levels: (90% of potential yield) and (100% of the water-limited yield).
- Low yield: Extensive cropping: the lowest yield of the following two: (50% of the potential yield) and (80% of the water limited yield).

These 3 yield levels have been used in the storyline studies to estimate the final total yield levels of miscanthus, Switchgrass and Reed Canary Grass (RCG) (see Annex 7 for results).

Since miscanthus and switchgrass are both C_4 crops the results will show the same yield levels for both crops. However, based on the literature review (Table 4) there is a clear difference in yield level for both crops. In the final yield level estimates we have therefore applied a final yield correction factor to switchgrass as follows:

- High yield (Modern fully irrigated yield): switchgrass reaches 70% of miscanthus yield
- Medium yield: Modern rainfed: switchgrass reaches 80% of miscanthus yield
- Low yield: Extensive cropping: switchgrass reaches 90% of miscanthus yield

BIOMASS FUTURES

Annex 7 Estimated yield and cost levels for perennial crops per region

Table 1 Yield and cost levels for perennials (Yields calculated according to methodology described in Annex 6). Cost levels estimated according to Annex 5.

	Yield in ton/ha										Cost estimate (Euro/ton DM)								
Code	NUTS_NAAM	high yield RCG	medium yield RCG	low yield RCG	high yield miscanthus	medium yield miscanthus	low yield miscanthus	high yield switchgrass	medium yield switchgrass	low yield switchgrass	high yield RCG	medium yield RCG	low yield RCG	high yield miscanthus	medium yield miscanthus	low yield miscanthus	high yield switchgrass	medium yield switchgrass	low yield switchgrass
AT110000	Burgenland (A)	17	15	8	18	15	8	13	12	8	38	42	65	81	64	83	94	66	77
AT120000	Nieder÷sterreich	15	15	8	15	15	8	11	12	8	41	42	65	94	64	83	110	66	77
AT210000	KŽrnten	12	12	7	10	10	6	7	8	5	49	49	80	136	91	119	160	95	111
AT220000	Steiermark	14	14	8	13	13	7	9	11	7	43	43	69	105	71	92	123	74	86
AT310000	Ober÷sterreich	13	13	7	12	12	7	8	10	6	45	45	73	116	79	102	136	82	95
AT320000	Salzburg	11	11	6	9	9	5	6	7	4	51	51	84	154	103	135	181	107	126
AT330000	Tirol	10	10	6	7	7	4	5	5	3	57	57	94	203	135	178	240	140	165
AT340000	Vorarlberg	10	10	5	6	6	3	4	5	3	58	58	96	220	146	192	260	152	179
BG010000	Severozapaden	18	16	10	20	18	11	14	15	10	36	39	57	72	63	78	98	65	73
BG020000	Severen tsentralen	19	15	10	22	18	12	15	14	11	35	40	55	68	63	73	92	66	69
BG030000	Severoiztochen	19	14	11	23	16	13	16	13	11	34	44	53	66	69	71	90	72	67
BG040000	Yugoiztochen	20	14	11	23	16	13	16	13	11	34	44	53	66	72	71	89	74	67
BG050000	Yugozapaden	17	14	10	19	15	11	13	12	10	37	44	59	76	74	82	103	76	76
BG060000	Yuzhen tsentralen	19	14	10	21	16	12	15	13	11	35	43	55	70	70	75	95	73	70
BL210000	Prov. Antwerpen	15	15	8	13	13	7	9	11	7	42	42	67	107	72	94	125	75	88
BL220000	Prov. Limburg (B)	15	15	8	13	13	7	9	11	7	42	42	68	106	72	94	125	75	88
BL230000	Prov. Oost-Vlaanderen	15	15	8	13	13	7	9	10	7	42	42	67	108	73	95	126	76	88
BL240000	Prov. Vlaams Brabant	15	15	8	13	13	7	9	11	7	42	42	67	107	73	94	125	75	88
BL250000	Prov. West-Vlaanderen	15	15	8	13	13	7	9	10	6	41	41	67	109	74	96	127	76	89
BL310000	Prov. Brabant Wallon	15	15	8	13	13	7	9	11	7	42	42	67	107	73	94	125	75	88
BL320000	Prov. Hainaut	15	15	8	13	13	7	9	11	7	42	42	67	105	72	93	124	74	87
BL330000	Prov. LiÐge	14	14	8	13	13	7	9	10	6	42	42	68	109	74	96	127	76	89

INTELLIGENT ENERGY EUROPE

BIOMASS FUTURES

Yield in ton/ha		Cost estimate (Euro/ton DM)										Cost estimate (Euro/ton DM)								
Code	**NUTS_NAAM**	**high yield RCG**	**medium yield RCG**	**low yield RCG**	**high yield miscanthus**	**medium yield miscanthus**	**low yield miscanthus**	**high yield switchgrass**	**medium yield switchgrass**	**low yield switchgrass**		**high yield RCG**	**medium yield RCG**	**low yield RCG**	**high yield miscanthus**	**medium yield miscanthus**	**low yield miscanthus**	**high yield switchgrass**	**medium yield switchgrass**	**low yield switchgrass**
BL340000	Prov. Luxembourg (B)	14	14	8	13	13	7	9	10	6		43	43	69	110	75	97	129	77	90
BL350000	Prov. Namur	14	14	8	13	13	7	9	10	7		42	42	68	107	73	95	126	76	88
CZ010000	Praha	14	14	8	13	13	7	9	11	7		43	44	70	105	72	93	124	74	87
CZ020000	Stredn‰ Cechy	14	14	8	13	13	7	9	11	7		43	43	69	105	71	92	123	74	86
CZ030000	Jihozßpad	14	14	8	13	13	7	9	10	6		44	44	71	109	74	96	127	76	89
CZ040000	Severozßpad	14	13	8	13	13	7	9	10	6		44	45	71	109	74	96	128	77	89
CZ050000	Severov?chod	14	14	8	14	14	8	9	11	7		43	43	69	104	71	91	122	73	85
CZ060000	Jihov?chod	15	15	8	15	15	8	10	12	7		41	42	66	95	65	84	111	67	78
CZ070000	Stredn‰ Morava	15	15	8	15	15	8	10	12	7		42	42	67	97	66	85	113	68	80
CZ080000	Moravskoslezsko	14	14	8	14	14	8	10	11	7		42	42	68	101	69	89	119	72	83
DE110000	Baden-W"rttemberg	14	14	8	13	13	7	9	10	7		43	43	69	107	73	94	126	75	88
DE120000		14	14	8	13	13	7	9	10	7		43	43	69	107	73	94	126	75	88
DE130000		14	14	8	13	13	7	9	10	7		43	43	69	107	73	94	126	75	88
DE140000		14	14	8	13	13	7	9	10	7		43	43	69	107	73	94	126	75	88
DE210000	Bayern	14	14	8	12	12	7	8	10	6		44	44	72	115	78	101	135	81	94
DE220000	Berlin, Bremen, Hamburg	14	14	8	13	13	7	9	10	7		43	43	69	108	73	95	126	76	89
DE230000		14	14	8	13	13	7	9	10	7		43	43	69	108	73	95	126	76	89
DE240000		14	14	8	13	13	7	9	10	7		43	43	69	108	73	95	126	76	89
DE250000		14	14	8	13	13	7	9	10	7		43	43	69	108	73	95	126	76	89
DE260000		14	14	8	13	13	7	9	10	7		43	43	69	108	73	95	126	76	89
DE270000		14	14	8	13	13	7	9	10	7		43	43	69	108	73	95	126	76	89
DE400000	Brandenburg	15	14	8	14	14	8	10	12	7		42	44	67	98	67	86	115	69	81
DE710000	Hessen	14	13	8	13	13	7	9	10	7		43	45	69	108	73	95	126	76	88
DE720000		14	13	8	13	13	7	9	10	7		43	45	69	108	73	95	126	76	88
DE730000		14	13	8	13	13	7	9	10	7		43	45	69	108	73	95	126	76	88

BIOMASS FUTURES

Yield in ton/ha		Cost estimate (Euro/ton DM)									Cost estimate (Euro/ton DM)								
Code	**NUTS_NAAM**	**high yield RCG**	**medium yield RCG**	**low yield RCG**	**high yield miscanthus**	**medium yield miscanthus**	**low yield miscanthus**	**high yield switchgrass**	**medium yield switchgrass**	**low yield switchgrass**	**high yield RCG**	**medium yield RCG**	**low yield RCG**	**high yield miscanthus**	**medium yield miscanthus**	**low yield miscanthus**	**high yield switchgrass**	**medium yield switchgrass**	**low yield switchgrass**
DE800000	Mecklenburg-Vorpommern	14	13	8	12	12	7	9	10	6	44	47	72	112	76	98	131	79	92
DE910000	Niedersachsen	14	14	8	12	12	7	9	10	6	44	44	70	112	76	98	131	79	92
DE920000		14	14	8	12	12	7	9	10	6	44	44	70	112	76	98	131	79	92
DE930000		14	14	8	12	12	7	9	10	6	44	44	70	112	76	98	131	79	92
DE940000		14	14	8	12	12	7	9	10	6	44	44	70	112	76	98	131	79	92
DEA10000	Nordrhein-Westfalen	14	14	8	13	13	7	9	10	6	43	43	70	110	75	97	129	77	90
DEA20000		14	14	8	13	13	7	9	10	6	43	43	70	110	75	97	129	77	90
DEA30000		14	14	8	13	13	7	9	10	6	43	43	70	110	75	97	129	77	90
DEA40000		14	14	8	13	13	7	9	10	6	43	43	70	110	75	97	129	77	90
DEA50000		14	14	8	13	13	7	9	10	6	43	43	70	110	75	97	129	77	90
DEB10000	Rheinland-Pfalz	15	14	8	14	14	8	10	11	7	42	44	68	104	71	91	121	73	85
DEB20000		15	14	8	14	14	8	10	11	7	42	44	68	104	71	91	121	73	85
DEB30000		15	14	8	14	14	8	10	11	7	42	44	68	104	71	91	121	73	85
DEC00000	Saarland	15	14	8	14	14	8	10	11	7	41	43	67	100	68	88	117	71	83
DED00000	Sachsen	14	14	8	13	13	7	9	11	7	43	43	69	105	71	92	123	74	86
DEE00000	Sachsen-Anhalt	14	14	8	14	14	8	10	11	7	42	44	68	103	70	91	120	73	85
DEF00000	Schleswig-Holstein	13	13	7	11	11	6	8	9	6	46	46	75	124	84	109	146	87	102
DEG00000	Th°ringen	14	13	8	13	13	7	9	10	6	44	46	71	112	76	98	131	78	92
DK000000	Hovedstaden	12	12	6	10	10	5	7	8	5	50	50	82	141	95	124	166	98	115
EE000000	Estonia	11	11	6	9	9	5	7	7	5	52	53	86	146	98	128	172	102	119
EL110000	Anatoliki Makedonia, Thraki	22	12	10	25	14	11	18	11	10	31	48	57	69	78	77	81	80	72
EL120000	Kentriki Makedonia	21	12	9	25	13	11	17	11	10	32	50	60	71	83	83	82	86	77
EL130000	Dytiki Makedonia	20	10	8	22	11	9	16	9	8	34	55	66	77	96	96	90	100	90
EL140000	Thessalia	22	10	8	26	11	9	18	9	8	31	57	68	69	98	98	80	102	91
EL210000	Ipeiros	21	10	8	24	11	9	17	9	8	32	55	66	72	98	98	84	102	91
EL220000	Ionia Nisia	25	11	9	29	11	9	20	9	8	29	52	63	63	97	96	73	100	90
EL230000	Dytiki Ellada	23	10	8	27	10	8	19	8	7	30	57	69	66	105	105	76	109	98

EL240000	Sterea Ellada	24	9	7	28	10	8	20	8	7		29	60	73	63	112	111	74	116	104
EL250000	Peloponnisos	26	9	7	30	9	7	21	7	7		28	60	72	60	114	113	69	118	106
EL300000	Attiki	29	10	8	33	9	7	23	7	7		26	58	70	55	113	112	64	117	105
EL410000	Voreio Aigaio	31	10	8	36	9	7	25	7	7		25	57	68	52	114	114	60	119	106
EL420000	Notio Aigaio	34	11	9	41	11	9	28	9	8		24	53	64	47	100	100	54	104	93
EL430000	Kriti	35	10	8	41	11	9	29	9	8		23	58	70	47	99	99	54	103	92
ES110000	Galicia	18	14	10	18	13	10	12	10	9		36	43	57	95	85	88	112	88	82
ES120000	Principado de Asturias	16	12	9	15	11	8	10	9	7		39	48	62	112	97	103	131	101	96
ES130000	Cantabria	18	13	10	17	13	9	12	10	8		37	45	58	100	86	92	117	89	86
ES210000	Pais Vasco	18	14	10	19	14	10	13	11	9		35	44	55	91	81	85	107	83	79
ES220000	Comunidad Foral de Navarra	19	13	10	19	13	11	13	11	10		35	45	55	89	82	82	104	85	77
ES230000	La Rioja	19	12	9	19	12	9	14	9	8		35	50	60	88	93	92	103	96	86
ES240000	Arag≤n	21	12	9	22	12	10	15	10	9		33	49	59	78	88	88	91	92	82
ES300000	Comunidad de Madrid	23	10	8	25	10	8	17	8	7		31	56	68	70	104	103	82	108	97
ES410000	Castilla y Le≤n	19	11	9	20	11	9	14	9	8		34	52	62	85	100	100	99	104	93
ES420000	Castilla-la Mancha	24	10	8	26	11	9	19	9	8		30	55	67	67	99	99	78	103	92
ES430000	Extremadura	25	11	9	29	11	9	20	9	8		29	53	64	63	96	96	73	100	90
ES510000	Catalu±a	21	12	10	23	13	10	16	10	9		32	49	58	75	84	84	88	87	78
ES520000	Comunidad Valenciana	25	11	9	28	12	10	20	10	9		29	52	63	64	88	88	74	91	82
ES530000	Illes Balears	30	12	10	34	14	11	24	11	10		26	48	57	55	77	77	63	80	72
ES610000	Andalucia	27	11	8	31	12	9	22	9	8		27	54	65	59	93	92	68	96	86
ES620000	Regi≤n de Murcia	27	10	8	31	11	9	22	9	8		27	57	69	59	95	94	68	98	88
ES700000	Regi≤n de Murcia	27	10	8	31	11	9	22	9	8		27	57	69	59	95	94	68	98	88
FI130000	Pohjois-Suomi	7	7	4	5	5	3	3	4	2		76	76	130	276	182	241	327	190	224
FI180000	ItΣ-Suomi	9	9	5	7	7	4	5	6	3		64	64	106	194	129	169	229	134	158
FI190000	EtelΣ-Suomi	10	10	6	8	8	5	6	7	4		56	56	93	162	108	142	191	112	132
FI1A0000	LΣnsi-Suomi	9	9	5	8	8	4	5	6	4		59	59	99	179	119	156	211	124	146
FI200000	┼land	9	8	5	8	7	4	5	5	4		59	70	99	179	133	157	211	138	146
FR100000	?le de France	16	14	9	16	15	9	11	12	8		39	43	62	90	65	80	106	68	75
FR210000	Champagne-Ardenne	16	14	9	15	15	9	11	12	8		39	43	63	93	67	82	109	69	77
FR220000	Picardie	15	15	9	14	14	8	10	12	7		40	42	64	98	67	87	115	70	81

FR230000	Haute-Normandie	15	15	9	14	14	8	10	11	7		40	42	64	101	69	89	119	72	83
FR240000	Centre	17	15	10	17	15	10	12	12	9		37	42	58	83	64	73	97	66	69
FR250000	Basse-Normandie	16	15	9	14	14	8	10	11	7		39	41	63	101	69	89	118	71	83
FR260000	Bourgogne	17	15	9	17	16	9	12	12	9		37	41	59	85	63	75	99	65	70
FR300000	Nord - Pas-de-Calais	15	15	8	13	13	7	9	11	7		41	42	66	105	72	93	124	74	87
FR410000	Lorraine	15	15	8	15	15	8	10	12	7		41	42	65	98	67	86	114	69	80
FR420000	Alsace	15	15	8	14	14	8	10	11	7		42	42	67	100	68	88	117	71	82
FR430000	Franche-ComtΘ	15	15	8	15	15	8	10	12	7		41	41	65	97	67	86	114	69	80
FR510000	Pays de la Loire	18	15	10	17	16	10	12	12	9		36	41	58	83	63	74	97	65	69
FR520000	Bretagne	16	16	9	14	14	8	10	12	7		38	39	61	98	67	86	115	69	81
FR530000	Poitou-Charentes	19	14	11	20	15	11	14	12	10		35	43	54	87	74	81	102	77	75
FR610000	Aquitaine	19	15	11	20	16	11	14	13	10		35	40	54	85	69	79	99	71	74
FR620000	Midi-PyrΘnΘes	18	14	10	18	15	10	13	12	9		37	43	58	92	76	85	108	79	80
FR630000	Limousin	18	15	10	18	16	10	12	13	9		36	40	58	96	72	88	112	74	83
FR710000	Rh⌠ne-Alpes	16	13	9	15	13	8	11	11	8		40	45	63	110	83	101	128	86	94
FR720000	Auvergne	17	15	9	17	15	9	12	12	8		37	41	59	100	74	93	117	76	87
FR810000	Languedoc-Roussillon	20	13	10	21	13	10	15	10	9		33	46	55	80	84	84	94	87	78
FR820000	Provence-Alpes-C⌠te d'Azur	18	11	9	19	11	9	13	9	8		36	52	62	90	98	98	105	102	91
FR830000	Corse	23	11	9	25	11	9	18	9	8		31	53	64	70	96	95	81	99	89
HU000000	K÷zΘp-Magyarorszßg	18	14	10	21	16	11	14	13	10		36	42	56	83	69	77	97	71	72
HU100000	K÷zΘp-Dunßnt·l	18	14	10	20	16	11	14	13	10		36	42	56	84	69	78	98	72	73
HU210000	Nyugat-Dunßnt·l	18	15	10	19	17	11	14	14	10		36	41	57	88	67	81	103	69	76
HU220000	DΘl-Dunßnt·l	19	14	10	21	16	12	15	13	11		35	43	55	82	70	76	95	72	71
HU230000	╔szak-Magyarorszßg	17	15	9	19	17	10	13	14	9		38	40	60	91	65	84	107	68	79
HU310000	╔szak-Alf÷ld	18	16	10	20	18	11	14	14	10		37	40	58	87	65	80	101	67	75
HU320000	DΘl-Alf÷ld	19	15	11	22	17	12	15	14	11		34	41	54	79	67	73	92	69	69
IR000000	Ireland	11	11	6	6	6	3	4	5	3		51	51	83	213	142	186	252	147	173
IT000000	Piemonte	15	12	8	15	12	8	11	10	8		41	48	65	109	89	101	128	92	94
IT110000	Valle d'Aosta/VallΘe d'Aoste	11	10	6	8	8	4	5	6	4		53	55	88	173	116	152	204	120	141
IT120000	Liguria	20	13	11	22	14	11	15	11	10		33	45	54	78	80	79	91	83	74
IT130000	Lombardia	16	13	9	16	14	9	11	11	8		40	45	64	91	70	80	106	73	75

IT200000	Provincia Autonoma Trento	13	13	7	12	12	6	8	9	6		46	47	75	119	81	105	140	84	98
IT310000	Veneto	16	15	9	16	15	9	11	12	8		40	41	64	89	65	79	104	67	74
IT320000	Friuli-Venezia Giulia	15	15	8	15	15	8	11	12	8		41	41	66	94	64	82	109	66	77
IT330000	Emilia-Romagna	20	15	11	23	17	13	16	13	11		34	41	53	77	68	71	89	71	67
IT400000	Toscana	21	14	11	24	15	12	17	12	11		33	44	52	74	74	74	86	77	69
IT510000	Umbria	20	14	11	23	15	12	16	12	11		34	44	53	76	76	76	89	79	71
IT520000	Marche	21	14	11	24	16	13	17	13	11		33	42	51	74	71	71	86	74	67
IT530000	Lazio	21	14	11	25	15	12	17	12	11		32	44	52	71	76	76	83	79	71
IT600000	Abruzzo	20	13	11	23	14	12	16	12	10		33	45	53	76	77	76	89	80	72
IT710000	Molise	21	14	11	25	15	12	17	12	11		32	44	52	71	75	75	83	78	70
IT720000	Campania	23	14	11	26	15	12	18	12	11		31	44	52	69	75	75	80	78	70
IT800000	Puglia	26	15	12	30	16	13	21	13	12		28	41	49	61	70	69	70	72	65
IT910000	Basilicata	24	14	11	27	15	12	19	12	11		30	43	51	65	73	73	76	76	68
IT920000	Calabria	26	14	11	29	15	12	21	12	11		28	43	52	61	73	73	71	76	68
IT930000	Sicilia	28	14	11	32	15	12	22	12	11		27	44	52	57	72	72	66	75	67
ITA00000	Sardegna	26	12	9	30	13	10	21	10	9		28	49	59	60	84	83	69	87	78
LT000000		13	13	7	12	12	6	8	8	5		47	76	50	120	138	129	141	143	120
LV000000		12	12	7	11	11	6	7	7	5		49	80	54	131	151	141	154	157	132
NL110000	Groningen	14	14	8	12	12	7	8	9	6		44	44	71	118	80	103	138	83	97
NL120000	Friesland (NL)	14	14	8	12	12	6	8	9	6		44	44	72	120	81	105	141	84	98
NL130000	Drenthe	14	14	8	12	12	7	8	9	6		44	44	71	118	80	104	138	83	97
NL210000	Overijssel	14	14	8	13	13	7	9	10	6		43	43	69	111	75	98	130	78	91
NL220000	Gelderland	14	14	8	13	13	7	9	10	7		42	42	68	107	73	94	126	76	88
NL230000	Flevoland	14	14	8	12	12	7	8	10	6		44	44	70	116	79	102	136	81	95
NL310000	Utrecht	14	14	8	13	13	7	9	10	6		42	42	68	109	74	96	128	77	90
NL320000	Noord-Holland	14	14	8	12	12	7	8	9	6		44	44	71	117	80	103	138	82	96
NL330000	Zuid-Holland	14	14	8	12	12	7	9	10	6		43	43	69	112	76	98	131	79	92
NL340000	Zeeland	14	14	8	13	13	7	9	10	6		42	42	68	110	75	97	129	77	90
NL410000	Noord-Brabant	14	14	8	13	13	7	9	10	6		42	42	68	108	74	95	127	76	89
NL420000	Limburg (NL)	14	14	8	13	13	7	9	11	7		42	42	68	107	73	94	125	75	88
PL110000	L≤dzkie	15	14	8	15	15	8	10	12	7		42	42	68	98	67	86	114	69	80

PL120000	Mazowieckie	14	14	8	14	14	8	10	11	7		42	43	68	99	68	87	116	70	82
PL210000	Malopolskie	14	14	8	14	14	8	10	11	7		43	43	70	104	71	91	121	73	85
PL220000	Slaskie	14	14	8	14	14	8	10	11	7		43	43	69	103	70	91	121	73	85
PL310000	Lubelskie	15	15	8	15	15	8	10	12	7		42	42	67	95	65	84	111	67	78
PL320000	Podkarpackie	15	15	8	15	15	8	10	12	7		42	42	68	96	66	85	113	68	79
PL330000	Swietokrzyskie	14	14	8	14	14	8	10	12	7		42	42	68	98	67	87	115	70	81
PL340000	Podlaskie	14	14	8	13	13	7	9	11	7		44	44	71	106	72	93	124	75	87
PL410000	Wielkopolskie	14	14	8	14	14	8	10	11	7		42	43	68	99	67	87	115	70	81
PL420000	Zachodniopomorskie	14	13	8	13	13	7	9	10	6		44	45	71	109	74	96	128	77	90
PL430000	Lubuskie	15	13	8	15	14	8	10	12	7		41	45	67	96	67	85	113	70	79
PL510000	Dolnoslaskie	14	14	8	14	14	8	10	11	7		43	43	69	101	69	89	118	71	83
PL520000	Opolskie	15	15	8	14	14	8	10	12	7		42	42	67	98	67	87	115	70	81
PL610000	Kujawsko-Pomorskie	14	14	8	13	13	7	9	11	7		43	44	70	105	71	92	123	74	86
PL620000	Warminsko-Mazurskie	13	13	7	13	13	7	9	10	6		45	46	73	112	76	98	131	79	92
PL630000	Pomorskie	13	13	7	11	11	6	8	9	6		47	47	76	122	82	107	143	85	100
PT110000	Norte	19	12	10	20	11	9	14	9	8		34	48	57	85	94	94	99	98	88
PT150000	Algarve	30	10	8	35	11	9	25	9	8		26	57	69	53	99	98	61	103	92
PT160000	Centro (PT)	23	12	10	25	12	10	17	10	9		31	48	57	71	89	88	82	92	82
PT170000	Lisboa	26	13	11	29	15	12	21	12	11		28	45	53	61	75	75	71	78	70
PT180000	Alentejo	27	11	9	31	12	10	22	10	9		28	51	61	59	88	88	68	91	82
RO010000	Nord-Vest	16	15	9	18	17	10	12	14	9		39	40	61	95	67	88	111	69	82
RO020000	Centru	19	15	11	22	18	12	16	14	11		34	41	53	78	65	72	91	67	67
RO030000	Nord-Est	18	16	10	21	18	12	15	15	10		36	40	56	82	63	76	96	65	71
RO040000	Sud-Est	18	16	10	20	18	11	14	15	10		37	39	58	86	63	79	100	65	74
RO050000	Sud - Muntenia	18	16	10	20	18	11	14	14	10		37	39	58	87	63	81	102	66	76
RO060000	Bucuresti - Ilfov	15	15	9	16	16	9	11	13	8		40	40	65	103	70	95	121	72	89
RO070000	Sud-Vest Oltenia	15	15	8	16	16	9	11	12	8		41	41	66	107	72	99	125	74	92
RO080000	Vest	19	16	10	21	18	12	15	15	11		35	39	55	81	62	75	94	64	70
SE010000	Stockholm	10	10	6	8	8	5	6	7	4		55	55	91	162	108	142	190	112	132
SE020000	¶stra Mellansverige	11	11	6	8	8	5	6	7	4		54	54	90	163	109	143	192	113	133
SE040000	Smøland med ÷arna	11	11	6	8	8	5	6	7	4		53	53	87	160	107	140	188	111	130
SE060000	Sydsverige	11	11	6	9	9	5	6	7	5		51	51	84	151	101	132	178	105	123

SE070000	VΣstsverige	11	11	6	8	8	5	6	7	4		53	53	88	163	109	142	192	113	133
SE080000	Norra Mellansverige	9	9	5	6	6	3	4	5	3		61	61	102	214	142	187	253	148	174
SE090000	Mellersta Norrland	7	7	4	4	4	2	3	3	2		75	75	128	330	217	288	391	226	267
SE0A0000	ΠvreNorrland	6	6	3	4	4	2	3	3	2		84	84	143	359	236	313	426	245	290
SI000000	Vzhodna Slovenija	16	16	9	17	17	10	12	14	9		38	40	61	83	58	73	97	60	69
SK010000	Bratislavsk? kraj	17	14	10	19	16	10	13	13	9		37	43	59	79	62	70	92	64	65
SK020000	ZßpadnΘ Slovensko	16	14	9	17	16	10	12	13	9		38	42	61	83	62	73	97	64	69
SK030000	StrednΘ Slovensko	15	14	8	15	15	8	10	12	7		42	43	67	96	66	84	112	69	79
SK040000	V?chodnΘ Slovensko	15	15	8	15	15	8	11	12	8		42	42	67	94	64	83	110	67	77
UKC00000	Tees Valley and Durham	11	11	6	6	6	4	4	5	3		52	52	86	210	140	184	248	145	171
UKD00000	Cumbria	11	11	6	17	13	8	12	10	7		53	53	87	85	74	85	100	77	79
UKE00000	East Yorkshire and Northern Lincolnshire	12	12	7	17	13	8	12	10	7		48	48	79	85	74	84	99	77	79
UKF00000	Derbyshire and Nottinghamshire	13	13	7	17	13	8	12	10	7		46	46	75	84	74	84	98	76	78
UKG00000	Herefordshire, Worcestershire and Warks	13	13	7	17	13	8	12	10	7		45	45	73	84	74	84	98	76	78
UKH00000	East Anglia	14	14	8	17	13	8	12	10	8		43	44	70	84	74	84	98	76	78
UKJ00000	Berkshire, Bucks and Oxfordshire	14	14	8	17	13	8	12	10	8		44	44	70	83	74	83	97	76	78
UKK00000	Gloucestershire, Wiltshire and Bristol/Bath area	14	14	8	17	13	8	12	10	7		44	44	71	84	74	84	98	77	79
UKL00000	West Wales and The Valleys	12	12	7	17	13	8	12	10	7		48	48	79	84	74	84	98	77	79
UKM00000	South Western Scotland	10	10	5	17	13	8	12	10	7		58	58	96	83	74	84	97	77	79
UKN00000	Northern Ireland	11	11	6	17	13	8	12	10	7		51	51	84	83	74	84	97	77	79

Table 2 Yield and cost levels for willow and poplar (yields based on GLOBIOM model, IIASA). Cost levels estimated according to Annex 5.

	Yield in ton/ha					Cost estimate (Euro/ton DM)			
Code	NUTS_NAAM	Willow high yield	Willow low yield	Poplar High yield	Poplar Low yield	Willow high yield	Willow low yield	Poplar high yield	Poplar low yield
AT110000	Burgenland (A)	11	7	12	8	48	63	30	40
AT120000	Nieder÷sterreich	10	7	7	4	64	69	32	49
AT210000	KŽrnten	11	7	5	3	50	64	45	69
AT220000	Steiermark	11	7	5	3	51	66	47	72
AT310000	Ober÷sterreich	10	6	5	3	57	75	43	66
AT320000	Salzburg	12	8	3	2	47	61	70	107
AT330000	Tirol	10	6	4	3	57	74	56	87
AT340000	Vorarlberg	10	7	4	3	53	70	52	81
BG010000	Severozapaden	12	8	9	6	44	58	24	36
BG020000	Severen tsentralen	12	8	10	6	48	62	22	34
BG030000	Severoiztochen	12	8	10	6	45	59	23	36
BG040000	Yugoiztochen	11	7	9	6	52	67	26	40
BG050000	Yugozapaden	11	7	9	6	50	65	25	38
BG060000	Yuzhen tsentralen	11	7	9	6	48	63	25	39
BL210000	Prov. Antwerpen	10	6	10	6	56	73		
BL220000	Prov. Limburg (B)	12	8	6	4	47	61	39	60
BL230000	Prov. Oost-Vlaanderen	11	7	11	7	51	66		
BL240000	Prov. Vlaams Brabant	13	9	13	9	42	55		
BL250000	Prov. West-Vlaanderen	11	7	11	7	49	64		
BL310000	Prov. Brabant Wallon	13	8	13	8	42	55		
BL320000	Prov. Hainaut	13	9	13	9	42	55		
BL330000	Prov. LiÐge	12	8	9	6	46	60	24	37
BL340000	Prov. Luxembourg (B)	12	8	9	6	47	61	24	36
BL350000	Prov. Namur	13	8	10	7	43	55	22	34

CZ010000	Praha	9	6	8	5		63	82	26	41
CZ020000	Strednф Cechy	9	6	7	5		64	83	30	47
CZ030000	Jihozßpad	9	6	6	4		64	83	37	58
CZ040000	Severozßpad	9	6	6	4		63	82	35	54
CZ050000	Severov?chod	9	6	6	4		61	79	35	54
CZ060000	Jihov?chod	9	6	6	4		62	80	36	56
CZ070000	Strednф Morava	10	7	8	5		53	69	27	41
CZ080000	Moravskoslezsko	10	6	8	5		56	72	26	40
DE110000	Baden-Wnrttemberg	11	7	8	5		52	67	26	41
DE120000		12	8	6	4		46	60	34	53
DE130000		11	7	6	4		51	67	37	57
DE140000		10	7	8	5		55	71	28	43
DE210000	Bayern	11	7	6	4		51	66	38	59
DE220000	Berlin, Bremen, Hamburg	10	7	6	4		53	70	39	60
DE230000		9	6	6	4		61	80	35	53
DE240000		8	5	7	4		69	89	34	52
DE250000		9	6	7	4		61	79	34	52
DE260000		10	7	8	5		53	69	29	45
DE270000		11	7	6	4		52	68	38	58
DE400000	Brandenburg	8	5	6	4		73	95	38	59
DE710000	Hessen	12	8	8	5		48	62	27	41
DE720000		12	8	8	5		48	62	27	42
DE730000		10	6	7	5		58	75	31	47
DE800000	Mecklenburg-Vorpommern	7	5	5	4		74	96	41	62
DE910000	Niedersachsen	9	6	7	5		59	77	30	47
DE920000		9	6	8	5		61	80	28	44
DE930000		7	5	5	4		76	100	41	63
DE940000		8	5	5	4		73	96	41	63
DEA10000	Nordrhein-Westfalen	12	8	12	8		46	60		
DEA20000		11	7	8	5		50	65	26	40
DEA30000		8	5	6	4		69	90	39	60
DEA40000		8	6	8	5		65	85	27	42

DEA50000		10	6	5	4		56	73	41	62
DEB10000	Rheinland-Pfalz	11	7	8	5		49	64	28	43
DEB20000		10	7	9	6		53	69	26	39
DEB30000		11	7	7	5		48	63	30	46
DEC00000	Saarland	12	8	9	6		46	60	25	38
DED00000	Sachsen	9	6	7	4		59	77	33	51
DEE00000	Sachsen-Anhalt	9	6	7	4		65	84	34	52
DEF00000	Schleswig-Holstein	7	5	5	4		77	100	41	63
DEG00000	Th^{n}ringen	9	6	7	5		62	81	30	46
DK000000	Hovedstaden	7	5	4	3		78	101	57	88
EE000000	Estonia	7	5	7	4		79	103	32	49
EL110000	Anatoliki Makedonia, Thraki	0	0	9	6				24	37
EL120000	Kentriki Makedonia	0	0	8	5				28	43
EL130000	Dytiki Makedonia	0	0	10	6				22	34
EL140000	Thessalia	0	0	9	6				24	37
EL210000	Ipeiros	0	0	11	7				20	31
EL220000	Ionia Nisia	0	0	14	9				16	25
EL230000	Dytiki Ellada	0	0	10	6				23	35
EL240000	Sterea Ellada	0	0	9	6				24	37
EL250000	Peloponnisos	0	0	10	7				22	33
EL300000	Attiki	0	0	10	6				23	35
EL410000	Voreio Aigaio	0	0	9	6				24	38
EL420000	Notio Aigaio	0	0	9	6				24	37
EL430000	Kriti	0	0	10	7				21	33
ES110000	Galicia	13	8	10	6		43	56	22	34
ES120000	Principado de Asturias	12	8	6	4		45	59	37	56
ES130000	Cantabria	12	8	8	5		44	58	26	41
ES210000	Pais Vasco	13	8	12	7		43	56	19	30
ES220000	Comunidad Foral de Navarra	0	0	11	7				21	32
ES230000	La Rioja	0	0	9	6				24	37
ES240000	Arag≤n	0	0	8	5				28	44
ES300000	Comunidad de Madrid	0	0	7	4				33	51

ES410000	Castilla y Le≤n	0	0	7	5				31	48
ES420000	Castilla-la Mancha	0	0	7	5				30	46
ES430000	Extremadura	0	0	10	7				22	33
ES510000	Catalu±a	0	0	8	5				26	41
ES520000	Comunidad Valenciana	0	0	9	6				25	38
ES530000	Illes Balears	0	0	11	7				20	31
ES610000	Andalucia	0	0	10	7				21	33
ES620000	Regi≤n de Murcia	0	0	8	5				27	42
ES700000	Regi≤n de Murcia	0	0	0	0					
FI130000	Pohjois-Suomi	4	3	3	2		129	168	85	132
FI180000	ItΣ-Suomi	5	3	3	2		122	158	71	109
FI190000	EtelΣ-Suomi	4	2	3	2		156	203	79	122
FI1A0000	LΣnsi-Suomi	0	0	0	0				1068	1644
FI200000	┼land	1	1	2	1		618	805	122	187
FR100000	?le de France	13	9	9	6		42	55	25	38
FR210000	Champagne-Ardenne	12	8	7	5		48	62	31	47
FR220000	Picardie	11	7	8	5		49	63	29	44
FR230000	Haute-Normandie	13	8	8	5		43	56	27	42
FR240000	Centre	12	8	10	6		44	58	23	35
FR250000	Basse-Normandie	14	9	9	6		39	50	25	38
FR260000	Bourgogne	13	8	9	6		43	56	23	36
FR300000	Nord - Pas-de-Calais	12	8	7	4		45	59	34	52
FR410000	Lorraine	12	8	6	4		47	62	40	61
FR420000	Alsace	11	7	5	3		48	63	46	72
FR430000	Franche-ComtΘ	13	8	6	4		43	56	37	56
FR510000	Pays de la Loire	14	9	10	7		40	52	22	33
FR520000	Bretagne	13	8	12	7		43	55	19	30
FR530000	Poitou-Charentes	14	9	11	7		39	50	20	30
FR610000	Aquitaine	14	9	10	7		39	50	22	33
FR620000	Midi-PyrΘnΘes	0	0	11	7				20	30
FR630000	Limousin	14	9	5	3		39	51	48	73
FR710000	Rh∫ne-Alpes	0	0	9	6				25	38

FR720000	Auvergne	12	8	7	4		47	61	34	52
FR810000	Languedoc-Roussillon	0	0	9	6				26	39
FR820000	Provence-Alpes-C⌠te d'Azur	0	0	9	6				25	39
FR830000	Corse	0	0	12	8				18	27
HU000000	K÷zΘp-Magyarorszßg	0	0	0	0					
HU100000	K÷zΘp-Dunßnt·l	12	8	10	6		47	61	23	35
HU210000	Nyugat-Dunßnt·l	11	7	11	7		52	68	21	32
HU220000	DΘl-Dunßnt·l	12	8	12	8		47	61	19	29
HU230000	╔szak-Magyarorszßg	12	8	11	7		46	60	20	30
HU310000	╔szak-Alf÷ld	12	8	10	7		45	58	22	33
HU320000	DΘl-Alf÷ld	10	7	8	5		53	69	29	45
IR000000	Ireland	0	0	0	0					
IT000000	Piemonte	0	0	0	0					
IT110000	Valle d'Aosta/VallΘe d'Aoste	0	0	10	6				23	35
IT120000	Liguria	0	0	2	1				114	175
IT130000	Lombardia	0	0	10	6				23	35
IT200000	Provincia Autonoma Trento	0	0	7	4				33	51
IT310000	Veneto	0	0	3	2				66	101
IT320000	Friuli-Venezia Giulia	0	0	6	4				36	56
IT330000	Emilia-Romagna	0	0	4	3				55	84
IT400000	Toscana	0	0	8	5				28	43
IT510000	Umbria	0	0	10	7				21	33
IT520000	Marche	0	0	10	6				23	36
IT530000	Lazio	0	0	9	6				26	40
IT600000	Abruzzo	0	0	11	7				20	31
IT710000	Molise	0	0	9	6				24	37
IT720000	Campania	0	0	9	6				23	36
IT800000	Puglia	0	0	9	6				25	38
IT910000	Basilicata	0	0	9	6				24	36
IT920000	Calabria	0	0	9	6				25	39
IT930000	Sicilia	0	0	10	7				22	34
ITA00000	Sardegna	0	0	8	5				28	44

LT000000		9	6	0	0		63	82		
LV000000		8	5	7	5		66	86	31	47
NL110000	Groningen	7	5	6	4		76	99	39	59
NL120000	Friesland (NL)	8	5	7	4		68	88	33	50
NL130000	Drenthe	7	5	6	4		75	97	36	55
NL210000	Overijssel	8	5	7	5		66	86	32	49
NL220000	Gelderland	11	7	6	4		52	68	40	61
NL230000	Flevoland	8	5	0	0		68	88		
NL310000	Utrecht	11	7	0	0		49	64		
NL320000	Noord-Holland	9	6	7	4		61	79	32	50
NL330000	Zuid-Holland	10	6	0	0		57	74		
NL340000	Zeeland	8	5	0	0		66	86		
NL410000	Noord-Brabant	9	6	6	4		60	79	35	54
NL420000	Limburg (NL)	11	7	0	0		49	64		
PL110000	L≤dzkie	9	6	6	4		63	82	38	59
PL120000	Mazowieckie	8	5	6	4		67	88	38	59
PL210000	Malopolskie	9	6	5	3		60	78	42	65
PL220000	Slaskie	10	6	6	4		56	73	39	60
PL310000	Lubelskie	10	6	6	4		57	74	39	60
PL320000	Podkarpackie	10	6	7	5		57	74	31	47
PL330000	Swietokrzyskie	10	7	6	4		53	69	37	57
PL340000	Podlaskie	8	5	6	4		66	86	40	62
PL410000	Wielkopolskie	8	5	6	4		70	91	36	55
PL420000	Zachodniopomorskie	9	6	6	4		63	82	36	56
PL430000	Lubuskie	8	5	7	4		66	85	33	50
PL510000	Dolnoslaskie	9	6	5	3		62	81	48	73
PL520000	Opolskie	9	6	10	7		59	77	22	34
PL610000	Kujawsko-Pomorskie	8	5	6	4		67	87	39	59
PL620000	Warminsko-Mazurskie	8	6	5	4		65	85	40	62
PL630000	Pomorskie	8	5	6	4		69	90	35	53
PT110000	Norte	15	10	9	6		38	49	24	37
PT150000	Algarve	0	0	11	7				20	31

PT160000	Centro (PT)	15	10	9	6		37	48	24	37
PT170000	Lisboa	0	0	0	0					
PT180000	Alentejo	0	0	10	6				22	34
RO010000	Nord-Vest	11	7	8	5		48	63	28	43
RO020000	Centru	11	7	7	4		48	63	33	51
RO030000	Nord-Est	12	8	8	5		47	61	28	42
RO040000	Sud-Est	12	8	9	6		46	60	25	39
RO050000	Sud - Muntenia	12	8	8	5		48	62	27	42
RO060000	Bucuresti - Ilfov	12	8	6	4		46	59	35	54
RO070000	Sud-Vest Oltenia	12	8	5	4		47	62	41	62
RO080000	Vest	13	8	10	6		43	56	23	35
SE010000	Stockholm	9	6	0	0		65	84		
SE020000	╥stra Mellansverige	8	5	6	4		70	91	40	62
SE040000	Smσland med ÷arna	9	6	7	4		62	81	33	51
SE060000	Sydsverige	5	3	3	2		115	150	64	99
SE070000	VΣstsverige	1	0	1	1		917	1194	152	234
SE080000	Norra Mellansverige	0	0	0	0				459	707
SE090000	Mellersta Norrland	8	5	5	4		67	87	41	62
SE0A0000	╥vre Norrland	8	5	5	4		68	89	41	62
SI000000	Vzhodna Slovenija	0	0	6	4				40	61
SK010000	Bratislavsk? kraj	11	7	11	7		51	66	21	32
SK020000	ZßpadnΘ Slovensko	11	7	10	6		51	67	23	35
SK030000	StrednΘ Slovensko	11	7	5	3		51	66	48	74
SK040000	V?chodnΘ Slovensko	11	7	6	4		49	63	37	58
UKC00000	Tees Valley and Durham	0	0	3	2				64	98
UKD00000	Cumbria	7	4	5	3		81	106	42	65
UKE00000	East Yorkshire and Northern Lincolnshire	7	5	5	3		76	100	44	68
UKF00000	Derbyshire and Nottinghamshire	8	5	7	4		68	88	34	52
UKG00000	Herefordshire, Worcestershire and Warks	10	7	7	4		54	70	34	52
UKH00000	East Anglia	10	6	7	5		56	73	32	49
UKJ00000	Berkshire, Bucks and Oxfordshire	11	7	10	6		48	63	23	35
UKK00000	Gloucestershire, Wiltshire and Bristol/Bath	11	7	7	5		49	64	30	46

	area									
UKL00000	West Wales and The Valleys	9	6	6	4		60	78	37	57
UKM00000	South Western Scotland	0	0	3	2				82	126
UKN00000	Northern Ireland	1	0	5	3		48	74	48	74

Annex 8 Cost supply tables per EU country

Austria		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
AT	manure	10	123	10	10	123	10
AT	straw	677	163	688	677	163	688
AT	verge gras	19	32	706	19	32	706
AT	prunings	48	103	754	48	103	754
AT	landscape care wood	264	83	1018	254	83	1008
AT	animal waste	110	119	1128	110	119	1118
AT	MSW (Not landfill, composting, recycling)	0	119	1128	0	119	1118
AT	MSW landfill	180	57	1308	180	57	1298
AT	used fats and oils	35	368	1343	35	368	1333
AT	paper cardboard	191	323	1534	191	323	1524
AT	post consumer wood	190	29	1724	190	29	1714
AT	common sludges	448	25	2172	448	25	2162
AT	maize/corn (bioethanol)	69	1252	2241	0	0	2162
AT	energy maize (biogas)	221	205	2462	0	0	2162
AT	rape	21	741	2483	0	0	2162
AT	sugarbeet	0	1063	2483	0	0	2162
AT	sunflower	0	837	2483	0	0	2162
AT	cereals	99	410	2582	0	0	2162
AT	perennials: woody	393	124	2975	180	107	2342
AT	perennials: grassy	362	127	3338	285	135	2627
AT	stemwood	2398	578	5736	2398	578	5025
AT	additional harvestable round wood	2503	463	8238	2291	463	7317
AT	primary forestry residues	1779	231	10018	780	231	8096
AT	sawmill by-products (excl sawdust)	840	130	10858	840	130	8937
AT	sawdust	420	104	11278	420	104	9356
AT	other industrial wood residues	342	130	11620	342	130	9699
AT	black liquor	772	0	12392	772	0	10471
AT	grass cuttings abandoned grassland	50	163	12442	50	163	10521

Bulgaria		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
BG	black liquor	84	0	84	84	0	84
BG	common sludges	76	25	161	76	25	161
BG	verge gras	8	32	168	8	32	168
BG	Landscape care wood	261	34	429	251	34	419
BG	post consumer wood	44	35	472	44	35	463
BG	prunings	242	42	714	242	42	704
BG	MSW landfill	360	57	1074	360	57	1064
BG	perennials: grassy	184	72	1258	558	65	1623
BG	perennials: woody	1206	73	2464	1156	62	2779
BG	saw-dust	33	80	2497	33	80	2812
BG	sawmill by-products (excl sawdust)	61	100	2557	61	100	2872
BG	other industrial wood residues	62	100	2619	62	100	2935
BG	animal waste	2	119	2622	2	119	2937
BG	MSW (Not landfill, composting, recycling)	258	119	2880	258	119	3195
BG	total manure	0	123	2880	0	123	3195
BG	straw	1396	136	4276	1396	136	4591
BG	grass cuttings abandoned grassland	35	136	4311	35	136	4626
BG	primary forestry residues	544	165	4855	288	165	4914
BG	Forrage maize (biogas)	0	214	4855	0	0	4914
BG	paper cardboard	41	323	4896	41	323	4955
BG	cereals	94	329	4990	0	0	4955
BG	used fats and oils	44	368	5034	44	368	4999
BG	additional harvestable roundwood	330	397	5364	325	397	5324
BG	roundwood	675	496	6039	675	496	5999
BG	rape	0	666	6039	0	0	5999
BG	sunflower	62	678	6101	0	0	5999
BG	sugarbeet	0	909	6101	0	0	5999
BG	maize/corn (bioethanol)	104	1248	6205	0	0	5999

Belgium/Luxembourg		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
BL	sunflower	0	0	0	0	0	0
BL	black liquor	303	0	303	303	0	303
BL	common sludges	263	25	566	263	25	566
BL	verge gras	23	32	589	23	32	589
BL	post consumer wood	336	35	925	336	35	925
BL	total manure	3090	37	4015	3090	37	4015
BL	MSW landfill	540	57	4555	540	57	4555
BL	landscape care wood	139	70	4694	134	70	4689
BL	prunings	18	88	4711	18	88	4706
BL	perennials: grassy	110	90	4821	99	89	4806
BL	saw-dust	50	107	4871	50	107	4856
BL	perennials: woody	160	118	5031	160	59	5015
BL	animal waste	111	119	5142	111	119	5126
BL	MSW (Not landfill, composting, recycling)	86	119	5228	86	119	5212
BL	sawmill by-products (excl sawdust)	106	133	5334	106	133	5318
BL	other industrial wood residues	140	133	5474	140	133	5458
BL	forrage maize (biogas)	0	167	5474	0	0	5458
BL	straw	334	184	5808	334	184	5792
BL	grass cuttings abandoned grassland	2	184	5810	2	184	5794
BL	primary forestry residues	224	231	6035	109	231	5903
BL	paper cardboard	644	323	6678	644	323	6546
BL	used fats and oils	34	368	6713	34	368	6581
BL	cereals	11	440	6724	0	0	6581
BL	additional harvestable roundwood	86	463	6810	65	463	6645
BL	roundwood	612	578	7421	612	578	7257
BL	rape	1	892	7423	0	0	7257
BL	sugarbeet	0	1013	7423	0	0	7257
BL	maize/corn (bioethanol)	0	1352	7423	0	0	7257

Cyprus		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
CY	straw	0	0	0	0	0	0
CY	maize/corn (bioethanol)	0	0	0	0	0	0
CY	Rape	0	0	0	0	0	0
CY	Sugarbeet	0	0	0	0	0	0
CY	Sunflower	0	0	0	0	0	0
CY	Perennials: woody	0	0	0	0	0	0
CY	Perennials: grassy	0	0	0	0	0	0
CY	black liquor	0	0	0	0	0	0
CY	common sludges	4	25	4	4	25	4
CY	Landscape care wood	23	31	28	22	31	27
CY	verge gras	1	32	29	1	32	28
CY	Post consumer wood	15	33	44	15	33	43
CY	prunings	31	38	74	31	38	73
CY	MSW landfill	0	57	74	0	57	73
CY	saw-dust	0	80	74	0	80	73
CY	sawmill by-products (excl sawdust)	0	99	74	0	99	73
CY	other industrial wood residues	0	99	74	0	99	73
CY	animal waste	5	119	79	5	119	78
CY	MSW (Not landfill, composting, recycling)	0	119	79	0	119	78
CY	total manure	238	123	317	238	123	316
CY	grass cuttings abandoned grassland	0	142	317	0	142	316
CY	primary forestry residues	2	165	319	1	165	317
CY	paper cardboard	34	323	352	34	323	351
CY	used fats and oils	31	368	383	31	368	382
CY	additional harvestable roundwood	4	463	387	3	463	385
CY	roundwood	1	578	388	1	578	387
CY	Cereals	0	1026	388	0	0	387
CY	Forrage maize (biogas)	0	5736	388	0	0	387

Czech Republic		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
CZ	black liquor	403	0	403	403	0	403
CZ	common sludges	273	25	676	273	25	676
CZ	Landscape care wood	306	27	981	294	27	970
CZ	verge gras	21	32	1002	21	32	991
CZ	prunings	10	33	1012	10	33	1001
CZ	MSW landfill	360	57	1372	360	57	1361
CZ	saw-dust	176	118	1548	176	118	1537
CZ	animal waste	16	119	1564	16	119	1553
CZ	MSW (Not landfill, composting, recycling	258	119	1822	258	119	1811
CZ	total manure	2003	123	3826	2003	123	3814
CZ	perennials: grassy	481	125	4307	506	95	4321
CZ	perennials: woody	33	129	4340	31	121	4352
CZ	post consumer wood	146	131	4486	146	131	4498
CZ	sawmill by-products (excl sawdust)	353	148	4839	353	148	4851
CZ	other industrial wood residues	156	148	4994	156	148	5006
CZ	straw	1448	161	6442	1448	161	6454
CZ	grass cuttings abandoned grassland	62	161	6505	62	161	6517
CZ	primary forestry residues	1402	165	7906	664	165	7181
CZ	paper cardboard	182	323	8089	182	323	7363
CZ	used fats and oils	44	368	8133	44	368	7407
CZ	additional harvestable roundwood	923	397	9056	882	397	8289
CZ	Cereals	27	421	9083	0	0	8289
CZ	roundwood	2318	496	11401	2318	496	10607
CZ	forrage maize (biogas)	0	528	11401	0	0	10607
CZ	rape	15	766	11415	0	0	10607
CZ	sunflower	0	1043	11415	0	0	10607
CZ	sugarbeet	0	1080	11415	0	0	10607
CZ	maize/corn (bioethanol)	22	1197	11437	0	0	10607

Germany		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
DE	black liquor	1432	0	1432	1432	0	1432
DE	common sludges	280	25	1712	280	25	1712
DE	verge gras	199	32	1911	199	32	1911
DE	MSW landfill	2160	57	4071	2160	57	4071
DE	landscape care wood	1144	69	5216	1103	69	5174
DE	perennials: woody	3024	81	8240	3881	74	9055
DE	total manure	7266	86	15506	7266	86	16321
DE	prunings	135	86	15641	135	86	16456
DE	post consumer wood	1373	105	17013	1373	105	17829
DE	perennials: grassy	2592	114	19606	2267	109	20095
DE	animal waste	84	119	19690	84	119	20180
DE	MSW (Not landfill, composting, recycl	1120	119	20810	1120	119	21300
DE	saw-dust	798	135	21608	798	135	22098
DE	straw	8883	147	30491	8883	147	30981
DE	grass cuttings abandoned grassland	687	147	31177	687	147	31667
DE	sawmill by-products (excl sawdust)	1723	168	32900	1723	168	33390
DE	other industrial wood residues	1183	168	34083	1183	168	34572
DE	forrage maize (biogas)	0	222	34083	0	0	34572
DE	primary forestry residues	6537	231	40619	3096	231	37669
DE	paper cardboard	1788	323	42408	1788	323	39457
DE	used fats and oils	356	368	42764	356	368	39813
DE	cereals	1457	431	44221	0	0	39813
DE	additional harvestable roundwood	5015	463	49236	4714	463	44527
DE	roundwood	8353	578	57589	8353	578	52880
DE	rape	698	846	58287	0	0	52880
DE	sunflower	0	1009	58287	0	0	52880
DE	sugarbeet	0	1156	58287	0	0	52880
DE	maize/corn (bioethanol)	0	1315	58287	0	0	52880

Denmark		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
DK	total manure	3300	0	3300	3300	84	3300
DK	sunflower	0	0	3300	0	0	3300
DK	black liquor	26	0	3326	26	0	3326
DK	common sludges	286	25	3612	286	25	3612
DK	verge gras	31	32	3643	31	32	3643
DK	Post consumer wood	190	35	3833	190	35	3833
DK	MSW landfill	0	57	3833	0	57	3833
DK	Landscape care wood	175	71	4008	169	71	4002
DK	perennials: woody	0	81	4008	0	0	4002
DK	prunings	6	89	4015	6	89	4008
DK	straw	1300	96	5314	1300	96	5308
DK	grass cuttings abandoned grassland	0	96	5314	0	96	5308
DK	saw-dust	11	107	5325	11	107	5319
DK	perennials: grassy	0	114	5325	0	0	5319
DK	animal waste	42	119	5367	42	119	5361
DK	MSW (Not landfill, composting, recyclir	172	119	5540	172	119	5533
DK	sawmill by-products (excl sawdust)	20	133	5560	20	133	5553
DK	other industrial wood residues	16	133	5575	16	133	5569
DK	primary forestry residues	252	231	5828	105	231	5674
DK	forrage maize (biogas)	0	251	5828	0	0	5674
DK	paper cardboard	73	323	5901	73	323	5746
DK	used fats and oils	23	368	5924	23	368	5770
DK	cereals	530	461	6454	0	0	5770
DK	additional harvestable roundwood	238	463	6692	221	463	5991
DK	roundwood	220	578	6912	220	578	6211
DK	maize/corn (bioethanol)	0	821	6912	0	0	6211
DK	rape	446	1005	7358	0	0	6211
DK	sugarbeet	0	1118	7358	0	0	6211

Estonia		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
EE	maize/corn (bioethanol)	0	0	0	0	0	0
EE	forrage maize (biogas)	0	0	0	0	0	0
EE	sunflower	0	0	0	0	0	0
EE	black liquor	84	0	84	84	0	84
EE	common sludges	22	25	107	22	25	107
EE	post consumer wood	234	32	341	234	32	341
EE	verge gras	2	32	342	2	32	342
EE	landscape care wood	102	36	444	98	36	441
EE	prunings	2	44	446	2	44	442
EE	MSW landfill	0	57	446	0	57	442
EE	perennials: woody	0	65	446	0	0	442
EE	saw-dust	105	79	551	105	79	547
EE	sawmill by-products (excl sawdust)	222	99	773	222	99	769
EE	other industrial wood residues	62	99	835	62	99	831
EE	animal waste	14	119	849	14	119	846
EE	MSW (Not landfill, composting, recycling	0	119	849	0	119	846
EE	total manure	0	123	849	0	123	846
EE	straw	285	161	1134	285	161	1131
EE	grass cuttings abandoned grassland	6	161	1141	6	161	1137
EE	primary forestry residues	414	165	1554	185	165	1322
EE	perennials: grassy	0	182	1554	0	0	1322
EE	cereals	0	306	1554	0	0	1322
EE	paper cardboard	43	323	1597	43	323	1365
EE	used fats and oils	6	368	1603	6	368	1371
EE	additional harvestable roundwood	1139	397	2742	1097	397	2468
EE	roundwood	700	496	3442	700	496	3168
EE	rape	0	863	3442	0	0	3168
EE	sugarbeet	0	896	3442	0	0	3168

Greece		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
EL	rape	0.0	0.00	0	0.0	0.00	0
EL	perennials: woody	0.0	0.00	0	0.0	0.00	0
EL	black liquor	0.0	0.00	0	0.0	0.00	0
EL	common sludges	74.0	24.84	74	74.0	24.84	74
EL	Post consumer wood	146.0	31.87	220	146.0	31.87	220
EL	verge gras	21.0	32.35	241	21.0	32.35	241
EL	MSW landfill	180.0	57.14	421	180.0	57.14	421
EL	landscape care wood	78.7	61.19	500	75.8	61.19	497
EL	prunings	801.2	76.24	1301	801.2	76.24	1298
EL	saw-dust	10.9	105.30	1312	10.9	105.30	1309
EL	animal waste	72.6	119.39	1384	72.6	119.39	1382
EL	MSW (Not landfill, composting, recycling)	430.7	119.39	1815	430.7	119.39	1812
EL	total manure	0.0	122.75	1815	0.0	122.75	1812
EL	sawmill by-products (excl sawdust)	20.2	131.62	1835	20.2	131.62	1832
EL	other industrial wood residues	46.7	131.62	1882	46.7	131.62	1879
EL	straw	438.5	161.70	2321	438.5	161.70	2318
EL	grass cuttings abandoned grassland	29.9	161.70	2350	29.9	161.70	2348
EL	perennials: grassy	2906.0	169.51	5257	1374.3	151.10	3722
EL	primary forestry residues	171.1	231.38	5428	87.7	231.38	3810
EL	paper cardboard	168.9	322.90	5597	168.9	322.90	3979
EL	forrage maize (biogas)	0.0	330.77	5597	0.0	0.00	3979
EL	used fats and oils	47.4	367.94	5644	47.4	367.94	4026
EL	additional harvestable roundwood	416.8	462.75	6061	389.9	462.75	4416
EL	roundwood	254.6	578.44	6315	254.6	578.44	4670
EL	cereals	0.0	653.17	6315	0.0	0.00	4670
EL	sugarbeet	0.0	992.32	6315	0.0	0.00	4670
EL	sunflower	0.0	1632.93	6315	0.0	0.00	4670
EL	maize/corn (bioethanol)	0.0	1972.08	6315	0.0	0.00	4670

Spain		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
ES	black liquor	1175	0	1175	1175	0	1175
ES	common sludges	853	25	2028	853	25	2028
ES	landscape care wood	409	29	2437	394	29	2422
ES	verge gras	112	32	2549	112	32	2534
ES	prunings	4164	36	6713	4164	36	6698
ES	forrage maize (biogas)	0	46	6713	0	0	6698
ES	MSW landfill	900	57	7613	900	57	7598
ES	post consumer wood	701	57	8314	701	57	8299
ES	perennials: woody	44	65	8358	14	60	8313
ES	total manure	2622	100	10980	2622	100	10935
ES	saw-dust	140	115	11120	140	115	11075
ES	animal waste	412	119	11532	412	119	11486
ES	MSW (Not landfill, composting, recycling)	861	119	12393	861	119	12348
ES	straw	2153	122	14546	2153	122	14501
ES	grass cuttings abandoned grassland	272	142	14818	272	142	14772
ES	sawmill by-products (excl sawdust)	296	144	15114	296	144	15068
ES	other industrial wood residues	373	144	15487	373	144	15442
ES	perennials: grassy	10133	182	25620	6064	163	21506
ES	primary forestry residues	1460	231	27081	818	231	22324
ES	paper cardboard	671	323	27752	671	323	22995
ES	used fats and oils	177	368	27929	177	368	23172
ES	additional harvestable roundwood	1170	463	29099	1003	463	24176
ES	cereals	142	488	29241	0	0	24176
ES	roundwood	2128	578	31368	2128	578	26303
ES	rape	0	791	31368	0	0	26303
ES	sunflower	2	1063	31370	0	0	26303
ES	sugarbeet	0	1290	31370	0	0	26303
ES	maize/corn (bioethanol)	178	1620	31548	0	0	26303

Finland		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
FI	forrage maize (biogas)	0	0	0	0	0	0
FI	perennials: woody	0	0	0	0	0	0
FI	black liquor	4449	0	4449	4449	0	4449
FI	common sludges	432	25	4881	432	25	4881
FI	verge gras	28	32	4909	28	32	4909
FI	landscape care wood	510	49	5420	492	49	5401
FI	MSW landfill	180	57	5600	180	57	5581
FI	prunings	8	61	5607	8	61	5589
FI	post consumer wood	190	67	5797	190	67	5778
FI	saw-dust	763	93	6560	763	93	6541
FI	sawmill by-products (excl sawdust)	1400	116	7960	1400	116	7941
FI	other industrial wood residues	467	116	8427	467	116	8408
FI	animal waste	39	119	8465	39	119	8447
FI	MSW (Not landfill, composting, recyc	172	119	8638	172	119	8619
FI	total manure	789	123	9426	789	123	9408
FI	perennials: grassy	374	129	9800	229	129	9637
FI	straw	576	161	10376	576	161	10213
FI	grass cuttings abandoned grassland	69	161	10445	69	161	10282
FI	primary forestry residues	6189	165	16634	2487	165	12768
FI	paper cardboard	170	323	16805	170	323	12939
FI	used fats and oils	22	368	16827	22	368	12961
FI	additional harvestable roundwood	5083	397	21910	4656	397	17617
FI	cereals	2	431	21912	0	0	17617
FI	roundwood	6297	496	28209	6297	496	23914
FI	maize/corn (bioethanol)	0	821	28209	0	0	23914
FI	sunflower	0	867	28209	0	0	23914
FI	Rape	0	892	28209	0	0	23914
FI	sugarbeet	0	1264	28209	0	0	23914

France		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
FR	black liquor	1059	0	1059	1059	0	1059
FR	common sludges	701	25	1760	701	25	1760
FR	verge gras	212	32	1972	212	32	1972
FR	MSW landfill	1800	57	3772	1800	57	3772
FR	post consumer wood	964	64	4736	964	64	4736
FR	perennials: woody	5418	68	10154	8669	61	13405
FR	landscape care wood	2877	78	13031	2772	78	16177
FR	prunings	996	97	14027	996	97	17173
FR	total manure	7309	106	21336	7309	106	24482
FR	straw	11000	117	32336	11000	117	35482
FR	grass cuttings abandoned grassland	946	117	33282	946	117	36428
FR	saw-dust	365	118	33646	365	118	36792
FR	animal waste	403	119	34049	403	119	37196
FR	MSW (Not landfill, composting, recycling)	1034	119	35083	1034	119	38229
FR	sawmill by-products (excl sawdust)	787	148	35870	787	148	39016
FR	other industrial wood residues	467	148	36337	467	148	39483
FR	forrage maize (biogas)	0	151	36337	0	0	39483
FR	perennials: grassy	5008	169	41345	4070	164	43553
FR	primary forestry residues	4337	231	45682	1808	231	45361
FR	paper cardboard	1433	323	47115	1433	323	46794
FR	used fats and oils	259	368	47373	259	368	47052
FR	cereals	2938	438	50311	0	0	47052
FR	additional harvestable roundwood	3691	463	54002	3239	463	50292
FR	roundwood	7788	578	61790	7788	578	58080
FR	sugarbeet	1671	854	63461	0	0	58080
FR	rape	857	867	64318	0	0	58080
FR	sunflower	0	938	64318	0	0	58080
FR	maize/corn (bioethanol)	288	1290	64607	0	0	58080

Hungary		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
HU	black liquor	10	0	10	10	0	10
HU	common sludges	111	25	121	111	25	121
HU	landscape care wood	301	28	422	290	28	411
HU	verge gras	7	32	429	7	32	418
HU	prunings	255	34	684	255	34	673
HU	total manure	721	53	1406	721	53	1395
HU	MSW landfill	540	57	1946	540	57	1935
HU	perennials: woody	838	63	2784	599	51	2534
HU	straw	3182	119	5965	3182	119	5715
HU	grass cuttings abandoned grassland	37	119	6002	37	119	5752
HU	animal waste	60	119	6062	60	119	5812
HU	MSW (Not landfill, composting, recycling)	86	119	6149	86	119	5898
HU	saw-dust	15	121	6164	15	121	5914
HU	perennials: grassy	680	131	6844	599	155	6512
HU	post consumer wood	102	137	6946	102	137	6615
HU	sawmill by-products (excl sawdust)	31	151	6978	31	151	6646
HU	other industrial wood residues	47	151	7024	47	151	6693
HU	primary forestry residues	574	165	7598	285	165	6978
HU	forrage maize (biogas)	1853	205	9451	0	0	6978
HU	paper cardboard	152	323	9603	152	323	7130
HU	used fats and oils	43	368	9645	43	368	7173
HU	cereals	3	375	9649	0	0	7173
HU	additional harvestable roundwood	665	397	10314	633	397	7805
HU	roundwood	772	496	11086	772	496	8577
HU	rape	1	766	11087	0	0	8577
HU	maize/corn (bioethanol)	2	963	11089	0	0	8577
HU	sunflower	4	1009	11093	0	0	8577
HU	sugarbeet	0	1197	11093	0	0	8577

Ireland		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
IE	forrage maize (biogas)	0	0	0	0	0	0
IE	perennials: woody	0	0	0	0	0	0
IE	black liquor	0	0	0	0	0	0
IE	common sludges	51	25	51	51	25	51
IE	verge gras	10	32	61	10	32	61
IE	MSW landfill	180	57	241	180	57	241
IE	post consumer wood	88	80	329	88	80	329
IE	landscape care wood	259	97	588	250	97	579
IE	animal waste	56	119	644	56	119	634
IE	MSW (Not landfill, composting, recycling)	172	119	816	172	119	807
IE	prunings	0	121	816	0	121	807
IE	grass cuttings abandoned grassland	101	121	917	101	121	908
IE	total manure	0	123	917	0	123	908
IE	perennials: grassy	16	128	934	12	125	920
IE	straw	55	130	989	55	129	975
IE	saw-dust	60	164	1049	60	164	1035
IE	sawmill by-products (excl sawdust)	111	205	1160	111	205	1147
IE	other industrial wood residues	62	205	1223	62	205	1209
IE	paper cardboard	159	323	1382	159	323	1368
IE	primary forestry residues	148	331	1530	56	331	1425
IE	used fats and oils	17	368	1547	17	368	1442
IE	cereals	0	385	1547	0	0	1442
IE	additional harvestable roundwood	135	529	1682	124	529	1565
IE	sunflower	0	620	1682	0	0	1565
IE	roundwood	401	661	2083	401	661	1967
IE	maize/corn (bioethanol)	0	821	2083	0	0	1967
IE	rape	0	867	2083	0	0	1967
IE	sugarbeet	0	1285	2083	0	0	1967

Italy		Reference scenario			Sustainability scenario		
country	**type potential**	**Ktoe**	**Euro/Toe**	**Aq. Ktoe**	**Ktoe**	**Euro/Toe**	**Aq. Ktoe**
IT	perennials: woody	0	0	0	134	61	134
IT	black liquor	46	0	46	46	0	179
IT	common sludges	623	25	669	623	25	803
IT	verge gras	105	32	774	105	32	908
IT	Landscape care wood	230	41	1005	222	41	1130
IT	prunings	2067	51	3071	2067	51	3196
IT	MSW landfill	900	57	3971	900	57	4096
IT	post consumer wood	1051	76	5023	1051	76	5148
IT	total manure	4834	98	9856	4834	98	9981
IT	animal waste	36	119	9892	36	119	10017
IT	MSW (Not landfill, composting, recycling)	689	119	10581	689	119	10706
IT	saw-dust	62	123	10643	62	123	10768
IT	straw	3205	149	13848	3205	149	13973
IT	grass cuttings abandoned grassland	73	149	13920	73	149	14046
IT	sawmill by-products (excl sawdust)	125	154	14046	125	154	14171
IT	other industrial wood residues	373	154	14419	373	154	14544
IT	perennials: grassy	5535	188	19954	4358	221	18903
IT	primary forestry residues	971	231	20924	568	231	19471
IT	paper cardboard	2319	323	23244	2319	323	21790
IT	used fats and oils	248	368	23492	248	368	22039
IT	additional harvestable roundwood	2575	463	26067	2394	463	24433
IT	roundwood	984	578	27051	984	578	25417
IT	cereals	12	586	27063	0	0	25417
IT	rape	2	699	27064	0	0	25417
IT	sunflower	12	821	27077	0	0	25417
IT	sugarbeet	97	925	27174	0	0	25417
IT	forrage maize (biogas)	3435	1362	30609	0	0	25417
IT	maize/corn (bioethanol)	27	1495	30635	0	0	25417

Lithuania		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
LT	sunflower	0	0	0	0	0	0
LT	black liquor	0	0	0	0	0	0
LT	common sludges	28	25	28	28	25	28
LT	post consumer wood	58	25	87	58	25	87
LT	verge gras	4	32	91	4	32	91
LT	landscape care wood	212	35	304	205	35	296
LT	prunings	14	43	318	14	43	310
LT	MSW landfill	180	57	498	180	57	490
LT	saw-dust	76	76	574	76	76	567
LT	sawmill by-products (excl sawdust)	141	95	716	141	95	708
LT	other industrial wood residues	62	95	778	62	95	770
LT	straw	576	96	1354	576	96	1346
LT	grass cuttings abandoned grassland	27	96	1381	27	96	1374
LT	perennials: grassy	692	106	2073	588	106	1961
LT	animal waste	20	119	2094	20	119	1982
LT	MSW (Not landfill, composting, recycling)	0	119	2094	0	119	1982
LT	total manure	0	123	2094	0	123	1982
LT	perennials: woody	272	133	2365	382	133	2363
LT	primary forestry residues	632	165	2997	245	165	2609
LT	paper cardboard	25	323	3022	25	323	2633
LT	cereals	66	348	3088	0	0	2633
LT	used fats and oils	15	368	3103	15	368	2648
LT	additional harvestable roundwood	543	397	3647	522	397	3170
LT	roundwood	735	496	4381	735	496	3905
LT	rape	19	745	4400	0	0	3905
LT	sugarbeet	0	1038	4400	0	0	3905
LT	forrage maize (biogas)	0	1396	4400	0	0	3905
LT	maize/corn (bioethanol)	0	1650	4400	0	0	3905

Latvia		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
LV	maize/corn (bioethanol)	0	0	0	0	0	0
LV	Sunflower	0	0	0	0	0	0
LV	black liquor	0	0	0	0	0	0
LV	common sludges	48	25	48	48	25	48
LV	post consumer wood	44	25	91	44	25	91
LV	verge gras	4	32	95	4	32	95
LV	landscape care wood	194	34	289	187	34	282
LV	prunings	7	42	296	7	42	289
LV	MSW landfill	0	57	296	0	57	289
LV	saw-dust	229	76	525	229	76	518
LV	sawmill by-products (excl sawdust)	424	95	950	424	95	943
LV	other industrial wood residues	124	95	1074	124	95	1067
LV	perennials: grassy	0	106	1074	0	0	1067
LV	animal waste	9	119	1083	9	119	1076
LV	MSW (Not landfill, composting, recycling)	0	119	1083	0	119	1076
LV	total manure	0	123	1083	0	123	1076
LV	straw	275	129	1358	275	129	1351
LV	perennials: woody	0	133	1358	0	0	1351
LV	grass cuttings abandoned grassland	21	142	1379	21	142	1372
LV	primary forestry residues	945	165	2323	410	165	1782
LV	paper cardboard	1	323	2324	1	323	1783
LV	cereals	3	329	2327	0	0	1783
LV	used fats and oils	10	368	2337	10	368	1793
LV	additional harvestable roundwood	852	397	3189	797	397	2589
LV	roundwood	1364	496	4553	1364	496	3953
LV	forrage maize (biogas)	0	581	4553	0	0	3953
LV	rape	2	754	4555	0	0	3953
LV	Sugarbeet	0	1118	4555	0	0	3953

Netherlands		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
NL	black liquor	46	0	46	46	0	46
NL	common sludges	354	25	400	354	25	400
NL	verge gras	40	32	440	40	32	440
NL	Post consumer wood	380	45	819	380	45	819
NL	MSW landfill	180	57	999	180	57	999
NL	total manure	4574	69	5574	4574	69	5574
NL	Landscape care wood	149	80	5723	144	80	5717
NL	prunings	13	99	5736	13	99	5730
NL	saw-dust	0	110	5736	0	110	5730
NL	animal waste	128	119	5864	128	119	5859
NL	MSW (Not landfill, composting, recycling	258	119	6123	258	119	6117
NL	Perennials: grassy	55	120	6178	49	118	6166
NL	sawmill by-products (excl sawdust)	31	138	6209	31	138	6197
NL	other industrial wood residues	0	138	6209	0	138	6197
NL	Perennials: woody	25	143	6235	24	137	6221
NL	Forrage maize (biogas)	0	176	6235	0	0	6221
NL	primary forestry residues	71	231	6306	22	231	6243
NL	straw	195	247	6501	195	247	6438
NL	grass cuttings abandoned grassland	44	247	6545	44	247	6482
NL	paper cardboard	304	323	6850	304	323	6786
NL	used fats and oils	20	368	6869	20	368	6806
NL	Cereals	0	427	6869	0	0	6806
NL	additional harvestable roundwood	61	463	6930	56	463	6862
NL	Rape	0	574	6930	0	0	6862
NL	roundwood	138	578	7068	138	578	6999
NL	Sunflower	0	620	7068	0	0	6999
NL	Sugarbeet	0	1097	7068	0	0	6999
NL	maize/corn (bioethanol)	0	1809	7068	0	0	6999

Poland		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
PL	black liquor	614	0	614	614	0	614
PL	common sludges	284	25	899	284	25	899
PL	Landscape care wood	1100	30	1999	1060	30	1959
PL	verge gras	57	32	2056	57	32	2016
PL	prunings	323	37	2379	323	37	2339
PL	MSW landfill	1080	57	3459	1080	57	3419
PL	total manure	8177	88	11636	8177	88	11596
PL	saw-dust	130	119	11766	130	119	11726
PL	animal waste	509	119	12275	509	119	12235
PL	MSW (Not landfill, composting, recycling)	431	119	12706	431	119	12666
PL	Forrage maize (biogas)	0	121	12706	0	0	12666
PL	Perennials: grassy	2668	131	15374	2653	101	15319
PL	Post consumer wood	686	131	16061	686	131	16005
PL	sawmill by-products (excl sawdust)	275	148	16335	275	148	16280
PL	other industrial wood residues	498	148	16833	498	148	16778
PL	Perennials: woody	472	155	17305	392	147	17170
PL	straw	6142	156	23447	6142	156	23312
PL	grass cuttings abandoned grassland	179	156	23626	179	156	23491
PL	primary forestry residues	3220	165	26845	1262	165	24753
PL	maize/corn (bioethanol)	3	172	26849	0	0	24753
PL	paper cardboard	2611	323	29460	2611	323	27364
PL	Cereals	304	341	29764	0	0	27364
PL	used fats and oils	167	368	29931	167	368	27531
PL	Rape	43	374	29974	0	0	27531
PL	additional harvestable roundwood	2596	397	32569	2521	397	30052
PL	Sunflower	6	420	32576	0	0	30052
PL	roundwood	4939	496	37515	4939	496	34991
PL	Sugarbeet	0	594	37515	0	0	34991

Portugal		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
PT	Rape	0	0	0	0	0	0
PT	Perennials: woody	0	0	0	0	0	0
PT	black liquor	887	0	887	887	0	887
PT	common sludges	419	25	1305	419	25	1305
PT	verge gras	37	32	1343	37	32	1343
PT	Post consumer wood	131	45	1474	131	45	1474
PT	MSW landfill	360	57	1834	360	57	1834
PT	Landscape care wood	159	61	1994	154	61	1988
PT	prunings	586	76	2580	586	76	2574
PT	saw-dust	50	110	2630	50	110	2624
PT	animal waste	35	119	2665	35	119	2659
PT	MSW (Not landfill, composting, recycling)	345	119	3010	345	119	3004
PT	total manure	0	123	3010	0	123	3004
PT	straw	190	131	3199	190	131	3193
PT	sawmill by-products (excl sawdust)	106	138	3305	106	138	3299
PT	other industrial wood residues	109	138	3414	109	138	3408
PT	grass cuttings abandoned grassland	50	142	3464	50	142	3458
PT	Perennials: grassy	489	160	3953	252	151	3710
PT	maize/corn (bioethanol)	0	225	3953	0	0	3710
PT	primary forestry residues	583	231	4536	314	231	4024
PT	paper cardboard	371	323	4907	371	323	4395
PT	Forrage maize (biogas)	0	364	4907	0	0	4395
PT	used fats and oils	43	368	4950	43	368	4438
PT	Sugarbeet	0	422	4950	0	0	4438
PT	Cereals	0	436	4950	0	0	4438
PT	additional harvestable roundwood	484	463	5434	417	463	4855
PT	roundwood	857	578	6291	857	578	5712
PT	Sunflower	0	622	6291	0	0	5712

Romania		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
RO	black liquor	59	0	59	59	0	59
RO	Post consumer wood	336	19	394	336	19	394
RO	common sludges	113	25	508	113	25	508
RO	verge gras	17	32	525	17	32	525
RO	Landscape care wood	595	39	1120	573	39	1098
RO	prunings	314	48	1434	314	48	1412
RO	MSW landfill	720	57	2154	720	57	2132
RO	Perennials: woody	5949	70	8102	5418	58	7549
RO	saw-dust	195	74	8297	195	74	7744
RO	Perennials: grassy	3220	89	11518	2660	84	10405
RO	total manure	328	92	11845	328	92	10733
RO	sawmill by-products (excl sawdust)	412	92	12257	412	92	11144
RO	other industrial wood residues	187	92	12444	187	92	11331
RO	straw	3312	105	15756	3312	105	14643
RO	Sugarbeet	232	114	15988	0	0	14643
RO	animal waste	12	119	15999	12	119	14655
RO	MSW (Not landfill, composting, recycling)	345	119	16344	345	119	15000
RO	grass cuttings abandoned grassland	309	142	16652	309	142	15308
RO	primary forestry residues	1364	165	18017	768	165	16076
RO	maize/corn (bioethanol)	39	279	18056	0	0	16076
RO	Rape	154	286	18210	0	0	16076
RO	paper cardboard	125	323	18335	125	323	16201
RO	used fats and oils	97	368	18431	97	368	16298
RO	additional harvestable roundwood	2573	397	21005	2520	397	18818
RO	Cereals	76	430	21081	0	0	18818
RO	Sunflower	148	481	21229	0	0	18818
RO	roundwood	1783	496	23012	1783	496	20601
RO	Forrage maize (biogas)	0	1114	23012	0	0	20601

Sweden		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
SE	maize/corn (bioethanol)	0	0	0	0	0	0
SE	Sunflower	0	0	0	0	0	0
SE	black liquor	4684	0	4684	4684	0	4684
SE	common sludges	315	25	4998	315	25	4998
SE	verge gras	31	32	5030	31	32	5030
SE	MSW landfill	360	57	5390	360	57	5390
SE	Landscape care wood	859	71	6249	828	71	6218
SE	prunings	22	89	6271	22	89	6239
SE	Post consumer wood	161	92	6431	161	92	6400
SE	saw-dust	1021	103	7452	1021	103	7421
SE	Perennials: grassy	323	119	7776	274	115	7695
SE	animal waste	37	119	7812	37	119	7732
SE	MSW (Not landfill, composting, recycling)	86	119	7898	86	119	7818
SE	total manure	0	123	7898	0	123	7818
SE	sawmill by-products (excl sawdust)	2044	129	9942	2044	129	9862
SE	other industrial wood residues	311	129	10253	311	129	10173
SE	straw	596	165	10849	596	165	10769
SE	grass cuttings abandoned grassland	55	165	10905	55	165	10824
SE	primary forestry residues	7965	165	18870	3561	165	14385
SE	Perennials: woody	304	185	19174	277	141	14662
SE	Sugarbeet	0	284	19174	0	0	14662
SE	paper cardboard	555	323	19730	555	323	15217
SE	Cereals	307	339	20036	0	0	15217
SE	used fats and oils	38	368	20075	38	368	15255
SE	additional harvestable roundwood	4789	397	24863	4250	397	19506
SE	Forrage maize (biogas)	0	415	24863	0	0	19506
SE	Rape	51	416	24915	0	0	19506
SE	roundwood	9561	496	34475	9561	496	29066

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Slovenia		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
SI	Forrage maize (biogas)	0	0	0	0	0	0
SI	Perennials: woody	0	0	0	0	0	0
SI	black liquor	0	0	0	0	0	0
SI	common sludges	45	25	45	45	25	45
SI	verge gras	4	32	49	4	32	49
SI	Landscape care wood	38	41	87	37	41	86
SI	prunings	20	51	107	20	51	105
SI	MSW landfill	180	57	287	180	57	285
SI	Post consumer wood	29	69	316	29	69	315
SI	total manure	0	92	316	0	92	315
SI	saw-dust	25	94	341	25	94	339
SI	sawmill by-products (excl sawdust)	53	117	394	53	117	392
SI	other industrial wood residues	62	117	456	62	117	455
SI	animal waste	8	119	464	8	119	462
SI	MSW (Not landfill, composting, recycling)	0	119	464	0	119	462
SI	Perennials: grassy	96	135	560	38	131	500
SI	straw	127	161	687	127	161	627
SI	grass cuttings abandoned grassland	9	161	696	9	161	636
SI	primary forestry residues	228	165	924	132	165	768
SI	maize/corn (bioethanol)	0	191	924	0	0	768
SI	paper cardboard	55	323	979	55	323	823
SI	used fats and oils	9	368	988	9	368	831
SI	additional harvestable roundwood	825	397	1813	773	397	1604
SI	Rape	0	468	1813	0	0	1604
SI	roundwood	399	496	2211	399	496	2003
SI	Cereals	0	499	2211	0	0	2003
SI	Sunflower	0	575	2211	0	0	2003
SI	Sugarbeet	0	651	2211	0	0	2003

Slovakia		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
SK	black liquor	317	0	317	317	0	317
SK	Forrage maize (biogas)	0	21	317	0	0	317
SK	common sludges	109	25	426	109	25	426
SK	verge gras	14	32	440	14	32	440
SK	Landscape care wood	173	39	613	167	39	607
SK	prunings	9	48	622	9	48	616
SK	MSW landfill	180	57	802	180	57	796
SK	Post consumer wood	44	69	846	44	69	839
SK	saw-dust	84	94	929	84	94	923
SK	Perennials: grassy	549	112	1478	455	112	1378
SK	sawmill by-products (excl sawdust)	181	117	1659	181	117	1559
SK	other industrial wood residues	93	117	1752	93	117	1652
SK	animal waste	10	119	1762	10	119	1662
SK	MSW (Not landfill, composting, recycling)	86	119	1849	86	119	1748
SK	total manure	321	123	2170	321	123	2070
SK	Perennials: woody	63	139	2233	42	59	2112
SK	straw	827	161	3060	827	161	2939
SK	grass cuttings abandoned grassland	50	161	3109	50	161	2988
SK	primary forestry residues	548	165	3657	310	165	3298
SK	maize/corn (bioethanol)	4	175	3661	0	0	3298
SK	Rape	4	318	3665	0	0	3298
SK	paper cardboard	84	323	3749	84	323	3382
SK	Cereals	6	361	3755	0	0	3382
SK	used fats and oils	23	368	3778	23	368	3405
SK	additional harvestable roundwood	121	397	3900	95	397	3500
SK	roundwood	1275	496	5174	1275	496	4775
SK	Sugarbeet	0	572	5174	0	0	4775
SK	Sunflower	1	576	5176	0	0	4775

UK		Reference scenario			Sustainability scenario		
country	type potential	Ktoe	Euro/Toe	Aq. Ktoe	Ktoe	Euro/Toe	Aq. Ktoe
UK	maize/corn (bioethanol)	0	0	0	0	0	0
UK	black liquor	300	0	300	300	0	300
UK	common sludges	1866	25	2166	1866	25	2166
UK	verge gras	135	32	2301	135	32	2301
UK	Post consumer wood	1154	41	3455	1154	41	3455
UK	Landscape care wood	851	45	4306	820	45	4275
UK	prunings	15	56	4321	15	56	4290
UK	MSW landfill	1800	57	6121	1800	57	6090
UK	saw-dust	161	109	6282	161	109	6251
UK	animal waste	639	119	6921	639	119	6890
UK	MSW (Not landfill, composting, recycling)	1981	119	8903	1981	119	8872
UK	Forrage maize (biogas)	0	121	8903	0	0	8872
UK	Perennials: grassy	3101	122	12004	2489	120	11361
UK	total manure	942	123	12945	942	123	942
UK	straw	2114	129	15059	2114	129	3056
UK	grass cuttings abandoned grassland	536	129	15595	536	129	3592
UK	Perennials: woody	418	136	16013	383	141	3975
UK	sawmill by-products (excl sawdust)	322	136	16334	322	136	4297
UK	other industrial wood residues	187	136	16521	187	136	4483
UK	primary forestry residues	586	231	17107	358	231	4841
UK	paper cardboard	2089	323	19196	2089	323	6930
UK	used fats and oils	256	368	19453	256	368	7186
UK	Cereals	346	419	19799	0	0	7186
UK	additional harvestable roundwood	963	463	20761	896	463	8083
UK	Rape	115	466	20877	0	0	8083
UK	Sugarbeet	0	493	20877	0	0	8083
UK	roundwood	1128	578	22005	1128	578	9211
UK	Sunflower	0	605	22005	0	0	9211

Annex 9 Lower heating values used

Category	Modified category	LHV dry [GJ/ton]	Moisture Content [%]	LHV as received [GJ/ton]
dry manure				0.0
wet manure		3.6	0.9	1.9
straw_2004		18.0	0.1	16.2
verge gras		N/A	0.3	11.7
prunings		18.0	0.1	16.2
wood-waste		19.6	0.3	15.1
animal waste		N/A	0.6	N/A
	MSW	15.1	0.5	10.1
bmw	Landfill gas	N/A	N/A	21.0
paper cardboard		19.6	0.1	18.6
common sludges		N/A	N/A	21.0
Maize 2008		16.1	0.1	14.4
OSR 2008		24.4	0.1	21.5
		17.3	0.8	9.6
Sugarbeet 2008				0.0
Sunflower 2008		24.4	0.1	21.5
Cereals		18.0	0.1	16.2
Woody crops	Woody crops	18.4	0.3	14.2
Grassy crops	Grassy crops	17.8	0.3	13.4
Additional Harvestable Roundwood		18.4	0.3	14.2
Primary Forestry Residues		19.6	0.3	15.1
Black liquor		N/A	N/A	10.0
Used fats/oils		36.0	0.0	36.0
Palm oil		36.5	0.0	36.5